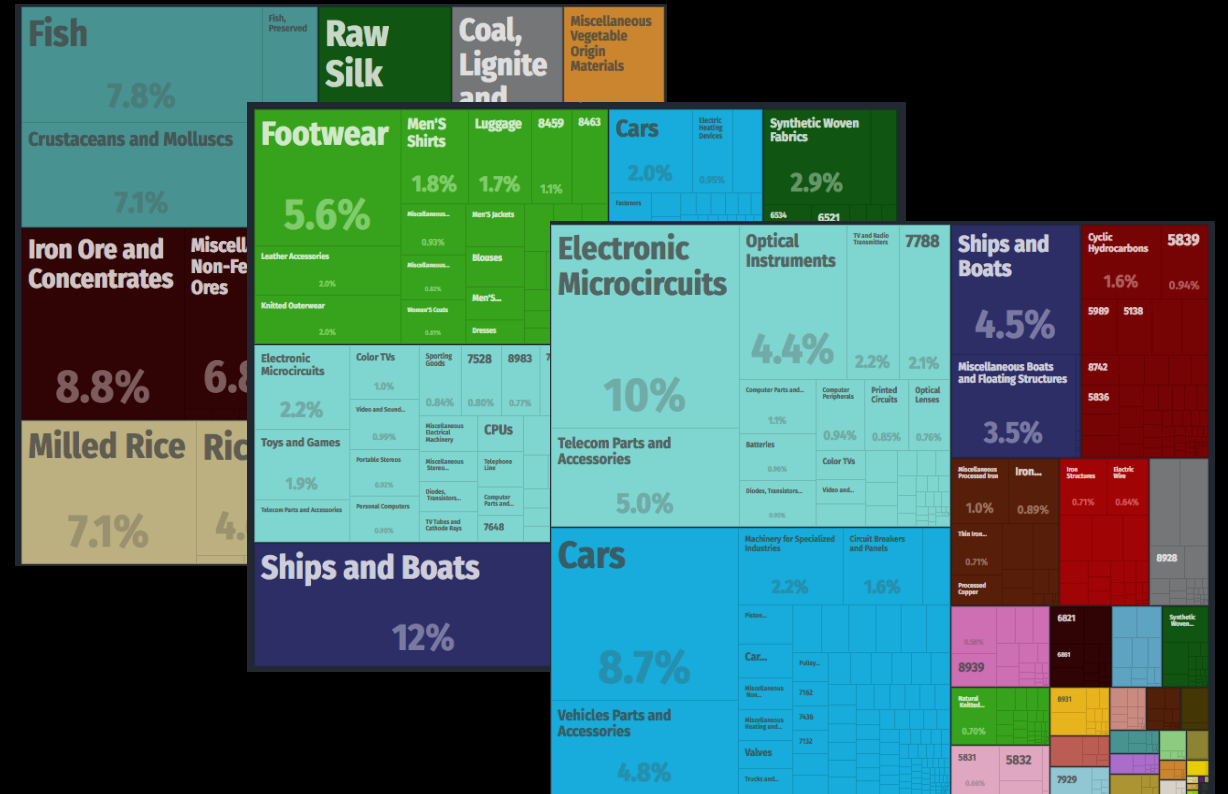
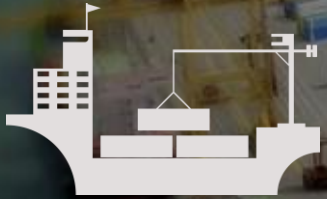


Different things



Korea (1962, 1985, 2015)

Economies learn



New
Products



New
Research
Areas



New
Occupations

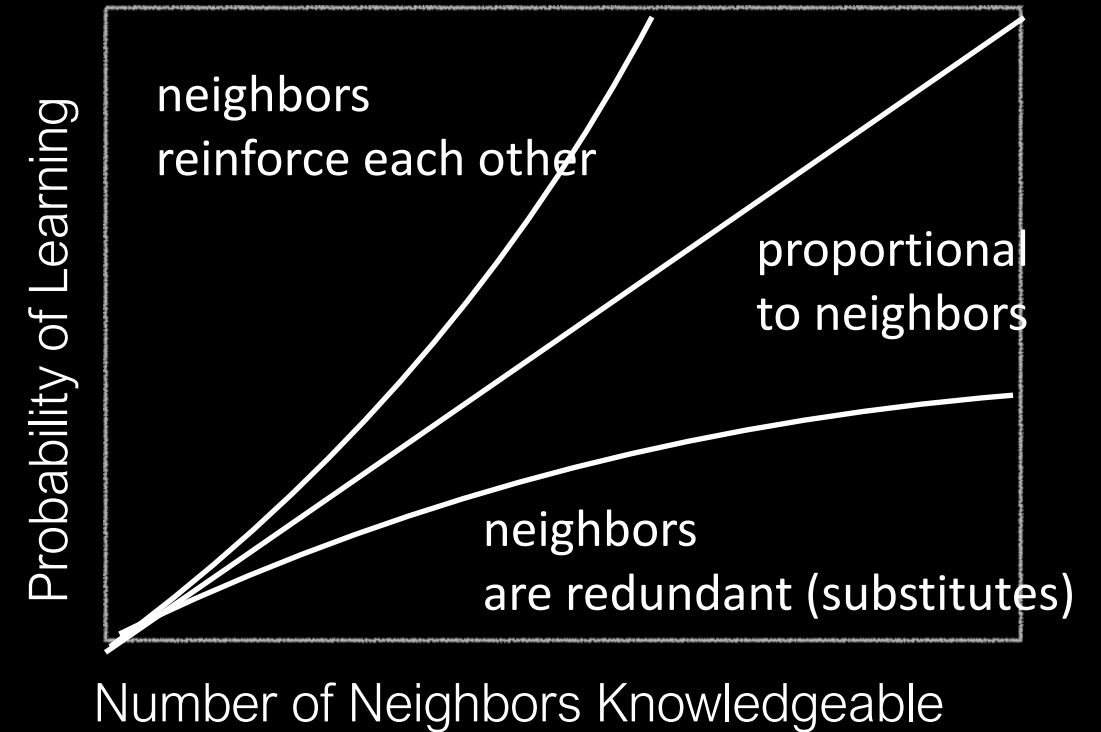
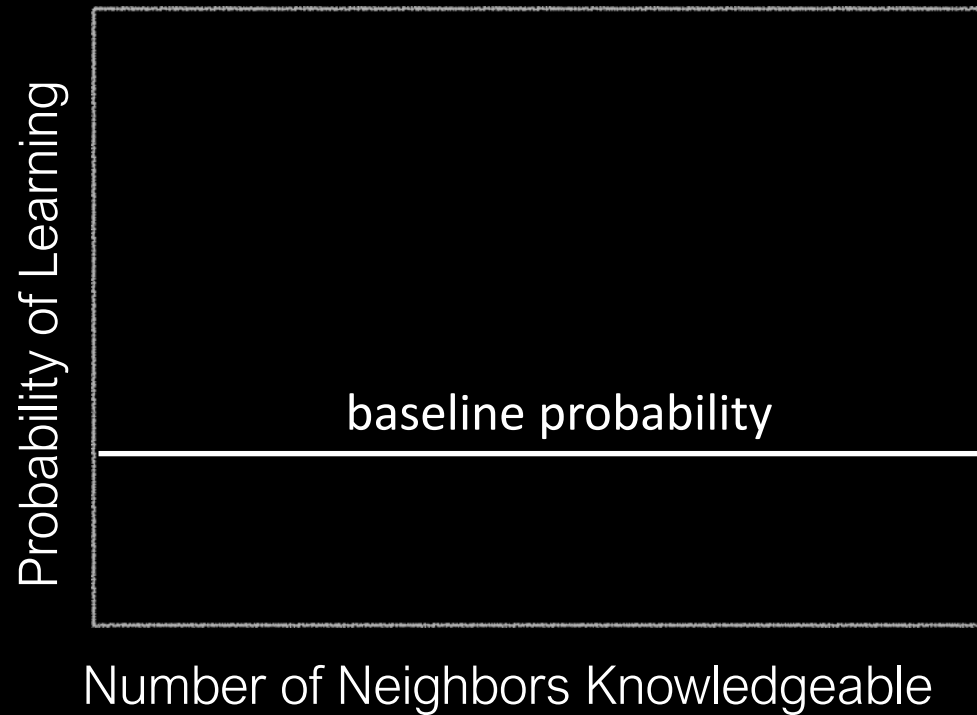
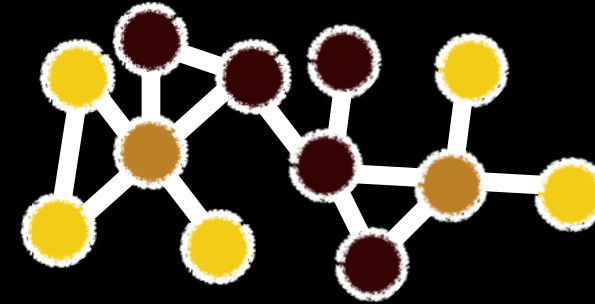
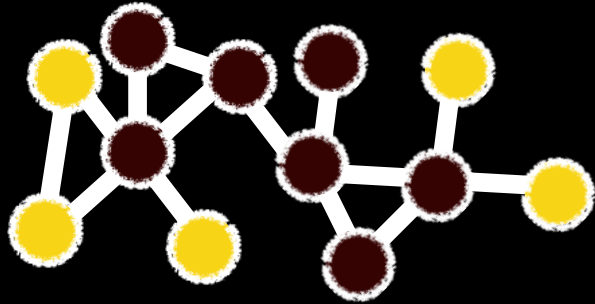


New
Skills



New
Industries

Collective Learning



WHO ARE THE KNOWLEDGE NEIGH

NEARBY LOCATIONS

(geographically, culturally, linguistically)



RELATED ACTIVITIES

(products, industries, patents, research areas)



WHO ARE THE KNOWLEDGE NEIGH

NEARBY LOCATIONS

(geographically, culturally, linguistically)

How far is Paris from
Toulouse?

680kms

How long is the train ride
from Shanghai to
Beijing?

4:30 hours
(1,200 kms)

How many people have
lived in Boston and New
York?

~1-2 million (40)

RELATED ACTIVITIES

(products, industries, patents, research areas)

How far is airplane manufacturing
from pharmaceuticals?

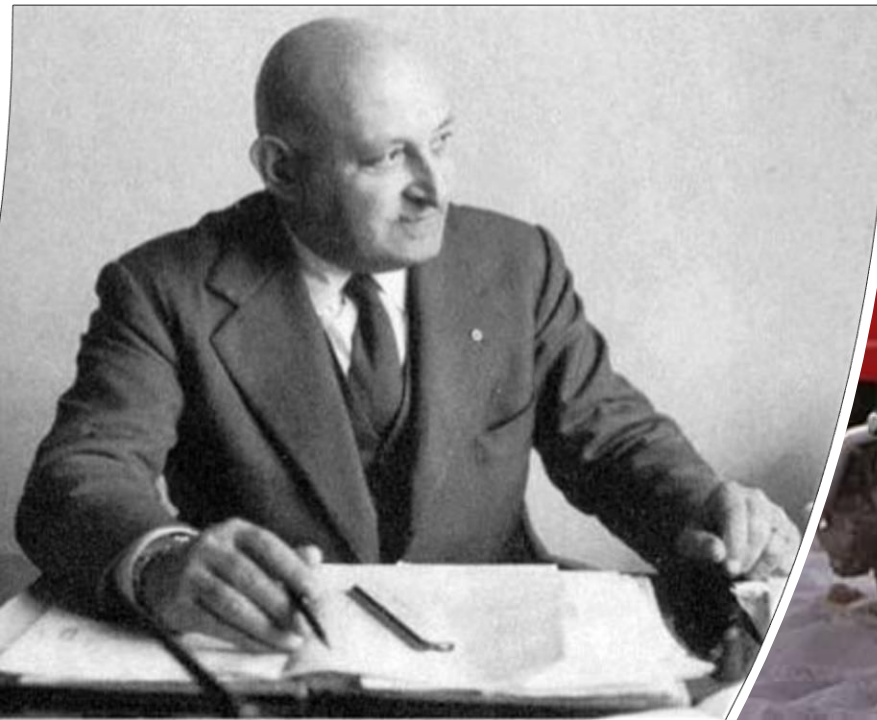
How far is Cleveland from
biotech?

How “easy” it would be for Peru
to export blueberries to Egypt?



Corradino D'Ascanio

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



AICHI



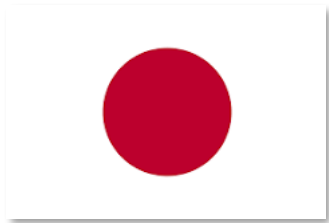
Aichi E13A (Jake)

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



Heinkel He 111

AICHI



Giant AA1 minicar (1947)



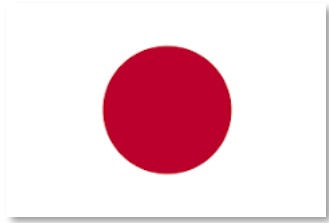
Aichi Machine Industry



Brands
Giant
Cony



The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



KAWANISHI

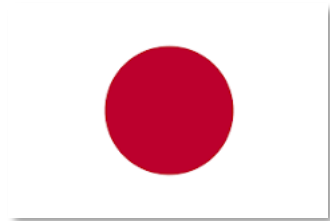


Kawanishi N1K fighter (Rex)



H8K, Flying boat (Emily)

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



KAWANISHI

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



Pointer/Super Lassie Motorcycle



HEINKEL



HE 178 (first to fly with a jet engine)

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)



HEINKEL



Foto: Hannes Denzel / MVCA

Heinkel Tourist

The Infinite Alphabet:
and the laws of knowledge. Hidalgo CA.
Penguin-Random House (2025)

Estimating Relatedness-Density

Three conceptual steps.

1. Estimate specialization:

an estimate of the overexpression of an economy in an activity

2. Define proximity:

an estimate of distance or closeness between pairs of economies or activities

3. Estimate Density:

Calculate the fraction of related specializations that are present in the economy

Example 1:

Estimating Relatedness-Density

Let X_{cp} be exports by country c and product p , value added in city c by industry p , etc.

1. Estimate specialization:

$$\text{RCA}_{cp} = \frac{\frac{x_{cp}}{\sum_{p'} x_{cp'}}}{\frac{\sum_{c'} x_{c'p}}{\sum_{c',p'} x_{c'p'}}}$$

Binary specialization matrix

$$M_{cp} = \begin{cases} 1 & \text{if } \text{RCA}_{cp} \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

2. Define proximity:

$$k_p = \sum_c M_{cp} \quad (\text{ubiquity of product } p)$$

Proximity matrix

$$\phi_{pp'} = \frac{\sum_c M_{cp} M_{cp'}}{\max(k_p, k_{p'})}$$

3. Estimate Density:

$$\omega_{cp} = \frac{\sum_{p'} \phi_{pp'} M_{cp'}}{\sum_{p'} \phi_{pp'}}$$

Estimating Entries and Exits

Relatedness is most often used in models to explain entries and exits. This is usually tested using regressions.

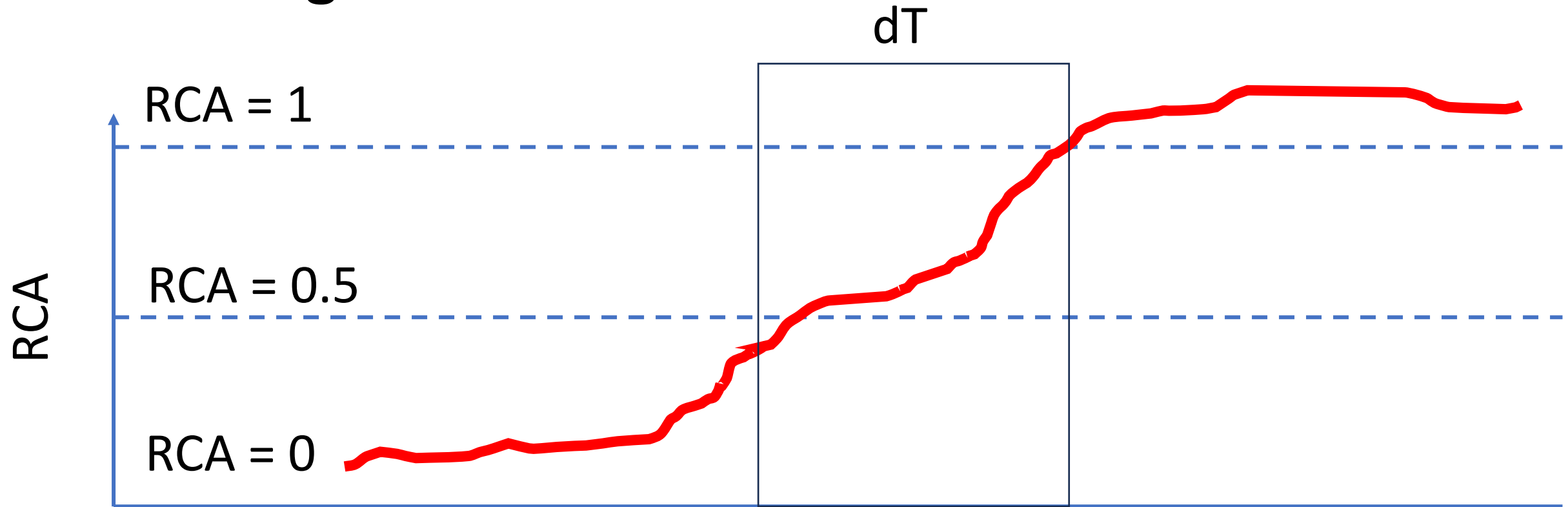
An “Entry” is the event of an economy becoming specialized in an activity. This usually involves a forward and backward condition.

For example, an economy is considered to have entered an activity if it has comparative advantage lower than a value for a pre-event window and higher than a value for the entirety of a post-event window.

E.g. $RCA < 0.5$ for the three years prior to the event and $RCA > 1$ for three years after the event.

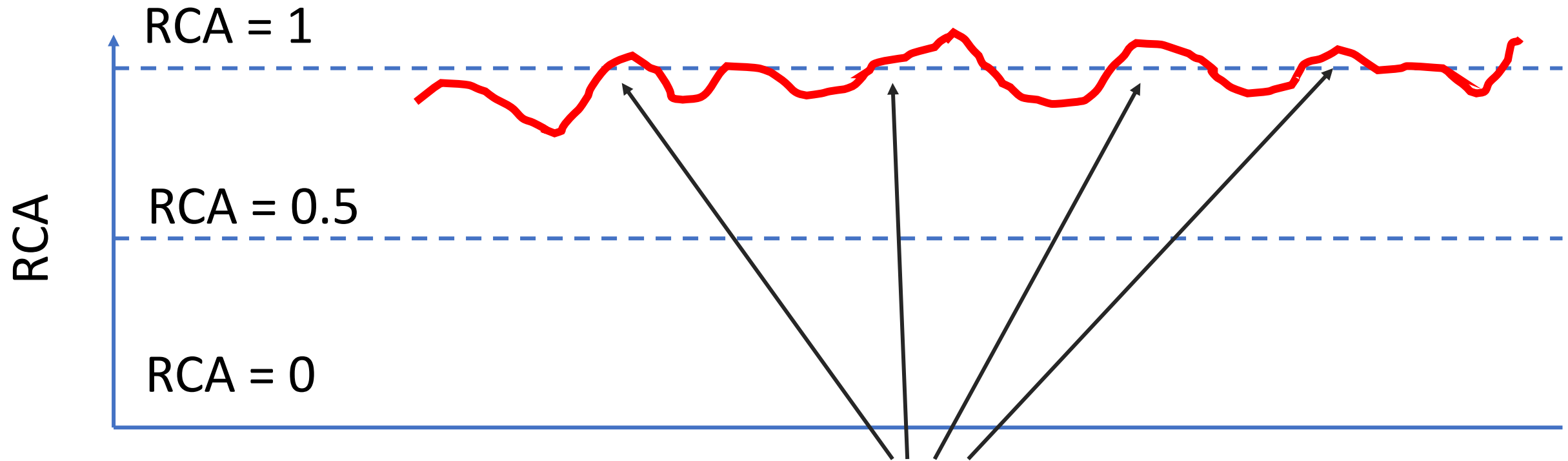
Exits follow the analogous criterion, but for losing an specialization.

Estimating Entries and Exits



This represents an entry over a time window dT

Estimating Entries and Exits



These events do not represent entries!

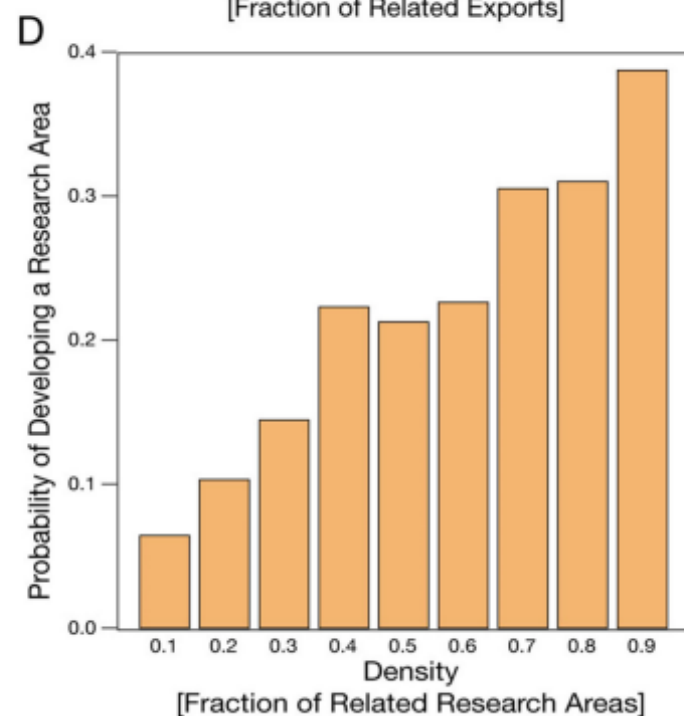
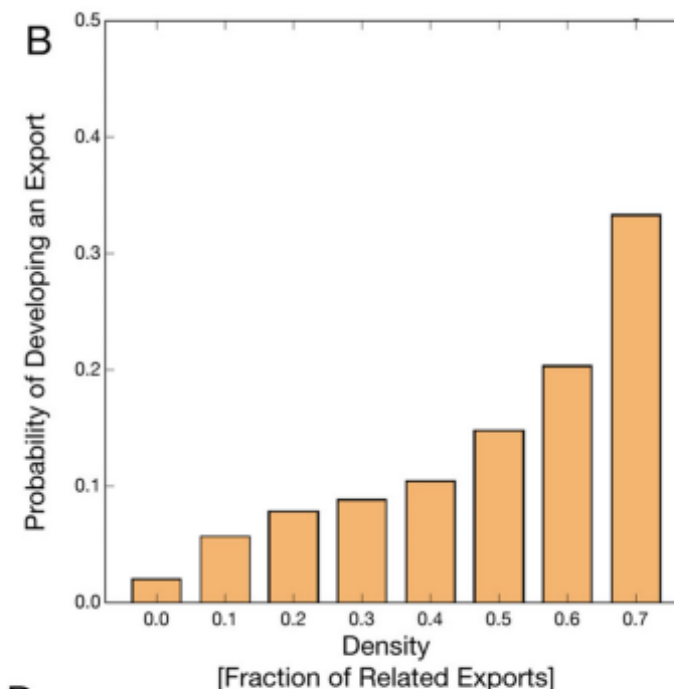
Testing the Impact of Relatedness on Entries and Exits

Entry Regression (Dependent Variable: $\text{Entry}_{cpt+1} \in \{0,1\}$)

(Indicator = 1 if country develops $\text{RCA} \geq 1$ in product in 10 years)

Density _{cpt}	0.135*** (0.015)
RCA _{cpt}	0.245*** (0.062)
Constant	0.182 (0.028)
Product Fixed Effects	Yes
Country Fixed Effects	Yes
Observations	480,000
R ²	0.152

Standard errors in parentheses. *** $p < 0.01$.



A key distinction!

Entry Regression (Dependent Variable: $\text{Entry}_{\text{cpt}+1} \in (0,1)$)	
(Indicator = 1 if country develops $\text{RCA} \geq 1$ in product in 10 years)	
Density _{cpt}	0.135*** (0.015)
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Constant	0.182 (0.028)
Product Fixed Effects	Yes
Country Fixed Effects	Yes
Observations	480,000
R ²	0.152

Standard errors in parentheses. *** $p < 0.01$.

Relatedness is an important component of a model of economic potential, but not an estimate of economic potential (that's the model).

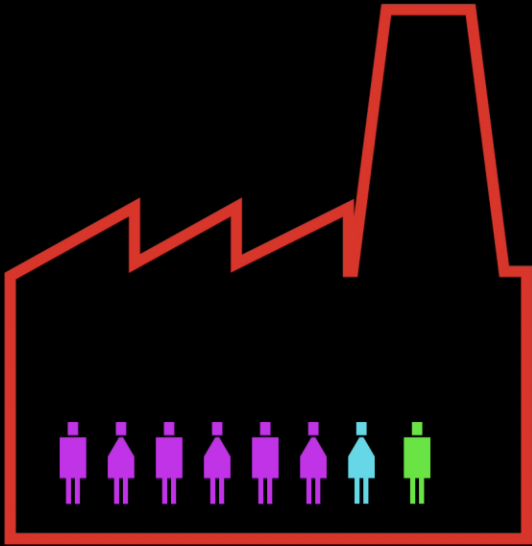
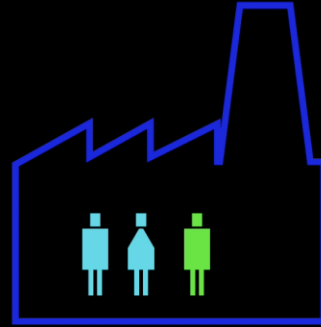
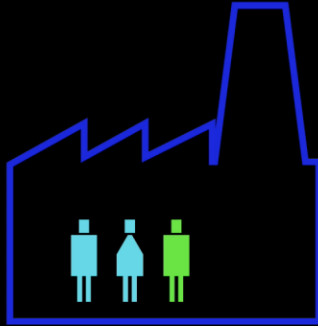
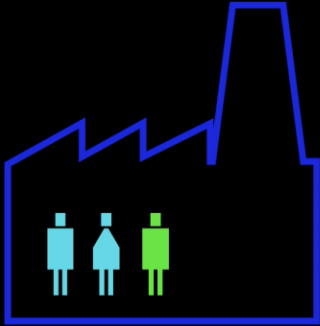
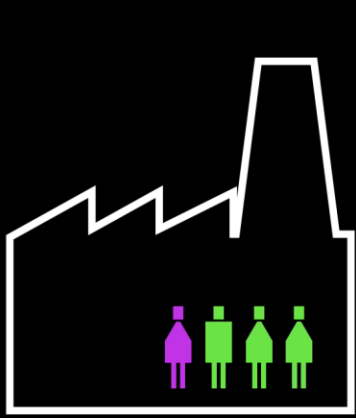




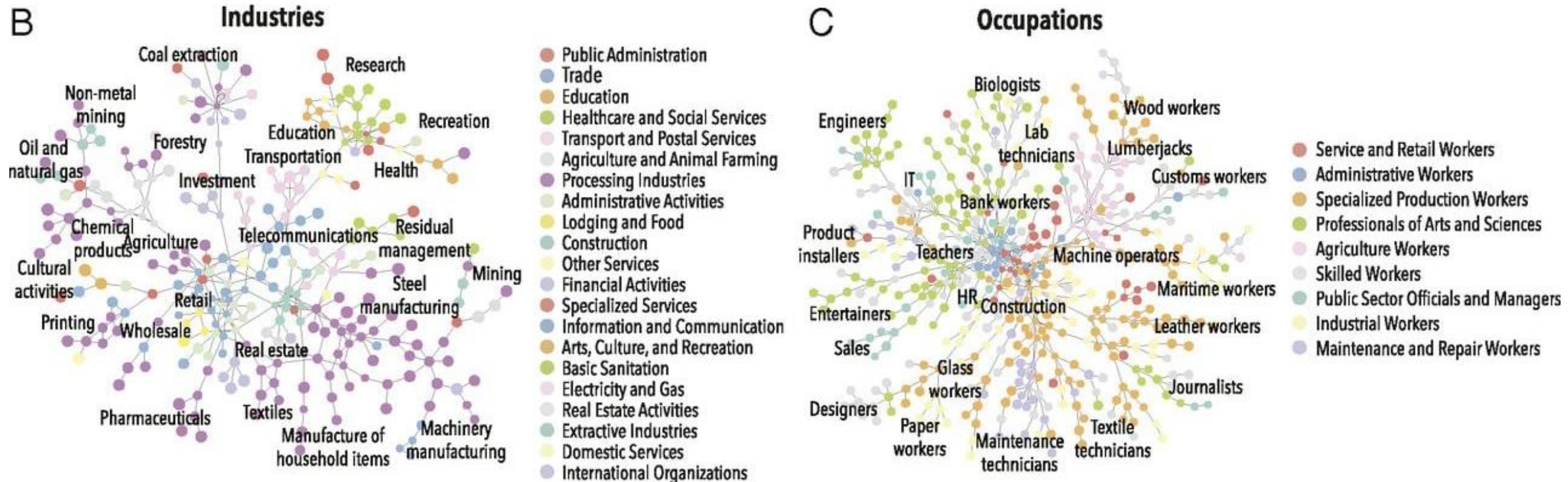
Extending & Unpacking Relatedness

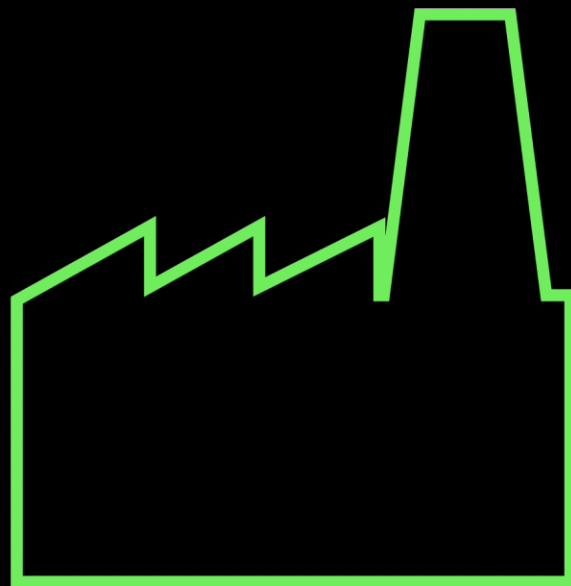
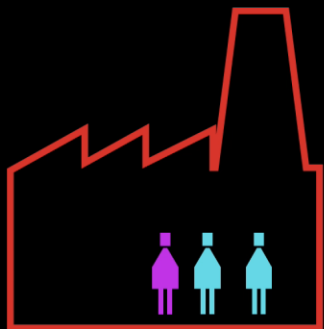
Microdiversification

An economy entering an activity is a macroeconomic outcome that can be decomposed into smaller units (such as firms and the individuals that compose them). Relatedness' impact on diversification can also be studied at this more granular scale.

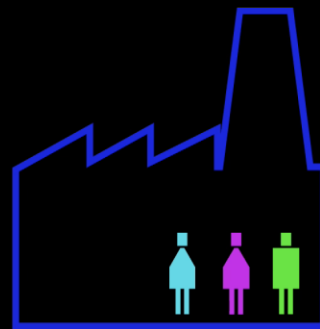


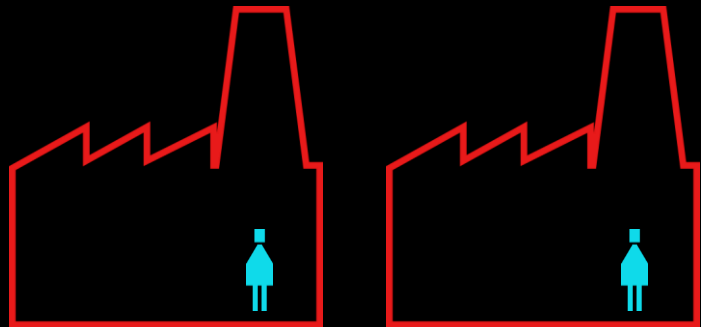
We can use work histories/career trajectories to create complementary maps of the relatedness of activities



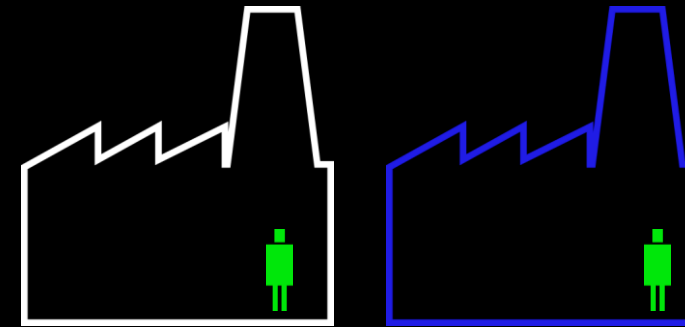


Pioneer Firm

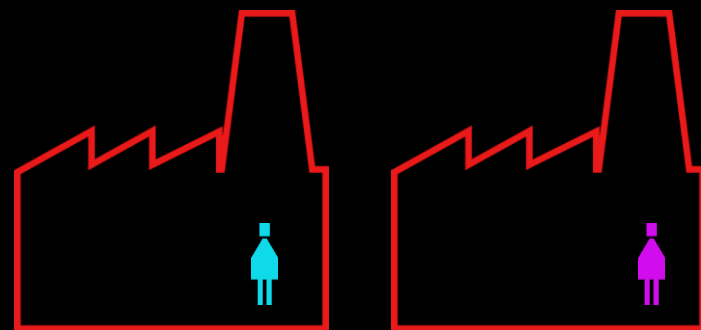




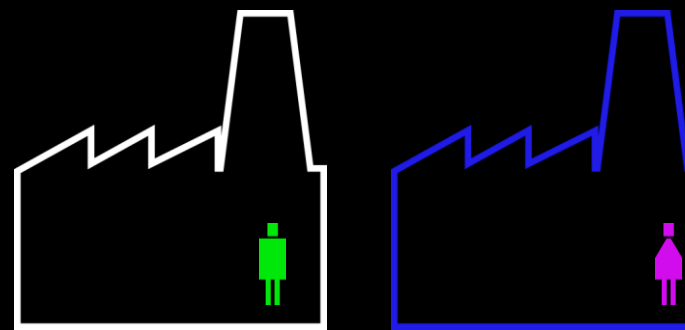
Same Industry & Occupation



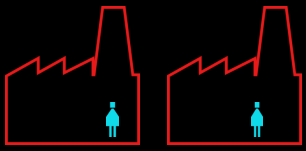
Different Industry, Same Occupation



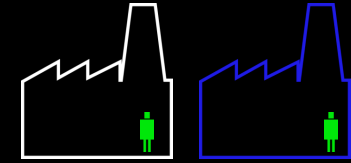
Same Industry, Different Occupation



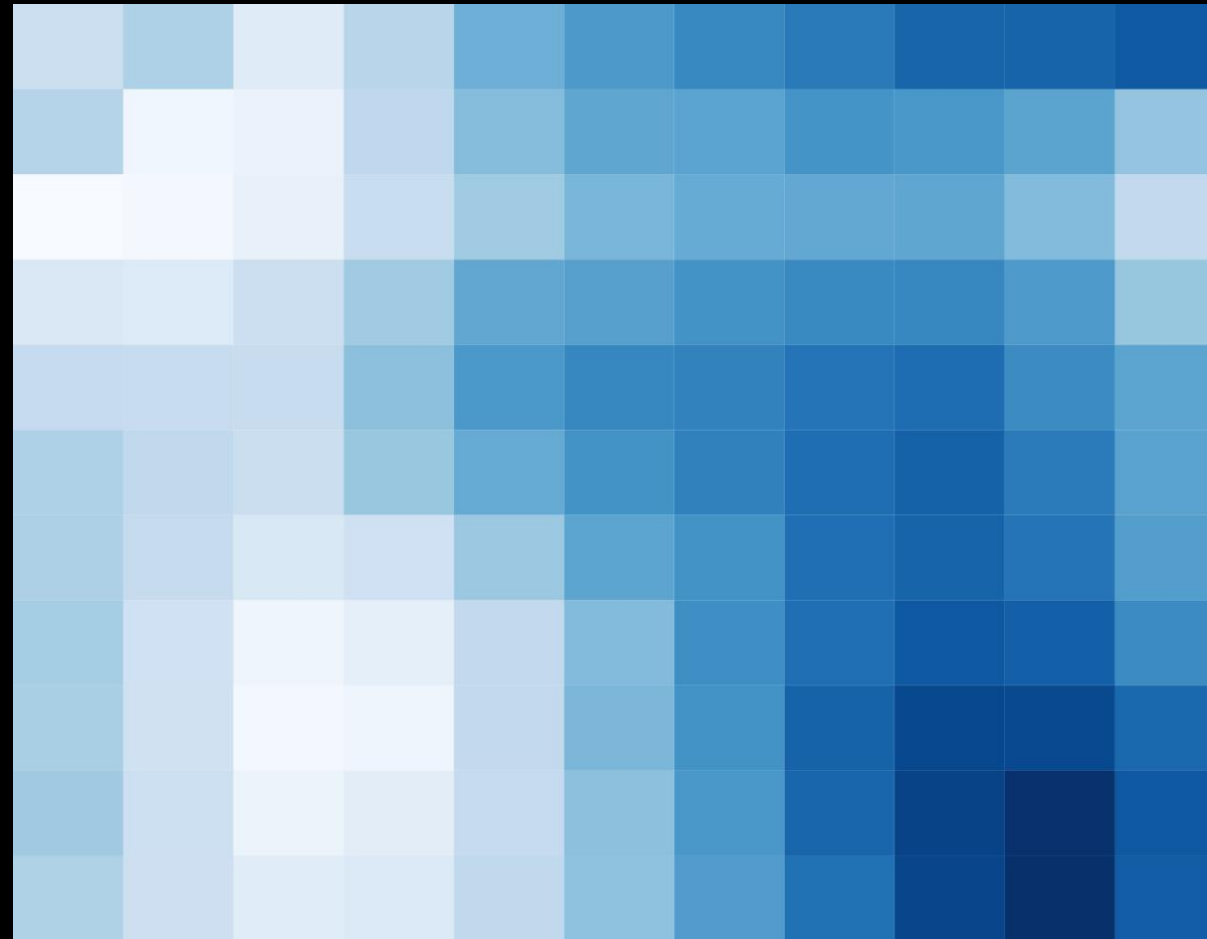
Different Industry & Occupation



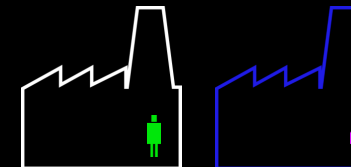
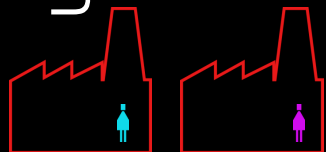
Who do pioneer companies hire?

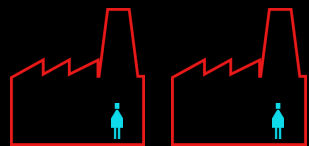


Unrelated Occupations, Related Occupation

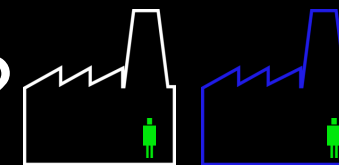


Related Industries - Unrelated Industries

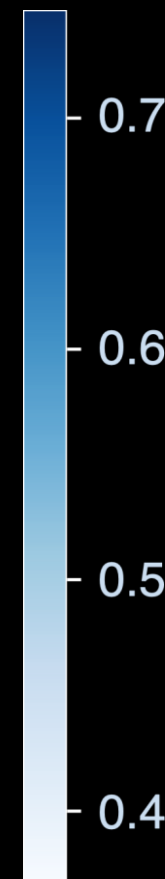
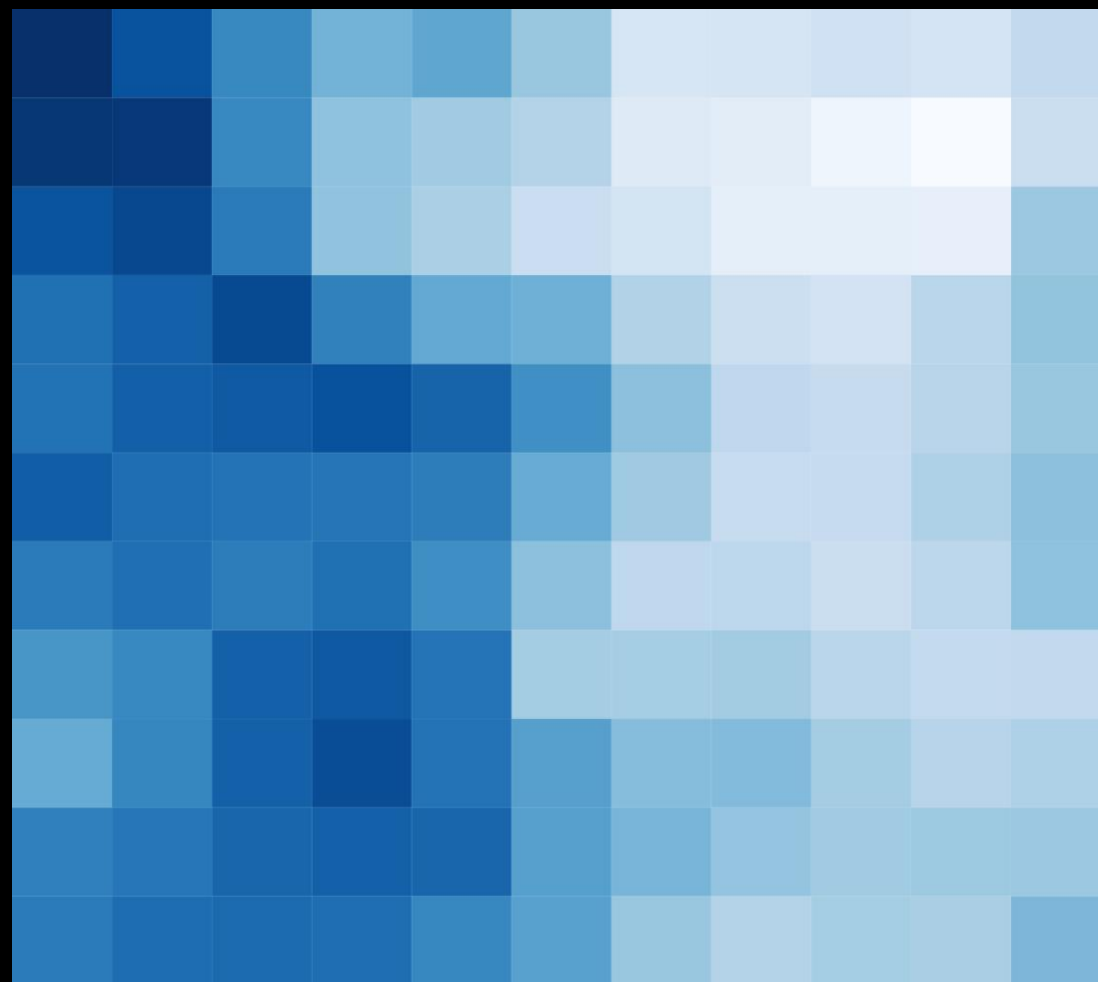




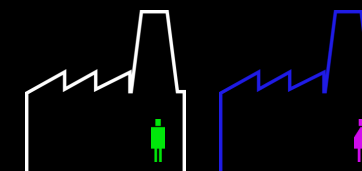
Which pioneer companies survive?



Unrelated Occupations, Related Occupation



Related Industries - Unrelated Industries



Dependent variable:

	Survival rate at third year, $S^{(t+3)}$						Three year growth rate, $G^{(t+3)}$					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Industry knowl. (Φ)		0.466*** (0.114)				0.457*** (0.123)		0.174*** (0.029)				0.185*** (0.031)
Occupation knowl. (Ψ)			0.184** (0.085)			0.035 (0.092)			0.033 (0.022)			−0.029 (0.023)
Years of schooling (edu)				0.163* (0.086)		0.134 (0.091)				0.023 (0.025)		0.012 (0.025)
Local knowledge (ρ)					0.238*** (0.071)	0.228*** (0.072)					0.014 (0.019)	0.007 (0.019)
Initial size ($\log(n_0)$)	−0.246*** (0.093)	−0.251*** (0.095)	−0.261*** (0.094)	−0.226** (0.092)	−0.235** (0.093)	−0.227** (0.096)	−0.393*** (0.031)	−0.394*** (0.030)	−0.395*** (0.031)	−0.391*** (0.031)	−0.393*** (0.031)	−0.391*** (0.030)
Average wage ($\log(w)$)	0.208 (0.220)	0.136 (0.233)	0.188 (0.221)	0.137 (0.224)	0.342 (0.235)	0.202 (0.257)	0.231*** (0.071)	0.209*** (0.069)	0.228*** (0.071)	0.221*** (0.072)	0.238*** (0.072)	0.208*** (0.071)
Year f.e.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Industry f.e.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Region f.e.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	1,632	1,632	1,632	1,632	1,632	1,632	1,376	1,376	1,376	1,376	1,376	1,376
McFadden	0.2128	0.2265	0.2161	0.2153	0.2212	0.2367						
AICc	1,635.9	1,619.1	1,633.9	1,635.0	1,626.6	1,612.7						
Log Likelihood	−558.1	−548.4	−555.8	−556.3	−552.1	−541.1						
R ²							0.324	0.343	0.325	0.324	0.324	0.344
Adjusted R ²							0.194	0.216	0.194	0.194	0.194	0.215
F Statistic							2.490*** (df = 222)	2.699*** (df = 223)	2.487*** (df = 223)	2.481*** (df = 223)	2.480*** (df = 223)	2.665*** (df = 226)

Relatedness

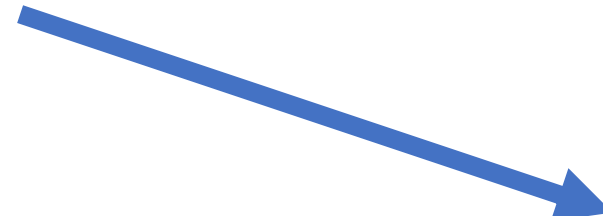
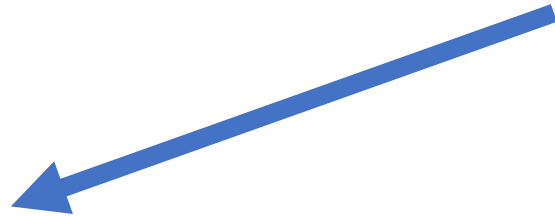
Country Product

$$\omega_{cp} = \frac{\sum_{p'} M_{cp'} \phi_{pp'}}{\sum_{p'} \phi_{pp'}}$$



Bilateral Relatedness

Country Product Destination



$$\omega_{opd} = \sum_{p'} \frac{\phi_{pp'}}{\phi_p} \cdot \frac{x_{op'd}}{x_{od}}$$

Bilateral Product Relatedness

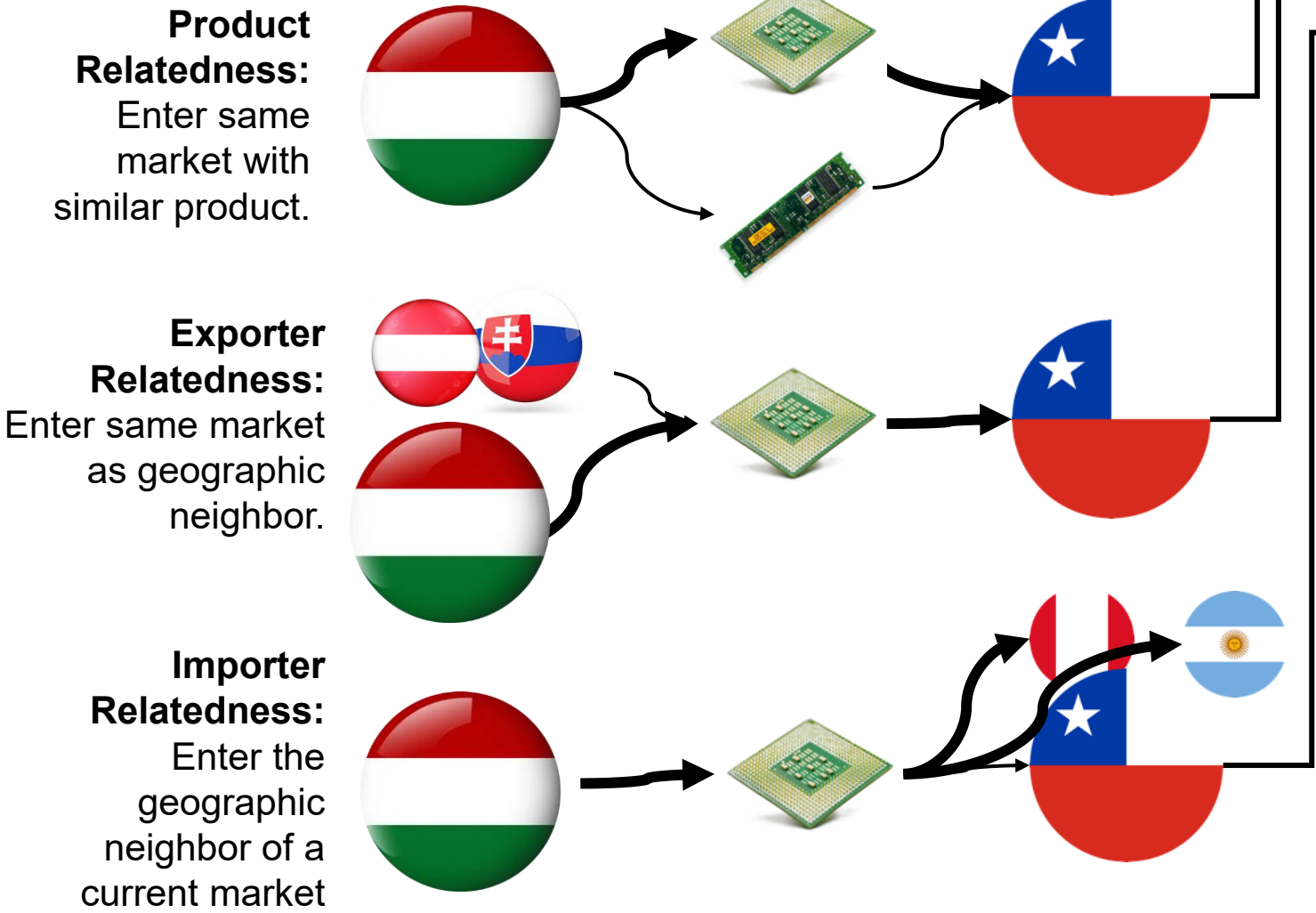
$$\Omega_{opd}^{(d)} = \sum_{d'} \frac{1/D_{dd'}}{1/D_d} \cdot \frac{x_{opd'}}{x_{op}},$$

Bilateral Importer Relatedness

$$\Omega_{opd}^{(o)} = \sum_{o'} \frac{1/D_{oo'}}{1/D_o} \cdot \frac{x_{o'pd}}{x_{pd}},$$

Bilateral Exporter Relatedness

Bilateral Relatedness



	Dependent variable: $\log x_{opd}^{t+2}$		
	(1) 2000–2006	(2) 2007–2012	(3) 2012–2015
ω_{opd}^t	0.209*** (0.017)	0.180*** (0.017)	0.152*** (0.015)
$\Omega_{opd}^{(d)}$	0.143*** (0.025)	0.143*** (0.047)	0.151*** (0.036)
$\Omega_{opd}^{(o)}$	0.077*** (0.020)	0.105*** (0.023)	0.107*** (0.020)
$\log x_{opd}^t$	1.371*** (0.055)	1.600*** (0.057)	1.769*** (0.057)
$\log x_{op}^t$	0.961*** (0.032)	0.971*** (0.035)	0.915*** (0.032)
$\log x_{pd}^t$	0.529*** (0.028)	0.421*** (0.030)	0.406*** (0.027)
$\log Distance$	−0.485*** (0.036)	−0.423*** (0.042)	−0.432*** (0.038)
$\log gdp_o^t$	0.165*** (0.035)	0.200*** (0.042)	0.203*** (0.042)
$\log gdp_d^t$	0.226*** (0.041)	0.218*** (0.037)	0.281*** (0.035)
$\log Population_o$	0.472*** (0.029)	0.457*** (0.047)	0.455*** (0.048)
$\log Population_d$	0.344*** (0.029)	0.368*** (0.030)	0.338 (0.028)
$Border_{od}$	0.712*** (0.066)	0.709*** (0.070)	0.611*** (0.061)
$Colony_{od}$	0.052 (0.070)	0.135 (0.083)	0.193*** (0.070)
$Language_{od}$	0.545*** (0.061)	0.436*** (0.070)	0.334*** (0.071)
$\log Lang.Proximity_{od}$	0.032* (0.000)	0.000* (0.000)	−0.001 (0.019)
Constant	9.653*** (0.045)	9.828*** (0.047)	9.793*** (0.045)
Observations	10,911,584	7,591,489	5,332,257
Adjusted R ²	0.495	0.516	0.558
Root MSE	2.5681	2.637	2.529

Three-way clustering robust standard errors are reported in parentheses and *p<0.1; **p<0.05; ***p<0.01

Dependent variable: $\log x_{opd}^{t+2}$

Bilateral
relatedness gravity
models
are a good example
of the difference
between
relatedness and
economic potential

	(1) 2000–2006	(2) 2007–2012	(3) 2012–2015
ω_{opd}^t	0.209*** (0.017)	0.180*** (0.017)	0.152*** (0.015)
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$\Omega_{opd}^{(o)}$	0.077*** (0.020)	0.105*** (0.023)	0.107*** (0.020)
$\log x_{opd}^t$	1.371*** (0.055)	1.600*** (0.057)	1.769*** (0.057)
$\log x_{op}^t$	0.961*** (0.032)	0.971*** (0.035)	0.915*** (0.032)
$\log x_{pd}^t$	0.529*** (0.028)	0.421*** (0.030)	0.406*** (0.027)
$\log Distance$	−0.485*** (0.036)	−0.423*** (0.042)	−0.432*** (0.038)
$\log gdp_o^t$	0.165*** (0.035)	0.200*** (0.042)	0.203*** (0.042)
$\log gdp_d^t$	0.226*** (0.041)	0.218*** (0.037)	0.281*** (0.035)
$\log Population_o$	0.472*** (0.022)	0.457*** (0.045)	0.455*** (0.042)

Example: growth potential in the OEC

Growth Potential



Growth Potential

Country

China 

Exports

Imports

Bar Chart

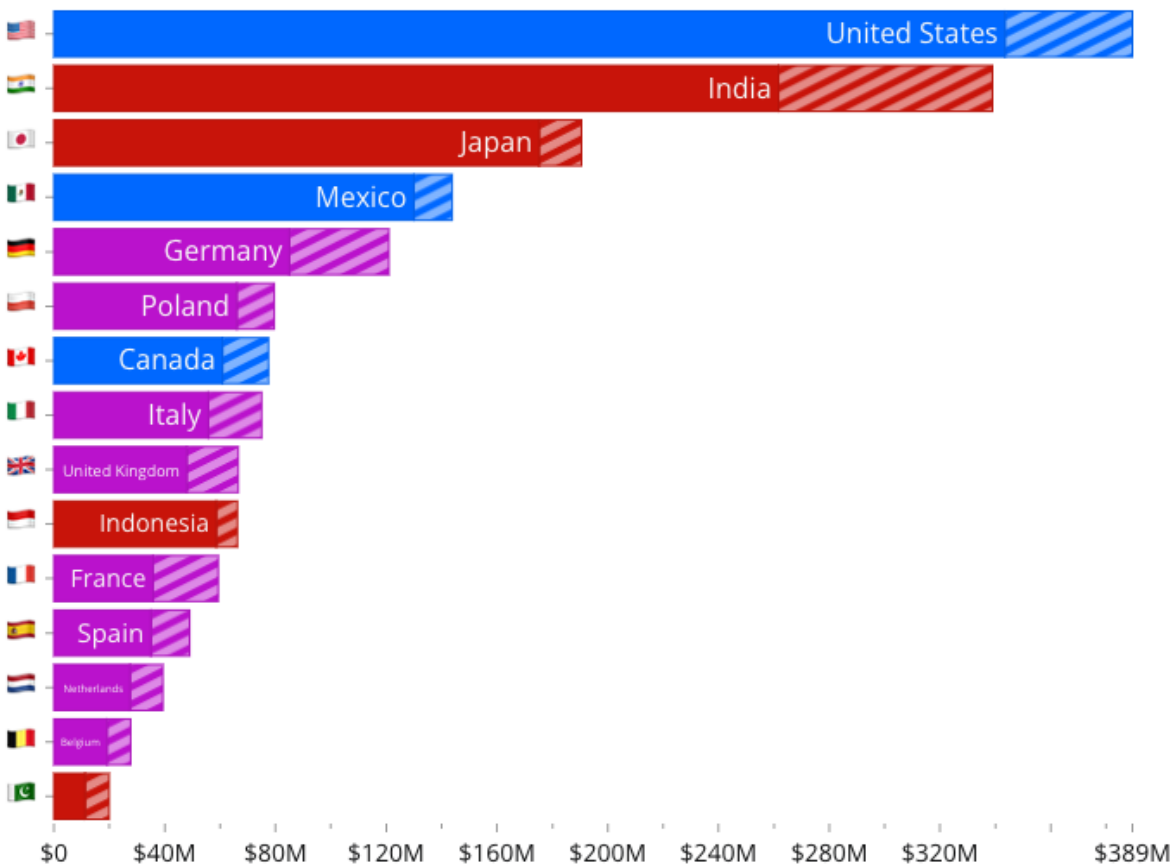
Map

In 2024, India represented the largest export potential for China's Glass Fibers. Exports from China to India amounted to \$262M, with a predicted export potential of \$77.3M. This could lead to total exports of \$339M, representing an increase of 29.5% compared to the current value.

The bar chart shows the countries with the highest export potential for China's Glass Fibers. The solid bar represents the current export value, while the hatched bar indicates the projected export value. Thus, the total length of each bar reflects the export potential to these countries.

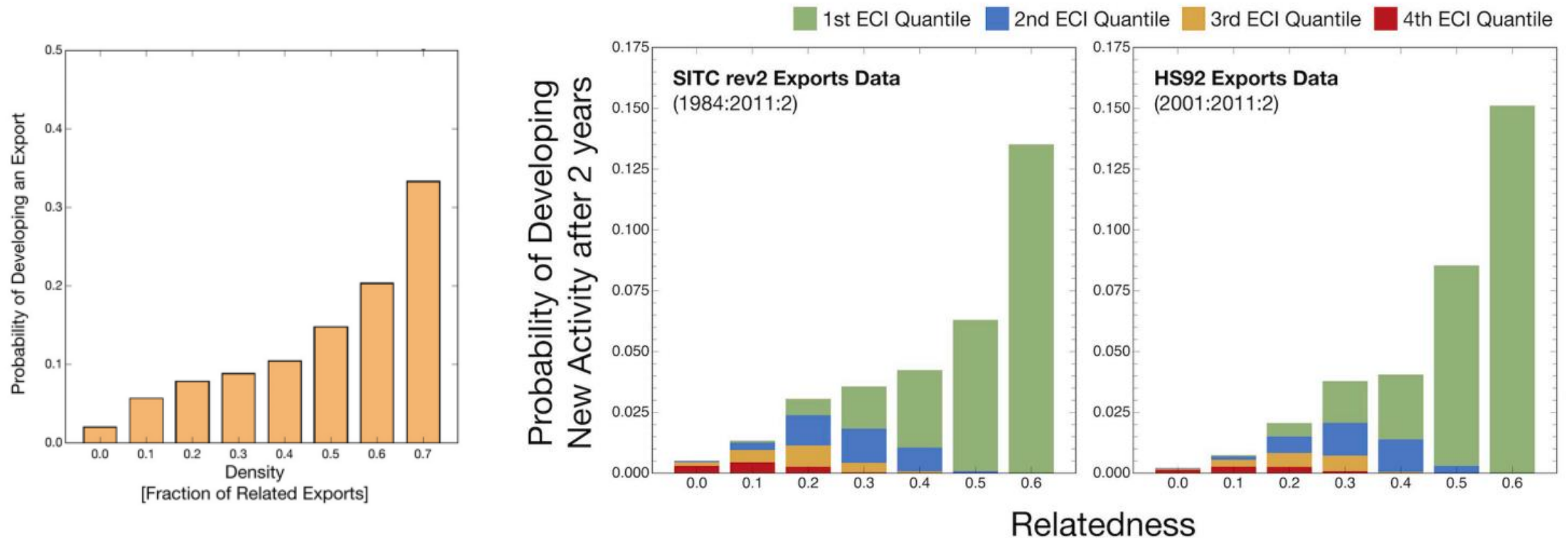
Data Source

Export Potential of Glass Fibers from China



Relative Relatedness

Relatedness, as estimated so far, is correlated with diversity, so the entries it predicts are those of high complexity economies. Relative relatedness attempts to correct for this.



Relative Relatedness

Relatedness, as estimated so far, is correlated with diversity, so the entries it predicts are those of high complexity economies. Relative relatedness attempts to correct for this.

$$\tilde{\omega}_{cp} = \frac{\omega_{cp} - \sum_{p'} \omega_{cp'} / N_{O_c}}{\sigma_{\omega_{cp'}}}$$



Probability of Developing
New Activity after 2 years

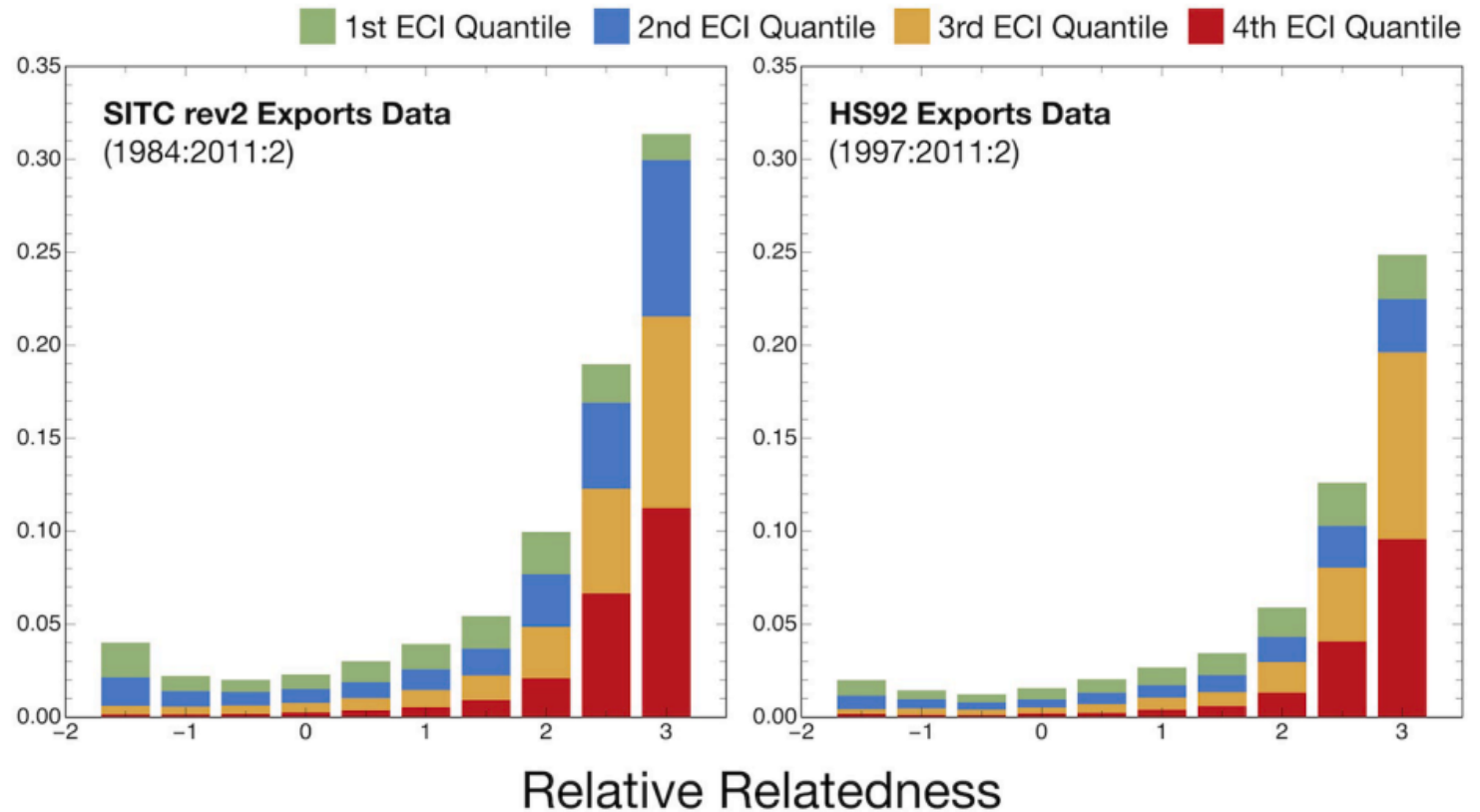
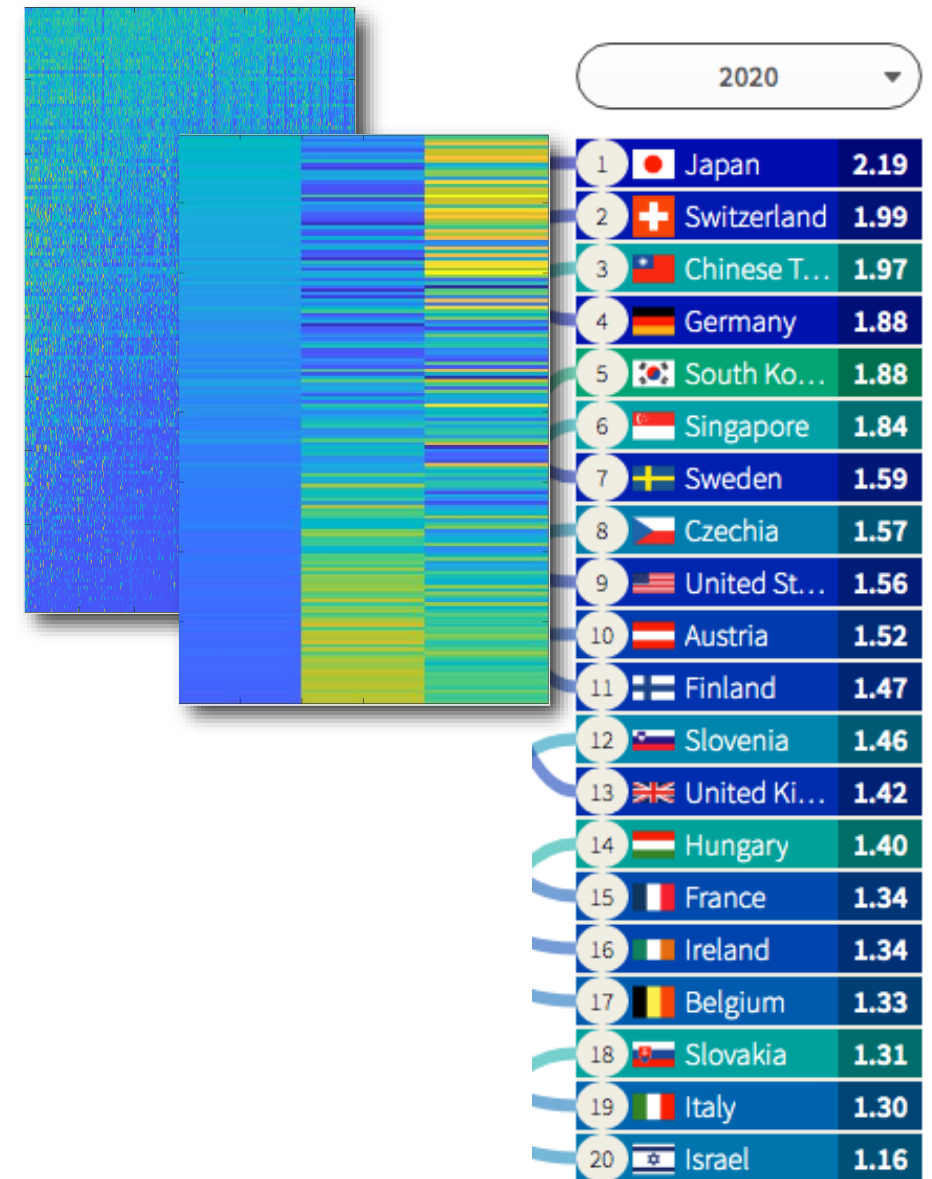


Table 1. Entry and Exit Model for Target Year 2022*Dependent variable: $\log(1 + RCA)$ (2022) [in 5 years]*

	Entry		Exit			
	(1)	(2)	(3)	(4)	(5)	(6)
log of RCA (2017)	0.648*** (0.003)		0.636*** (0.003)	0.889*** (0.005)		0.874*** (0.005)
log of RCA (2012)	0.283*** (0.004)		0.215*** (0.005)	0.007 (0.006)		0.022*** (0.006)
Relatedness (2012)		0.878*** (0.007)	0.180*** (0.007)		-0.653*** (0.042)	0.284*** (0.022)
Relative Relatedness (2012)		0.043*** (0.001)	0.014*** (0.001)		0.271*** (0.006)	0.023*** (0.003)
Observations	134948	134948	134948	24217	24217	24217
R ²	0.459	0.140	0.465	0.754	0.087	0.757
Adjusted R ²	0.459	0.140	0.465	0.754	0.087	0.757

Economic Complexity

Measures of the value of an activity (e.g. a product, industry, etc.) and of the portfolio of activities present in an economy.



Economic Complexity Index

The complexity (K_c) of an economy is the complexity (K_p) of its activities.

The complexity (K_p) of an activity is the complexity (K_c) of the economies where that activity is produced.



Complexity can be estimated as a solution to the following eigenproblem.

$$K_c = f(M_{cp}, K_p),$$

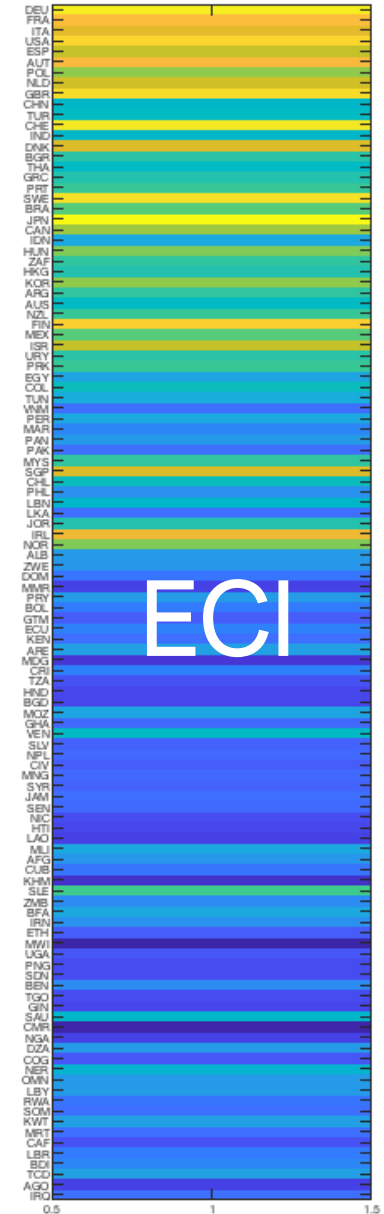
$$K_p = g(M_{cp}, K_c),$$



$$K_c = f(M_{cp}, g(M_{cp}, K_c)),$$

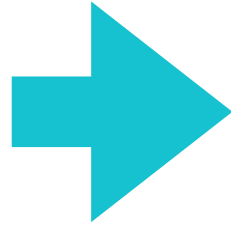


$$\tilde{M}_{cc'} K_{c'} = \lambda K_c$$



When f and g are defined as simple averages....

$$K_c = \frac{1}{M_c} \sum_p M_{cp} K_p$$
$$K_p = \frac{1}{M_p} \sum_c M_{cp} K_c,$$



$$\tilde{M}_{cc'} = \frac{1}{M_c} \sum_p \frac{M_{cp} M_{c'p}}{M_p}$$

The “easy way” to estimate K_c and K_p is to simply iterate the mapping, starting with $K_p=M_p$ and $K_c=M_c$. The mapping converges after about 20 iterations.

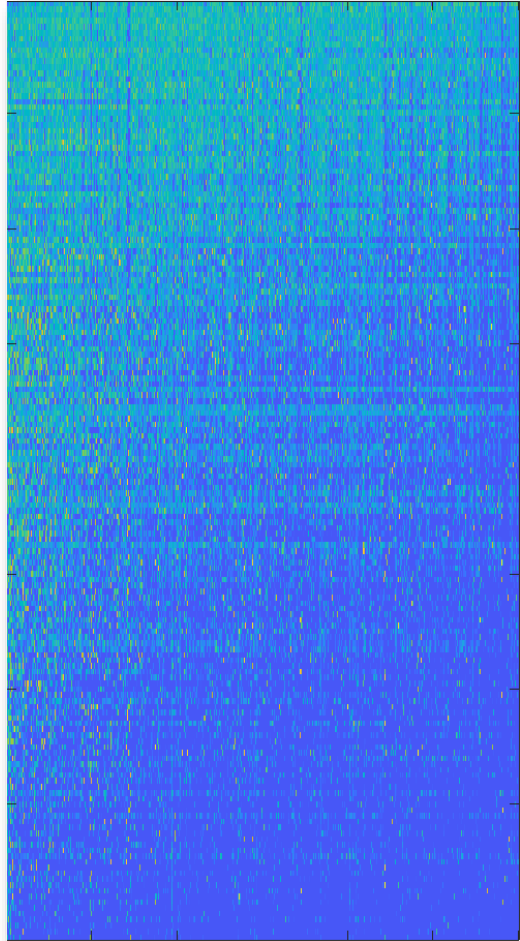
The Economic Complexity Index as an optimization problem

Let V_i be a vector whose entries describe a location (e.g. country) or an activity (e.g. product). Let A_{ij} be a matrix connecting locations and activities. Then ECI is a solution that minimizes the cost function U :

$$U(\vec{V}) = \frac{1}{4} \sum_{i,j} A_{ij} (V_i - V_j)^2$$

But is not that easy!

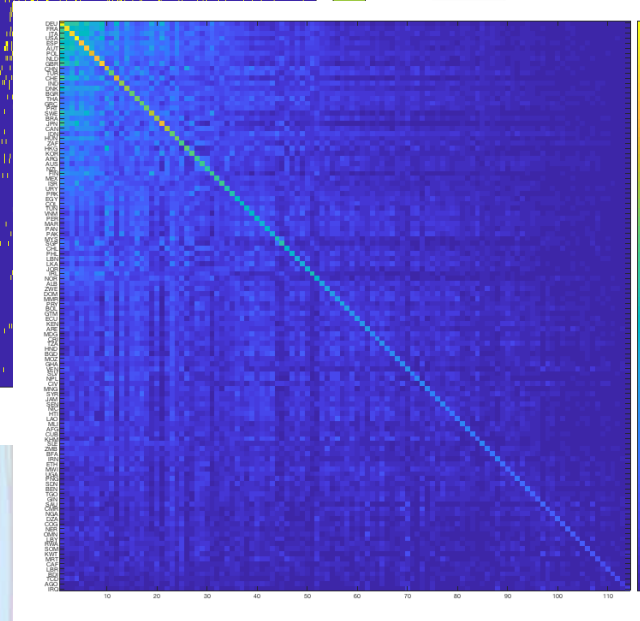
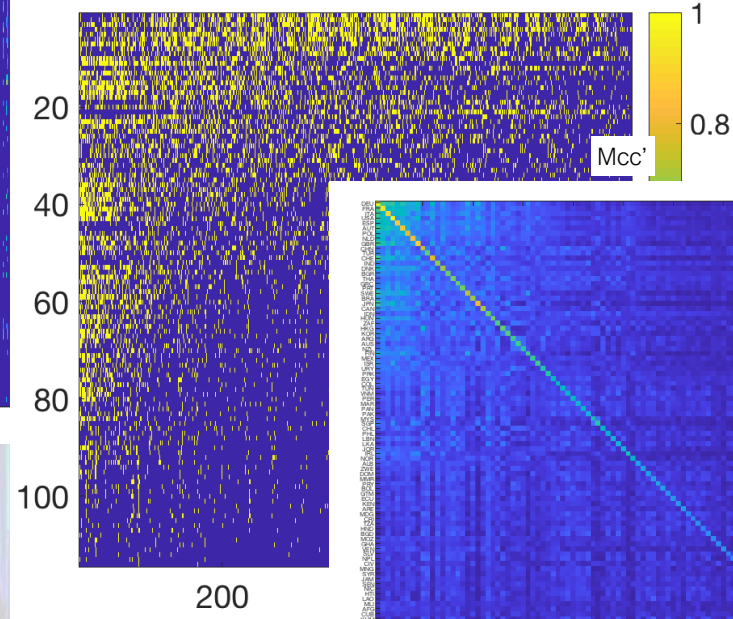
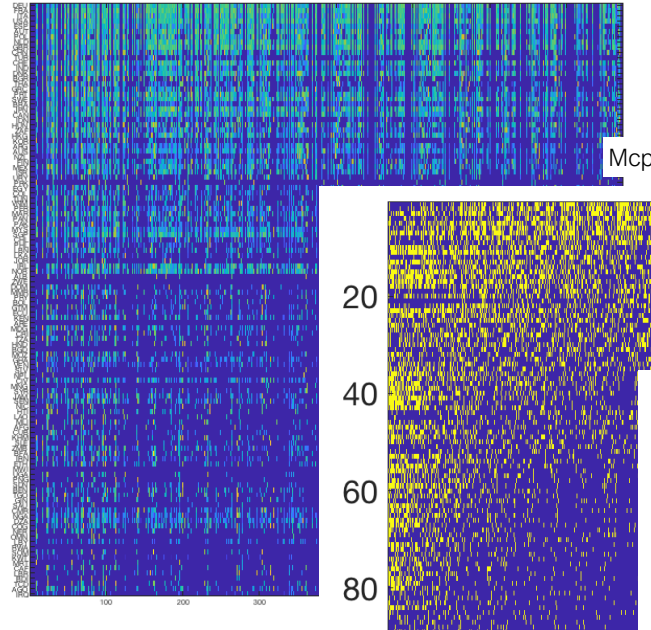
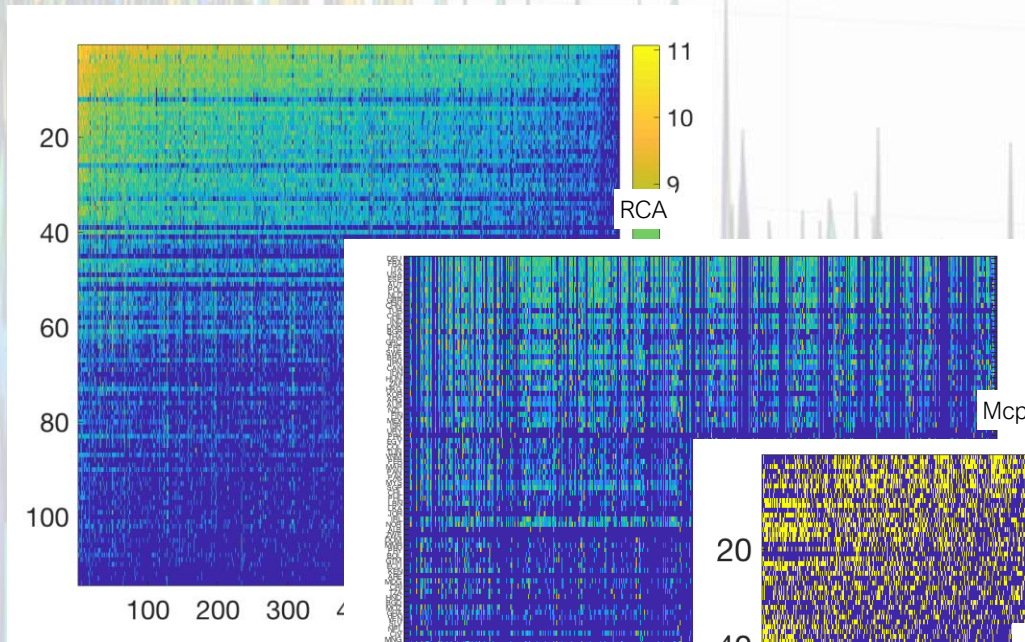
Units of observation are not comparable!



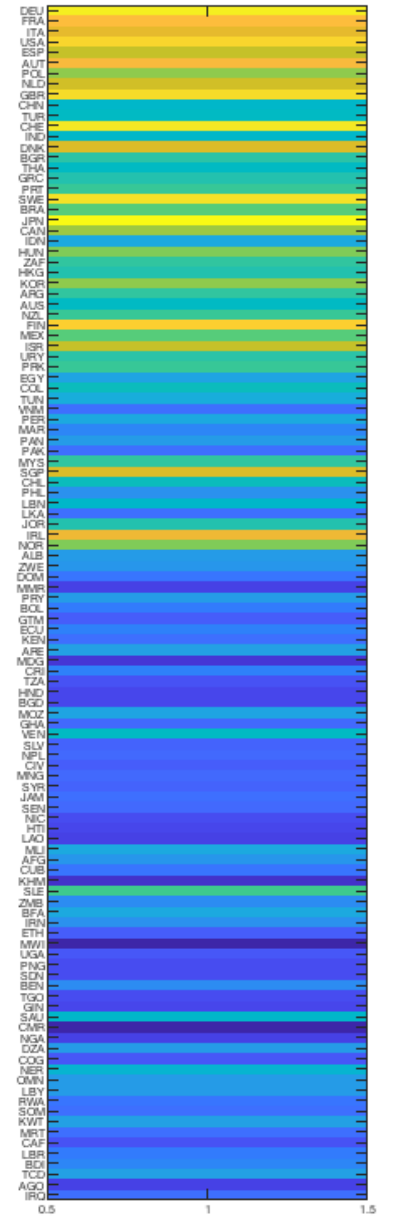
China & USA ~ 15 to 20 trillion GDP

Macedonia ~ 0.0012 trillion GDP

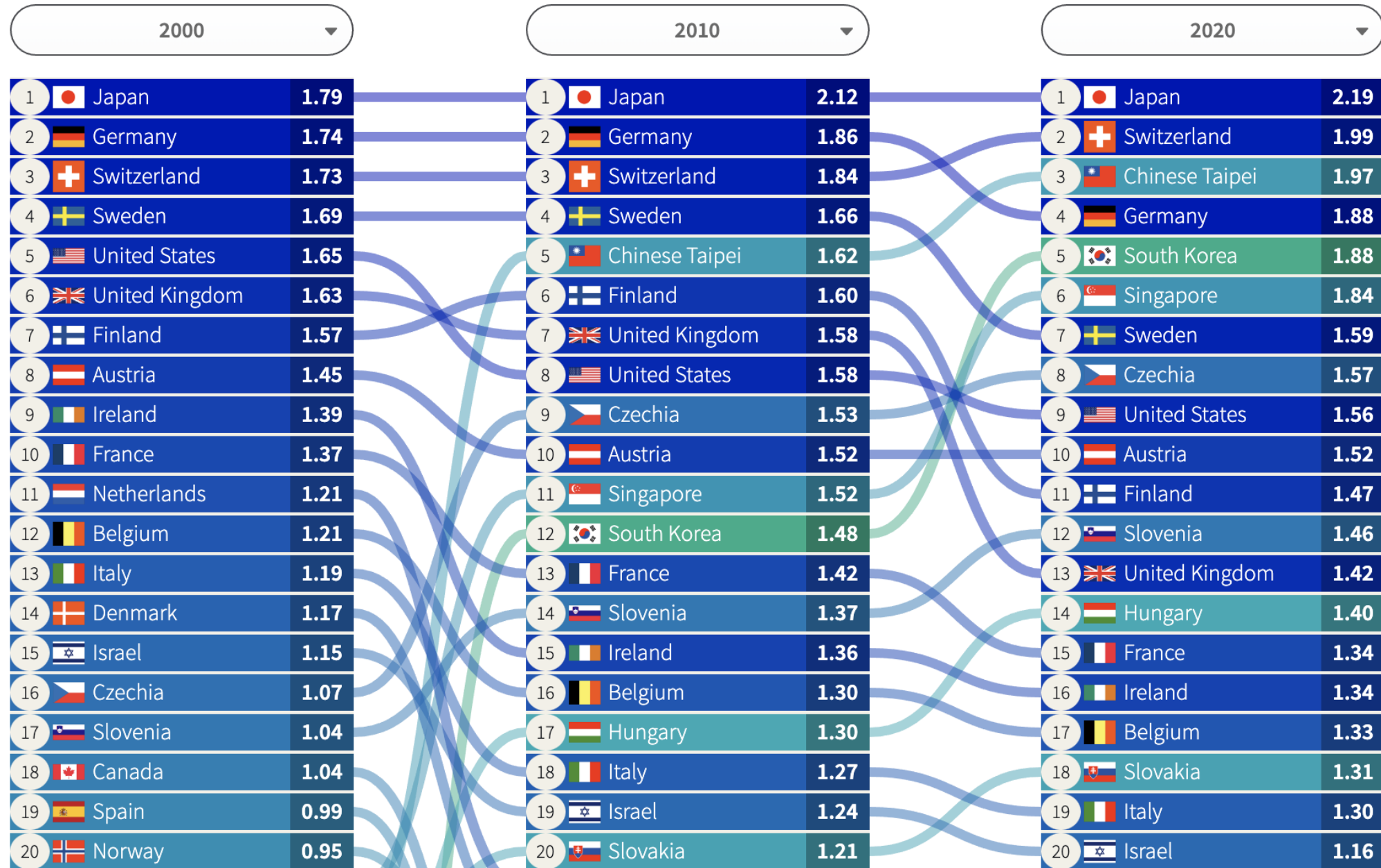
Exports



ECI



Economic Complexity Index (Trade)



Economic Complexity Explains

Economic Growth

Hidalgo and Hausmann, 2009; Chávez et al., 2017; Domini, 2019; Hausmann et al., 2014; Koch, 2021; Lo Turco and Maggioni, 2020; Ourens, 2012; Stojkoski et al., 2016

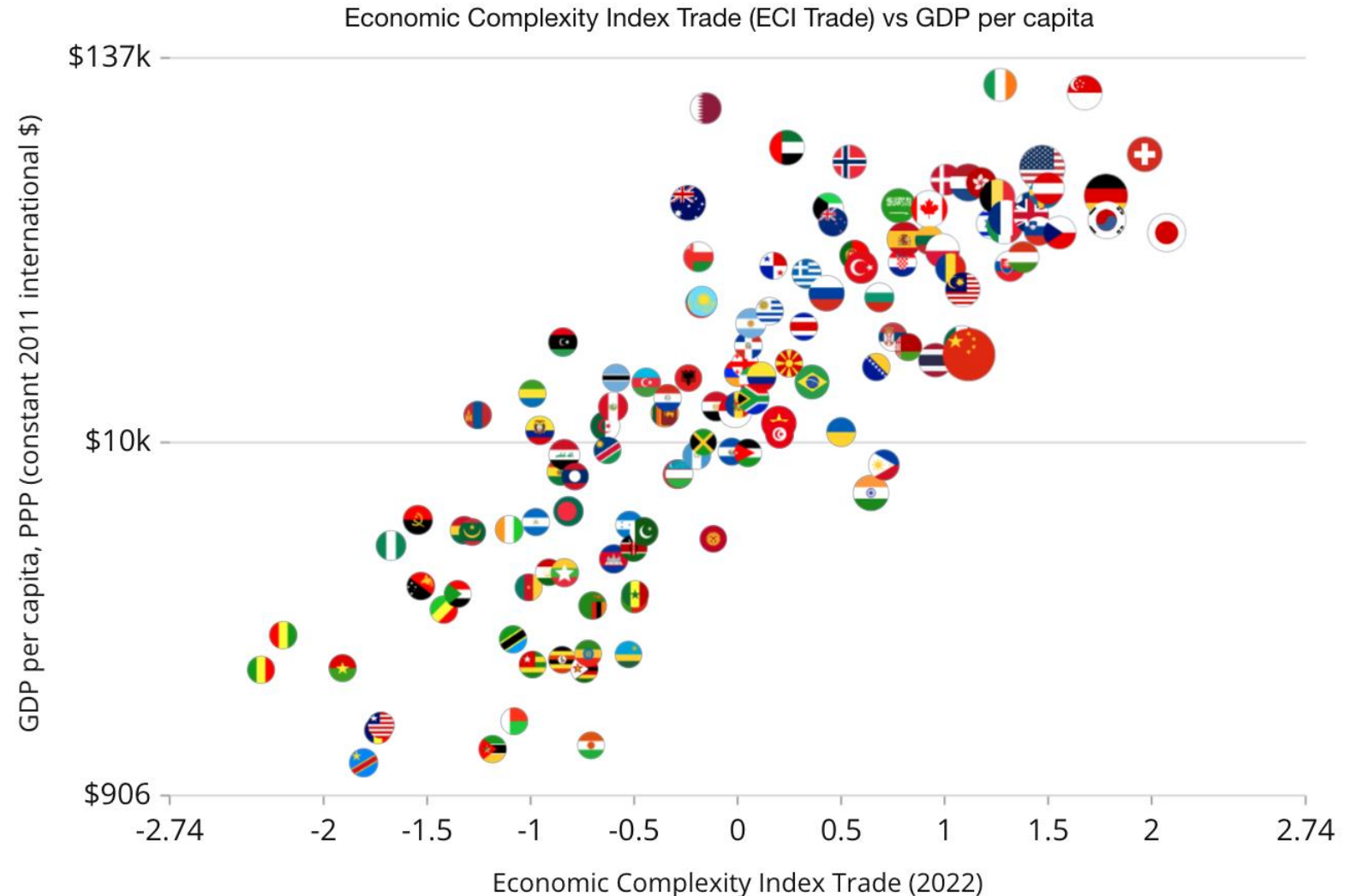
Inequality

Hartmann et al., 2017, Barza et al., 2020; Ben Saâd and Assoumou-Ella, 2019; Chu and Hoang, 2020; Fawaz and Rahnema-Moghadamm, 2019

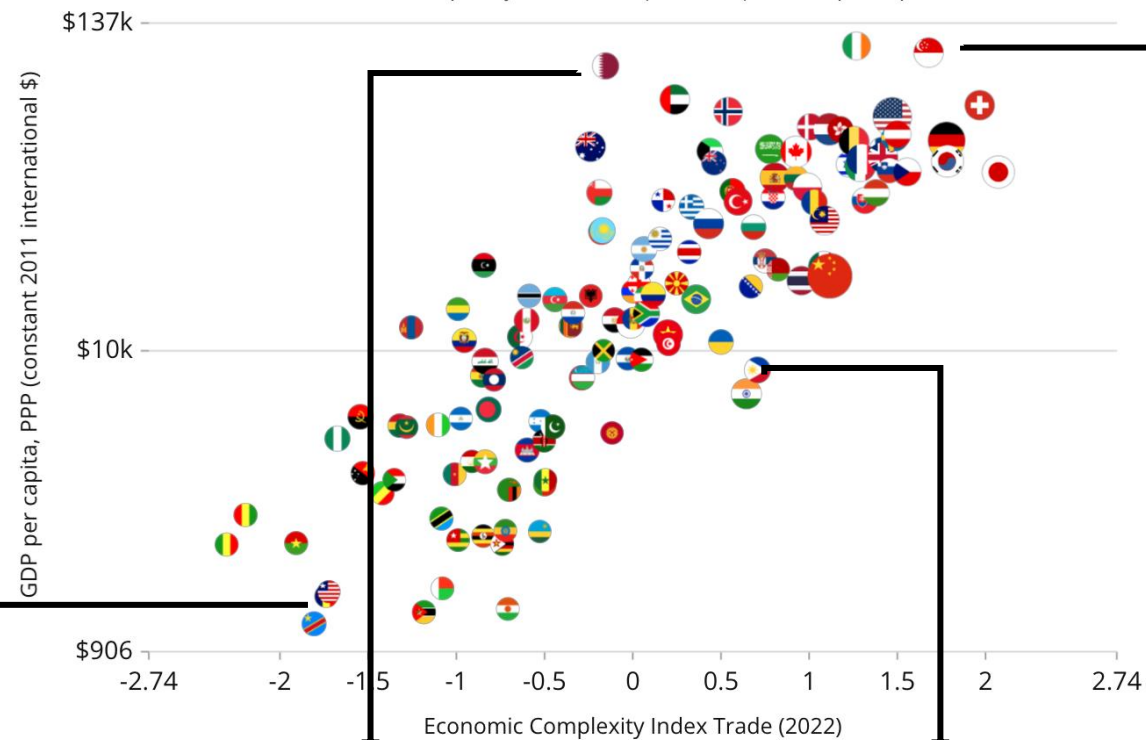
Emissions

Can and Gozgor, 2017; Dordmond et al., 2020; Fraccascia et al., 2018; Hamwey et al., 2013; Lapatinas et al., 2019; Mealy and Teytelboym, 2020; Neagu, 2019; Romero and Gramkow, 2021

Among other outcomes



Economic Complexity Index Trade (ECI Trade) vs GDP per capita



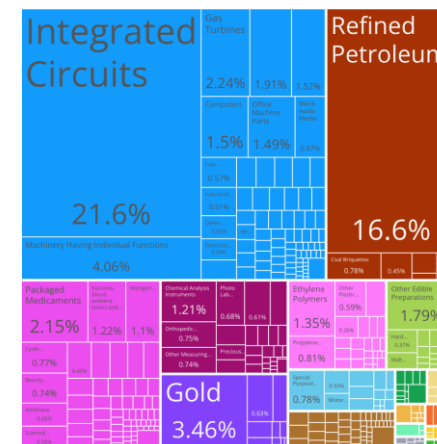
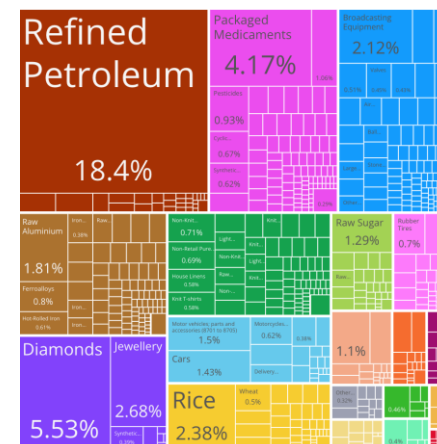
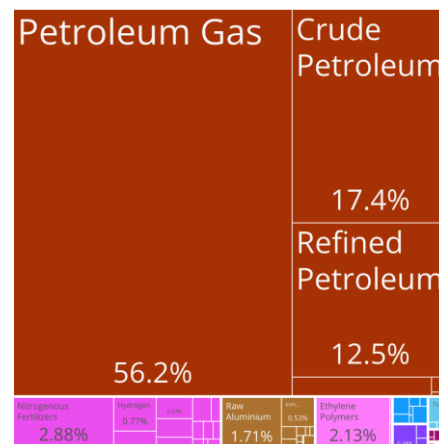
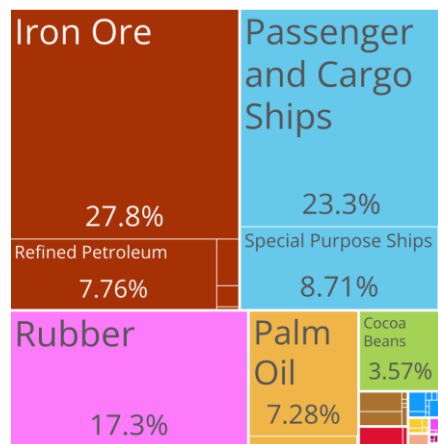
Liberia

Qatar

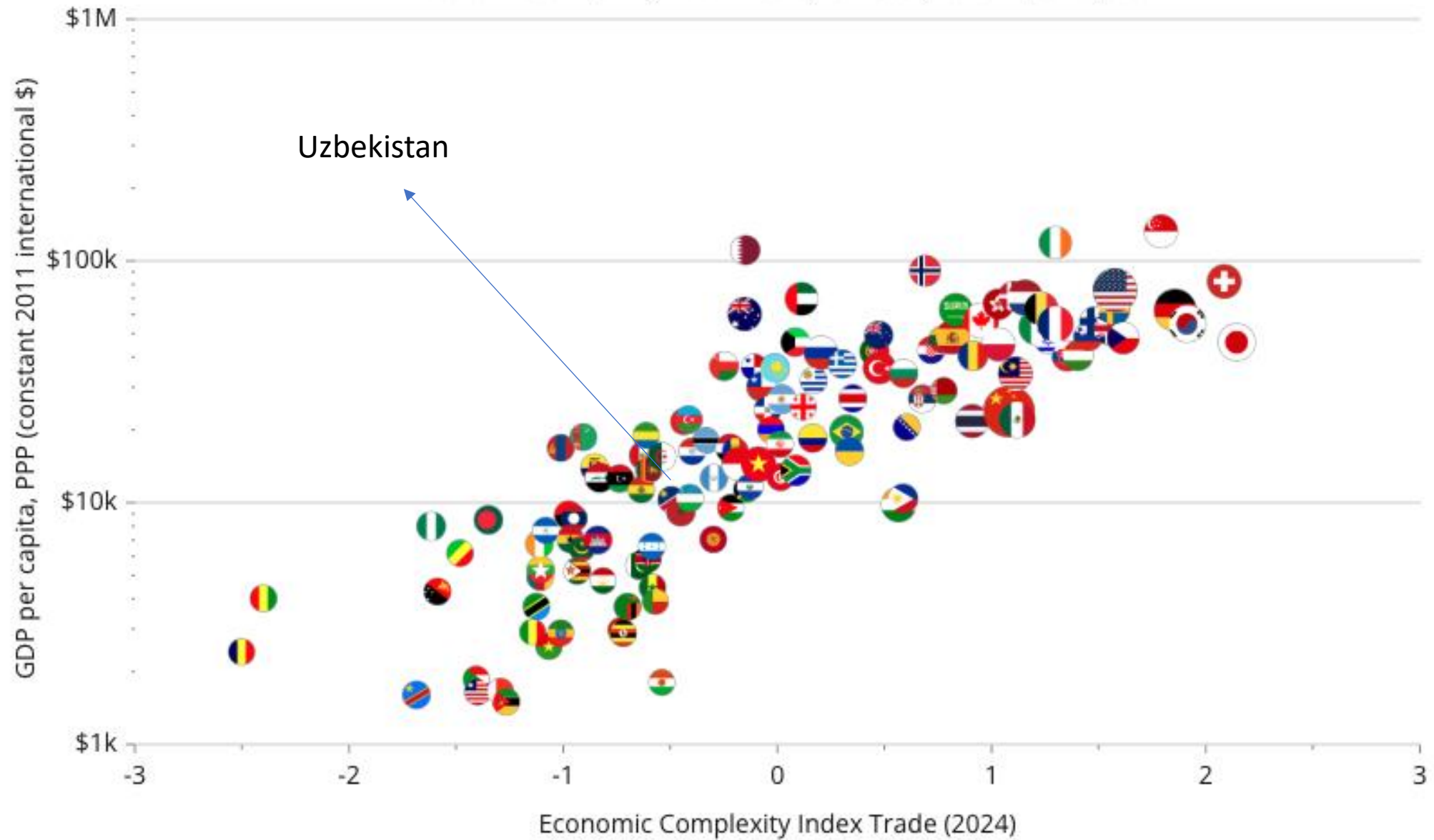
India

Singapore

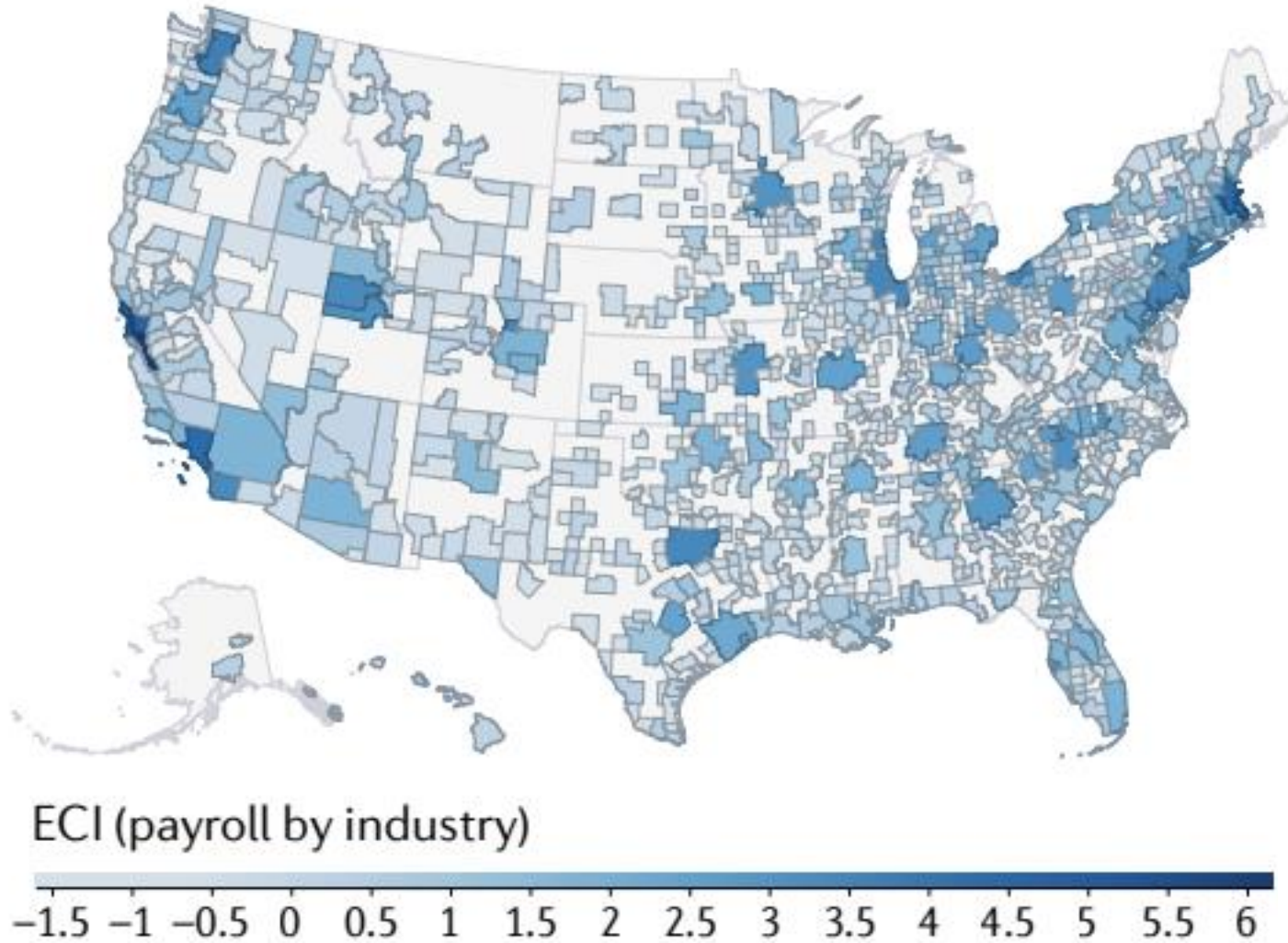
(Poor & not complex) (Rich & Not too Complex) (Poor & quite complex) (Rich & Complex)



Economic Complexity Index Trade (ECI Trade) vs GDP per capita

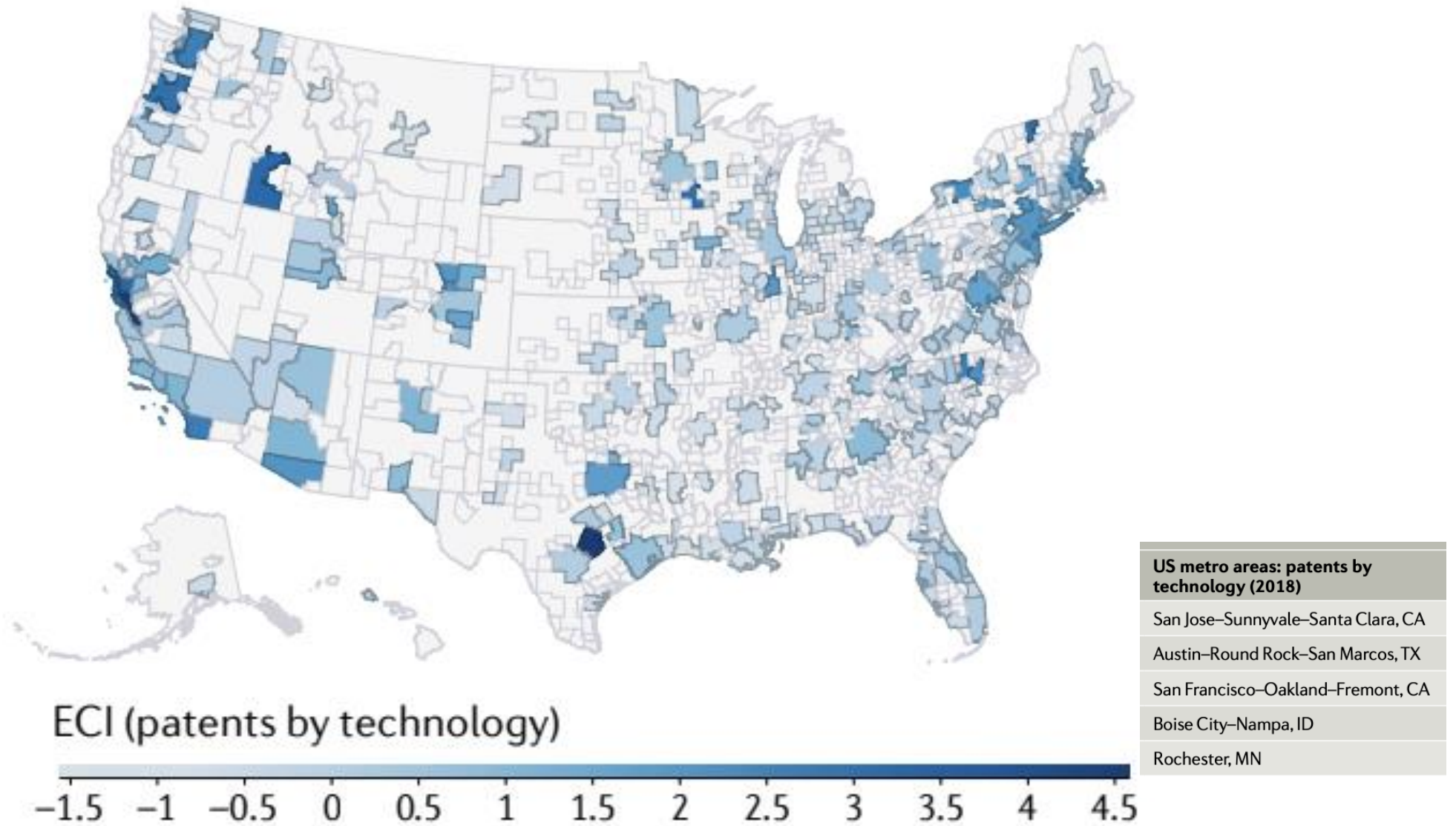


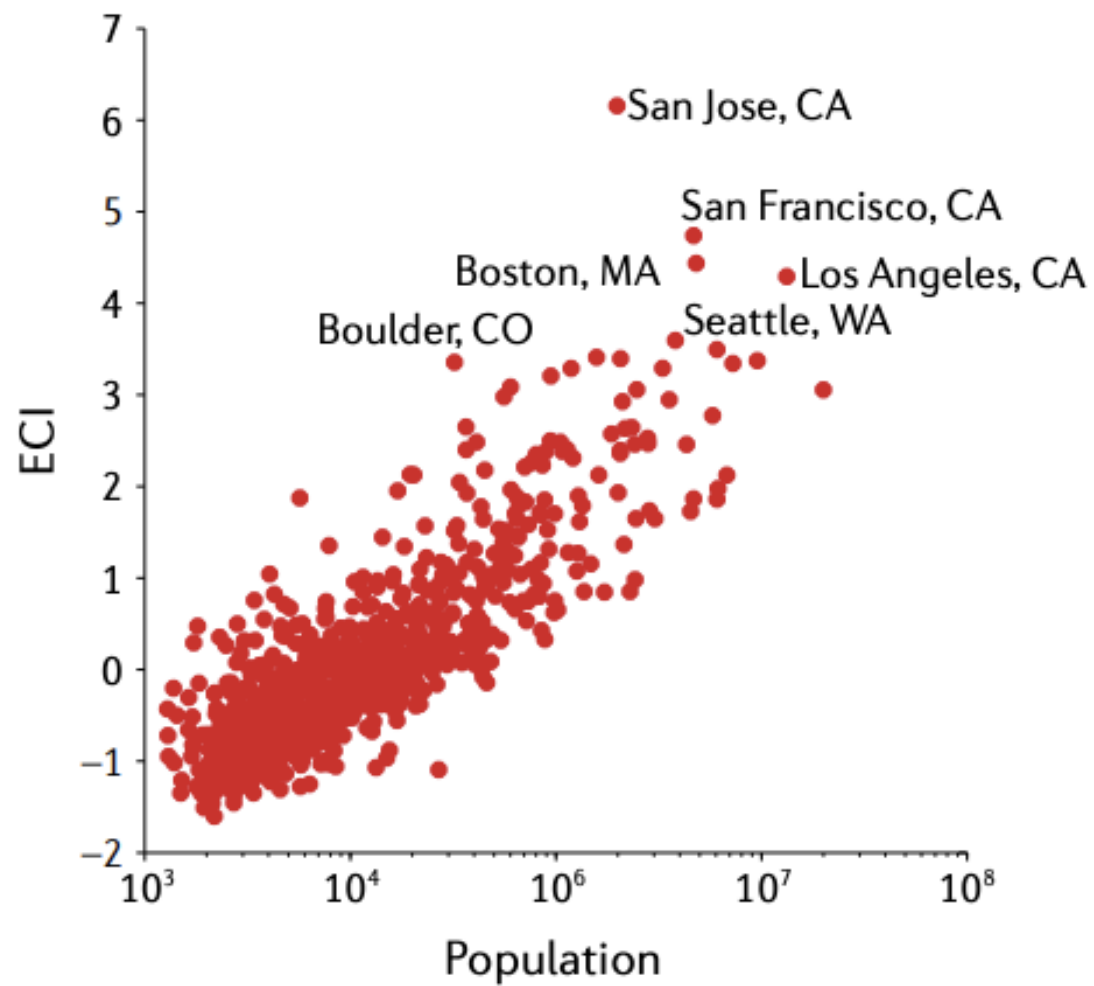
b Economic complexity of US MSAs
(industry payroll)



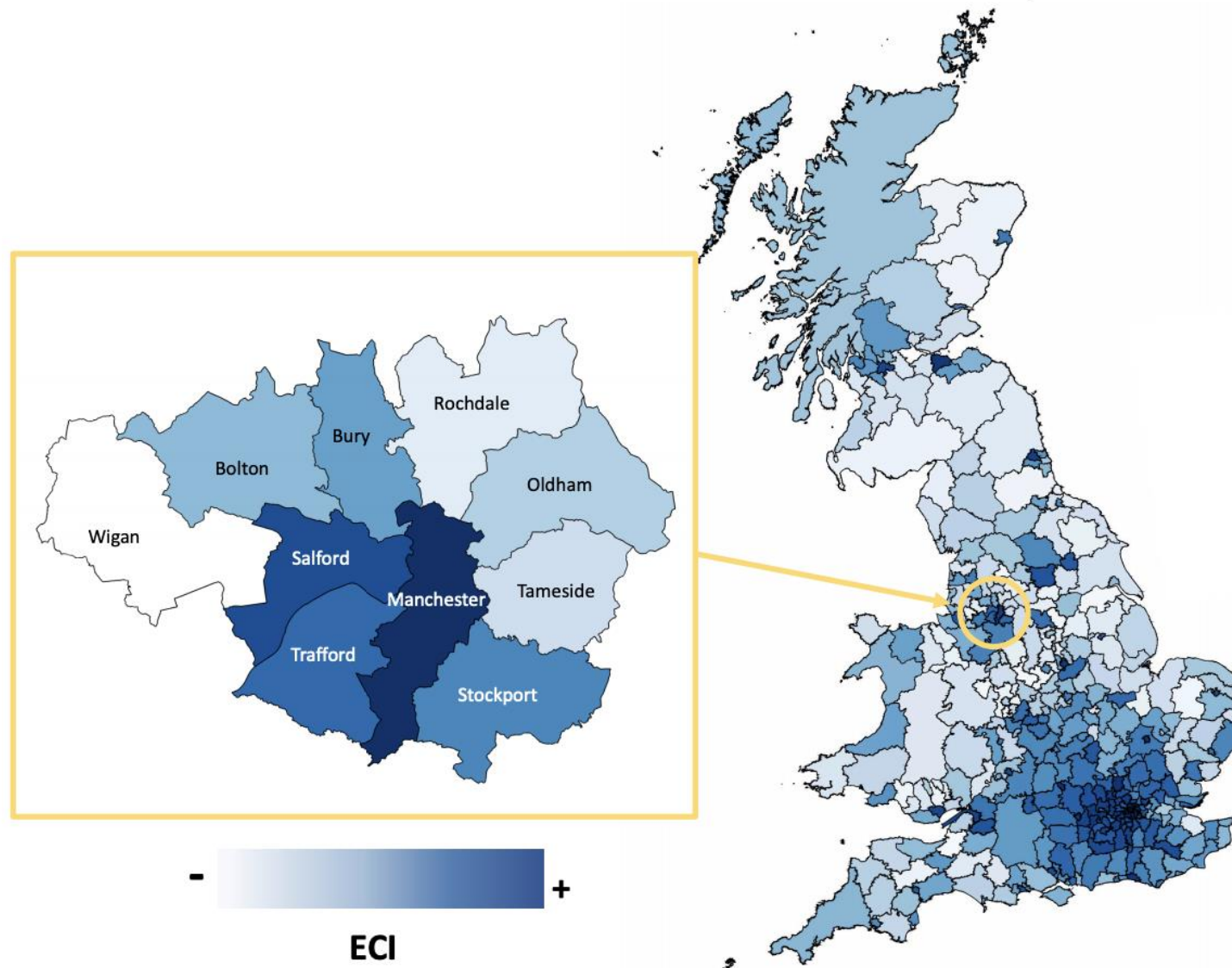
Rank	Economic complexity rankings US metro areas: payroll by industry (2018)
1	San Jose–Sunnyvale–Santa Clara, CA
2	San Francisco–Oakland–Hayward, CA
3	Boston–Cambridge–Newton, MA–NH
4	Los Angeles–Long Beach–Anaheim, CA
5	Seattle–Tacoma–Bellevue, WA

Economic complexity of US MSAs (patents by technology class)



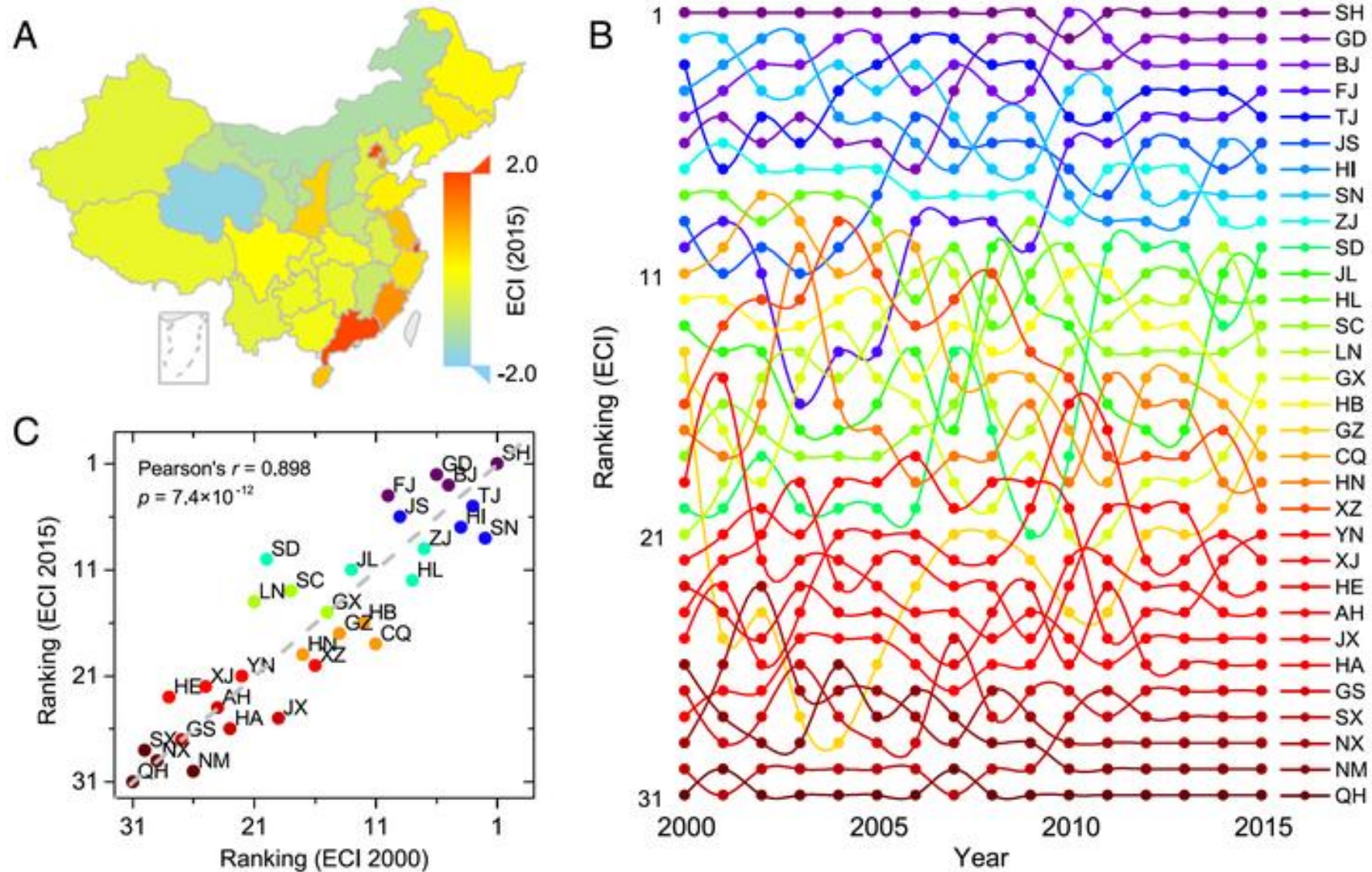


Economic Complexity of UK Local Authorities by Industry



Mealy, Penny, and Diane Coyle. "To them that hath: economic complexity and local industrial strategy in the UK." *International Tax and Public Finance* (2021): 1-20.

Economic Complexity of Chinese Provinces Using Data on Publicly Listed Firms



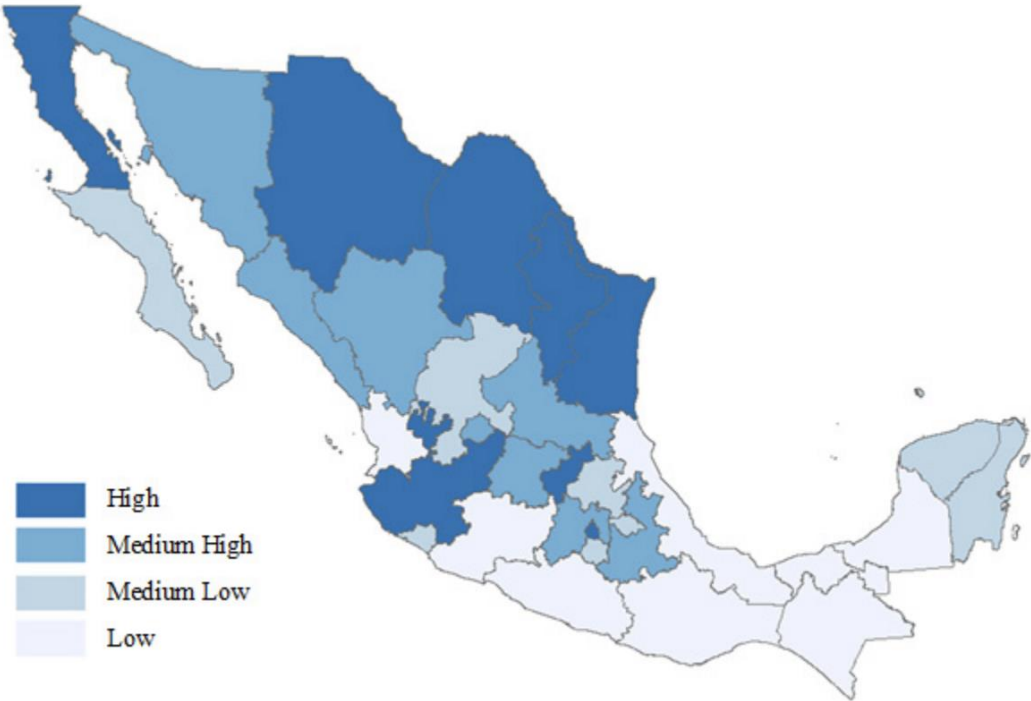
Gao, Jian, and Tao Zhou. "Quantifying China's regional economic complexity." *Physica A: Statistical Mechanics and its Applications* 492 (2018): 1591-1603.

Economic Complexity of Mexican States Using Industry Data

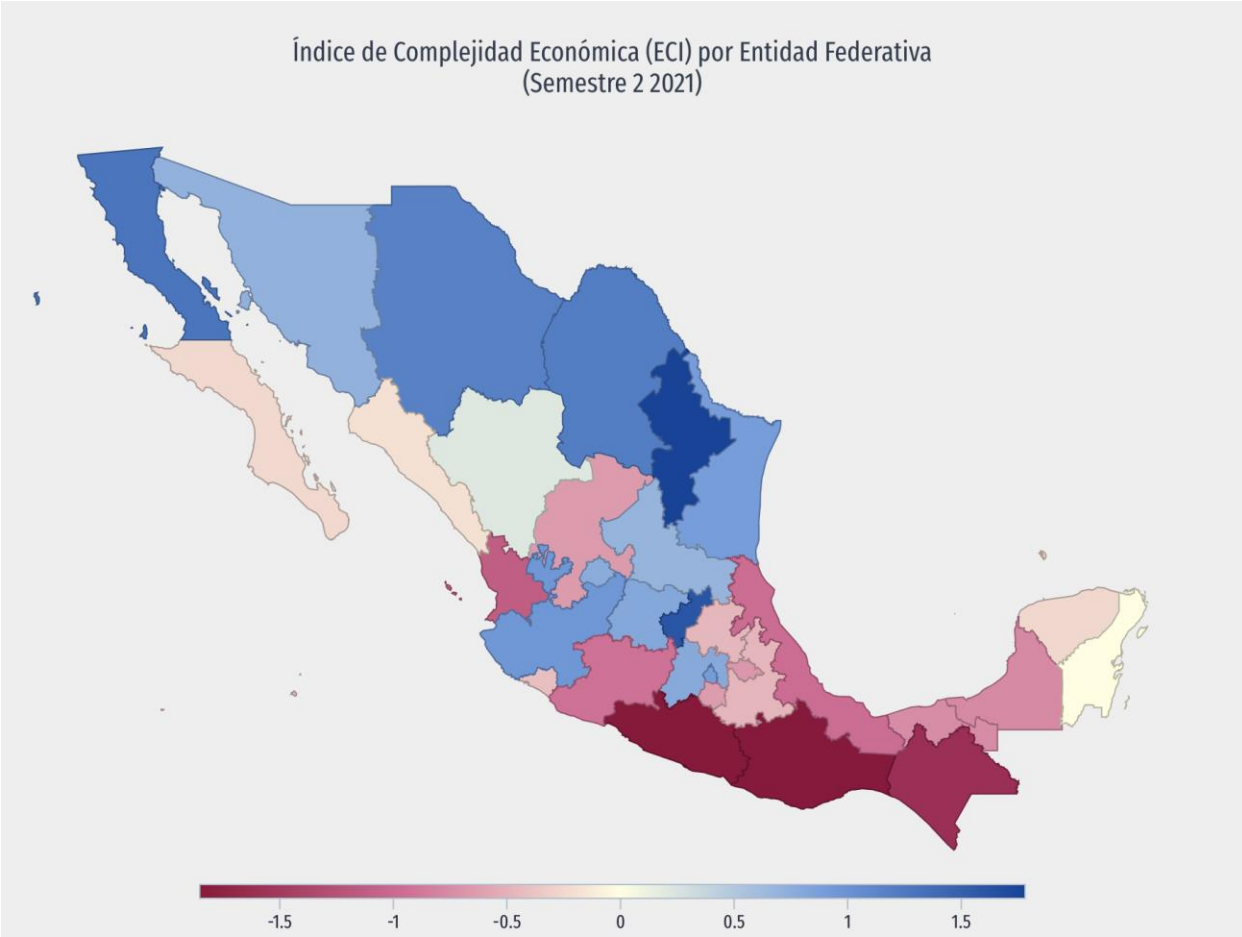
CHAVEZ, MOSQUEDA, & GOMEZ-ZALDIVAR: COMPLEXITY AND GROWTH

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Map 1: States' Level of Economic Complexity, 2013

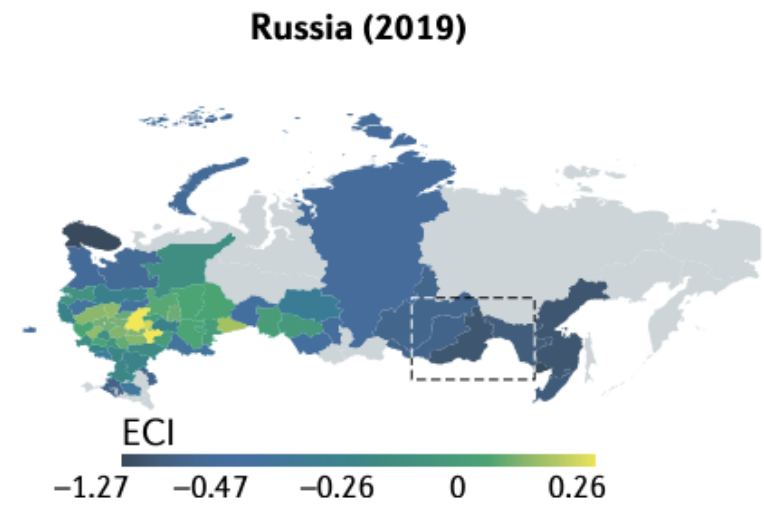
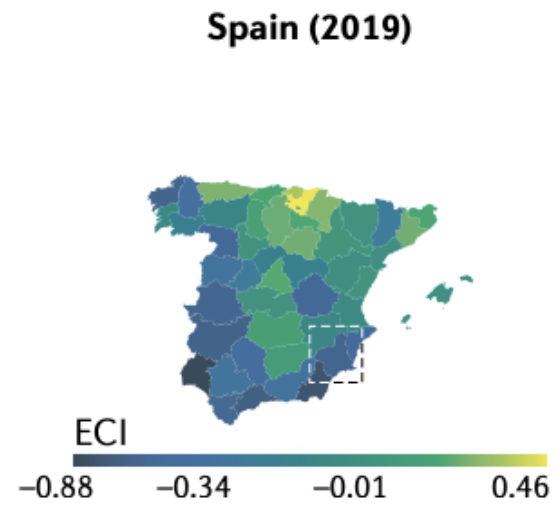
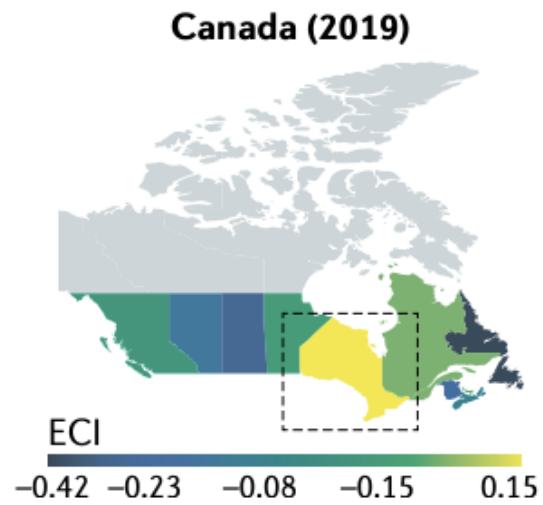
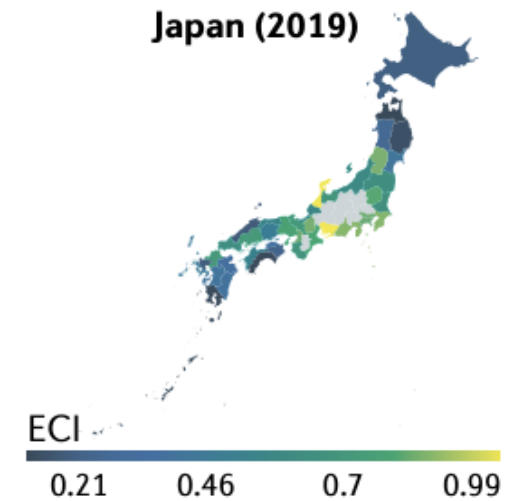
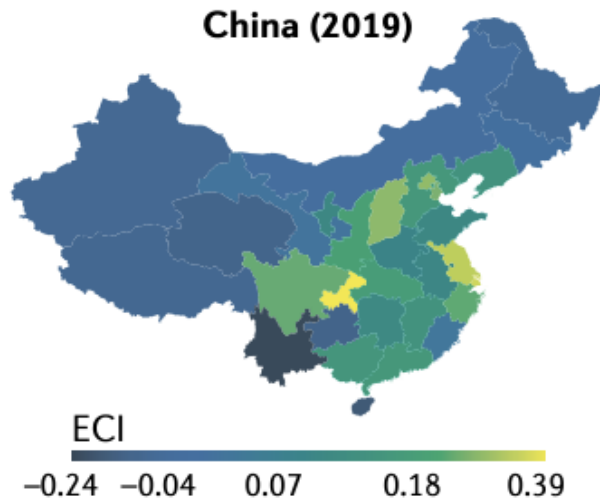
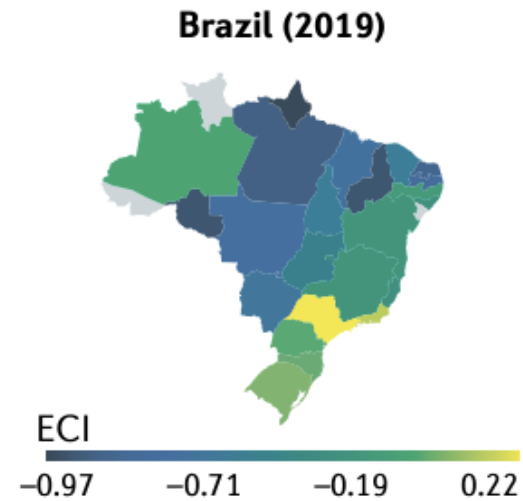


Central Bank of Mexico

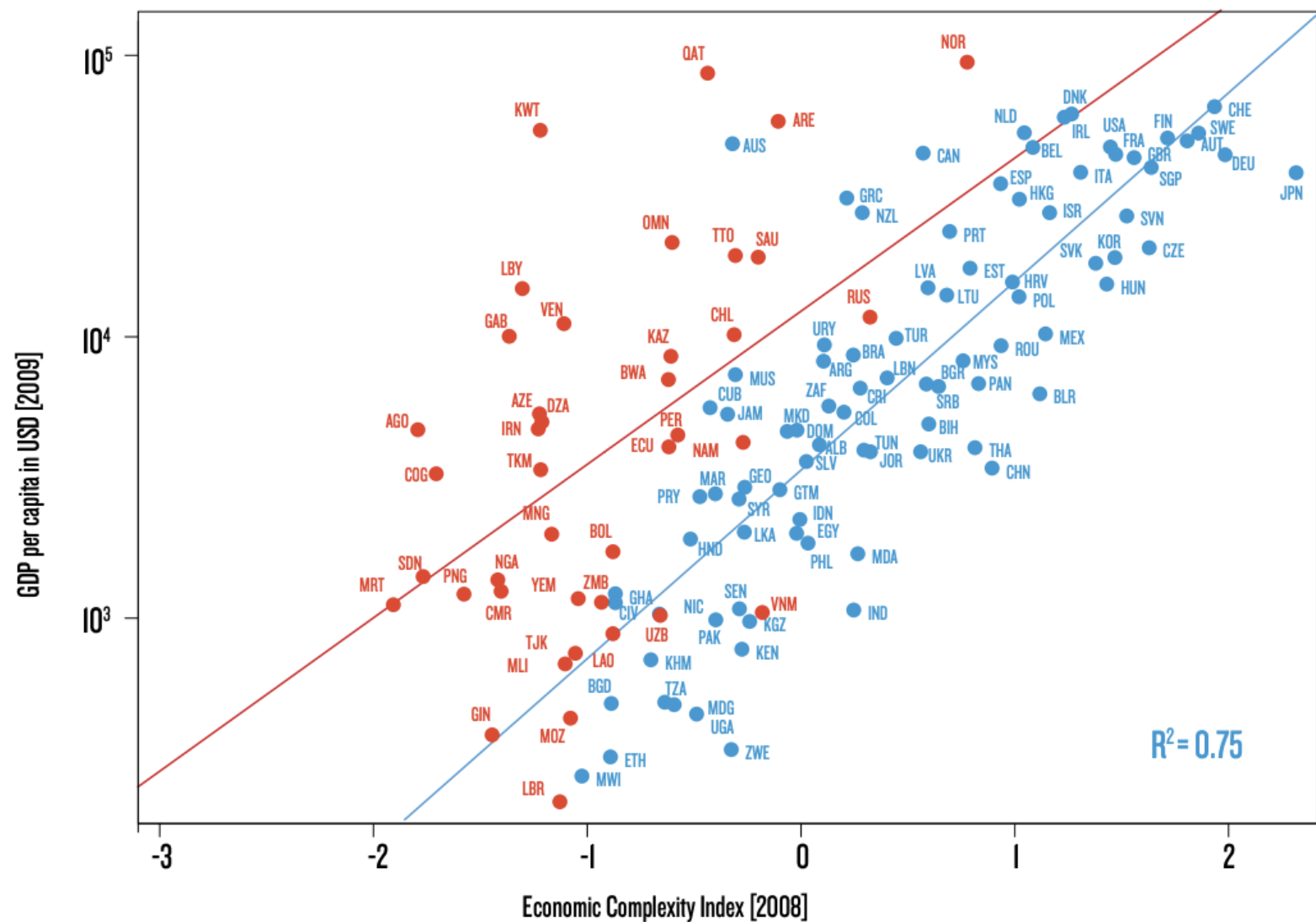


DataMexico.org

Economic Complexity of State/Provinces based on Exp

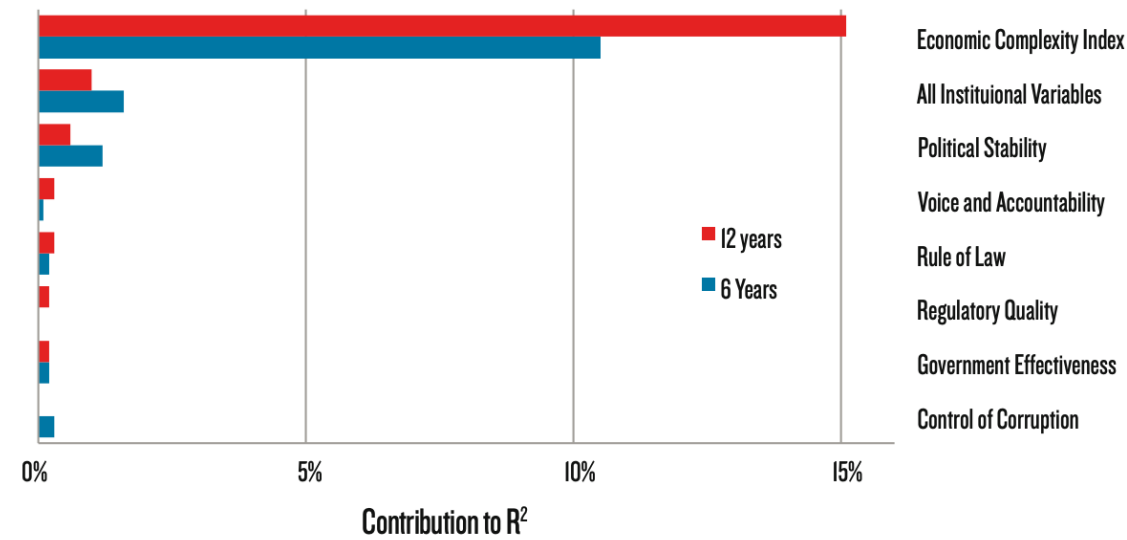


Economic Complexity Explains Future Economic Growth



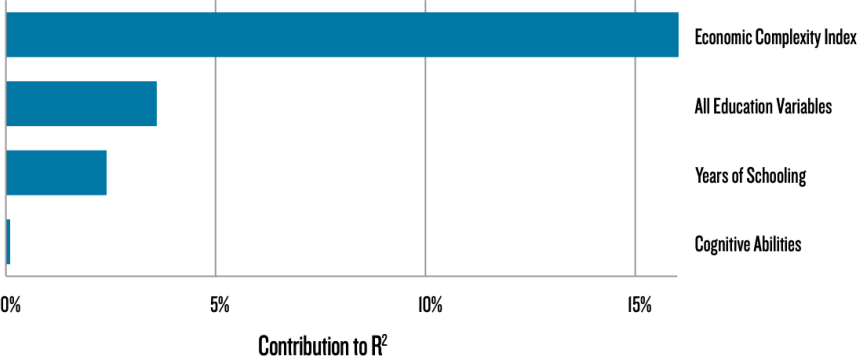
	Annualized growth in GDP pc (12 years)								
	1996-2008								
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Income per capita, logs	-0.00202***	-0.00127***	-0.00199***	-0.00213***	-0.00203***	-0.00196***	-0.00207***	-0.00201***	-0.00202***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Increase in natural resource exports	0.00228***	0.00255***	0.00226***	0.00233***	0.00228***	0.00227***	0.00233***	0.00232***	0.00235***
- in constant dollars (as a share of initial GDP)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Initial NR exports to GDP	0.00361***	0.00314**	0.00357***	0.00367***	0.00342***	0.00367***	0.00383***	0.00361***	0.00369***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Initial Economic Complexity Index [ECI]	0.00380**		0.00371***	0.00436***	0.00405**	0.00415***	0.00364**	0.00392**	0.00467***
	(0.002)		(0.001)	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
[ECI] x [GDP per capita, log]	-0.00017		-0.00015	-0.00026	-0.00020	-0.00020	-0.00014	-0.00019	-0.00026**
	(0.000)		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Initial Control of Corruption	0.00004	-0.00079		-0.00027	0.00001	0.00029	0.00005	-0.00003	
	(0.001)	(0.001)		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
Initial Government Effectiveness	-0.00097	-0.00040	-0.00117		-0.00080	-0.00069	-0.00107	-0.00087	
	(0.001)	(0.001)	(0.001)		(0.001)	(0.001)	(0.001)	(0.001)	
Initial Political Stability	0.00065*	0.00079*	0.00054	0.00057*		0.00078**	0.00069*	0.00059*	
	(0.000)	(0.000)	(0.000)	(0.000)		(0.000)	(0.000)	(0.000)	
Initial Rule of Law	0.00103	0.00237***	0.00121*	0.00073	0.00135*		0.00087	0.00094	
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)		(0.001)	(0.001)	
Initial Regulatory Quality	-0.00040	-0.00125*	-0.00036	-0.00051	-0.00051	-0.00024		-0.00053	
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)		(0.001)	
Initial Voice and Accountability	-0.00030	-0.00014	-0.00028	-0.00019	-0.00016	-0.00021	-0.00042		
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)		
Constant	0.01960***	0.01384***	0.01936***	0.02052***	0.01963***	0.01918***	0.01993***	0.01961***	0.01954***
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Observations	118	118	118	118	118	118	118	118	118
Adjusted R²	0.516	0.365	0.516	0.514	0.510	0.513	0.518	0.519	0.506
Difference in R²		15.1%	0.0%	0.2%	0.6%	0.3%	0.2%	0.3%	1.0%

Explains more growth than institutions



	Annualized growth in GDP pc (by decade)					
	(1978-1988, 1988-1998, 1998-2008)					
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Initial Income per capita, log	-0.009***	-0.006***	-0.008***	-0.009***	-0.009***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Increase in natural resource exports	0.055***	0.049**	0.055***	0.055***	0.055***	0.061***
- in constant dollars (as a share of initial GDP)	(0.018)	(0.021)	(0.018)	(0.018)	(0.018)	(0.017)
Initial Economic Complexity Index	0.044***		0.046***	0.044***	0.045***	0.052***
	(0.011)		(0.011)	(0.011)	(0.011)	(0.011)
Initial Economic Complexity Index X Income per capita, log	-0.004***		-0.004***	-0.004***	-0.004***	-0.004***
	(0.001)		(0.001)	(0.001)	(0.001)	(0.001)
Years of schooling (standardized)	0.006*	0.013***		0.008***	0.005	
	(0.004)	(0.004)		(0.003)	(0.003)	
Secondary school enrollment (standardized)	0.000	0.001	0.005**		0.002	
	(0.003)	(0.003)	(0.002)		(0.003)	
Tertiary school enrollment (standardized)	-0.003	-0.003	-0.000	-0.003		
	(0.002)	(0.002)	(0.001)	(0.002)		
Constant	0.097***	0.071***	0.091***	0.097***	0.099***	0.083***
	(0.012)	(0.012)	(0.011)	(0.012)	(0.012)	(0.010)
Observations	263	263	263	263	263	263
R ²	0.409	0.288	0.401	0.409	0.406	0.383
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Explains more growth than education



Patterns of specialization and economic complexity through the lens of universal exhibitions, 1855-1900

Giacomo Domini 



	(3)	(4)
	Next-century growth	Next-century growth
ECI	0.509*** (0.154)	0.400*** (0.128)
GDP per capita	-0.597*** (0.161)	-0.542** (0.219)
Constant	-0.128 (0.121)	-0.092** (0.043)
Country fixed effects	No	Yes
N of observations	96	96
N of countries	33	33
N of time periods	5	5
Adjusted R ²	0.221	0.770

Map 1: States' Level of Economic Complexity, 2013

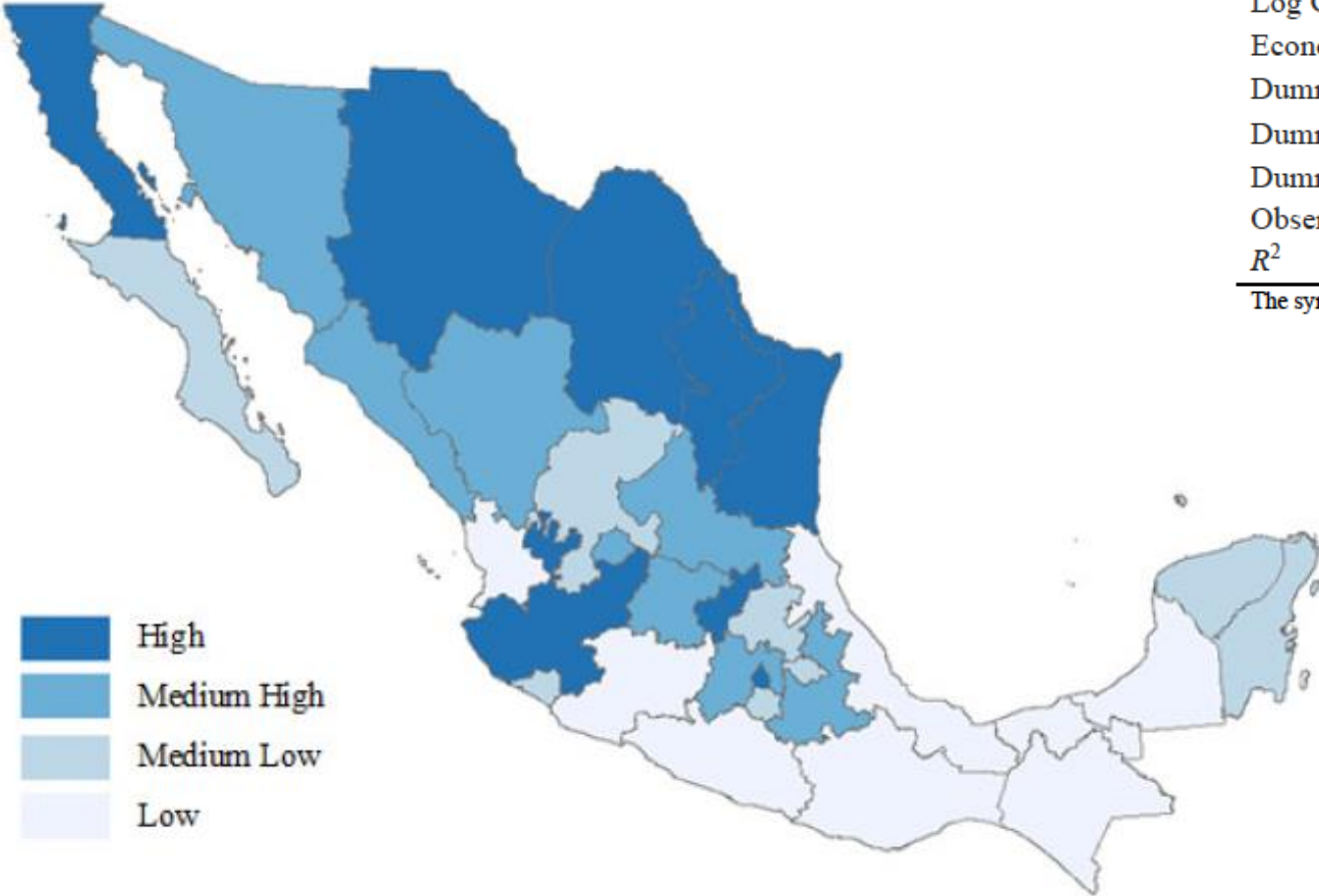


Table 3: Growth Rate of per capita GDP and Economic Complexity (1998–2003, 2003–2008, and 2008–2013)

Variable	Estimated Parameters
Constant	5.500 ***
Log GDP per capita, (β_0)	-0.991 ***
Economic Complexity, (β_1)	0.407 ***
Dummy variable for oil-mining states (α_1)	-0.351
Dummy variable for the period 2003–2008 (α_2)	1.257****
Dummy variable for the period 2008–2013 (α_3)	-0.251
Observations	96
R^2	.35

The symbols *** denote statistical significance at 1 percent level.

Chávez, Juan Carlos, Marco T. Mosqueda, and Manuel Gómez-Zaldívar. "Economic complexity and regional growth performance: Evidence from the Mexican Economy." *Review of Regional Studies* 47, no. 2 (2017): 201-219.



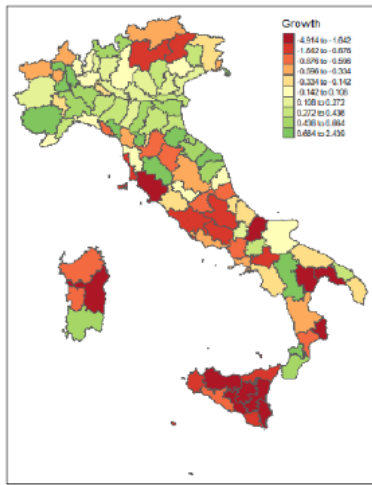
Table 2: GDP per capita growth (t+5) by province. Baseline specifications. Cross-sections between the first and the last year every t+5 years. OLS estimators.

Years-Thresholds	1995-2000, 2000-2005, 2005-2010, 2010-2015							
	M1	M2	M3	M4	M1	M2	M3	M4
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
VARIABLES	GDPpc/World GDPpc (t+5)				GDPpc (t+5)			
Total ECI	0.393*** (0.0528)		0.395*** (0.0581)	0.437*** (0.0556)	0.316*** (0.0460)		0.318*** (0.0541)	0.398*** (0.0533)
Physical cap.	0.317*** (0.0784)	0.320*** (0.0861)		0.369*** (0.0571)	0.376*** (0.0728)	0.378*** (0.0774)		0.474*** (0.0557)
Human cap.	0.117 (0.0827)	0.317*** (0.0886)	0.277*** (0.0604)		0.219*** (0.0769)	0.379*** (0.0809)	0.408*** (0.0560)	
Observations	200	200	200	200	200	200	200	200
R-squared	0.551	0.425	0.483	0.543	0.603	0.518	0.503	0.573
Change in R-squared		22.90%	12.30%	1.50%		14.10%	16.58%	4.98%

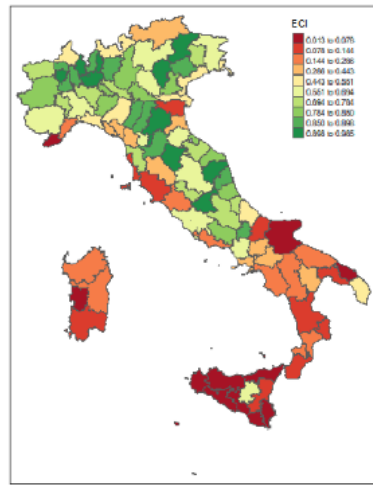
Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

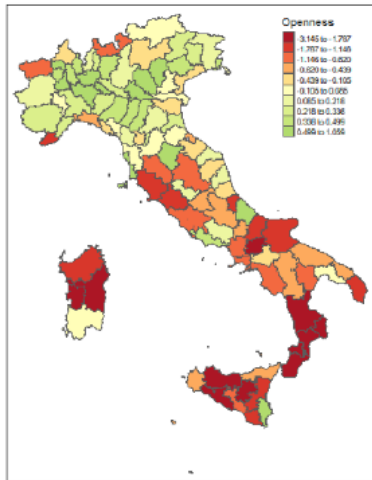
Balsalobre, S. (2019). Measuring subnational economic complexity: An application with Spanish data.



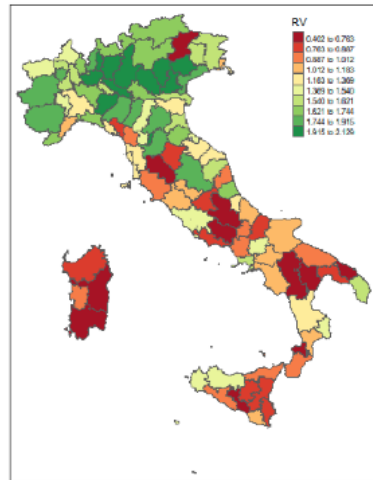
(a) Growth



(b) ECI



(c) Openness



(d) RV

Basile, Roberto, Gloria Cicerone, and Lelio Iapadre. "Economic complexity and regional labor productivity distribution: evidence from Italy." *Rev. Reg. Stud* 47 (2019): 201-219.

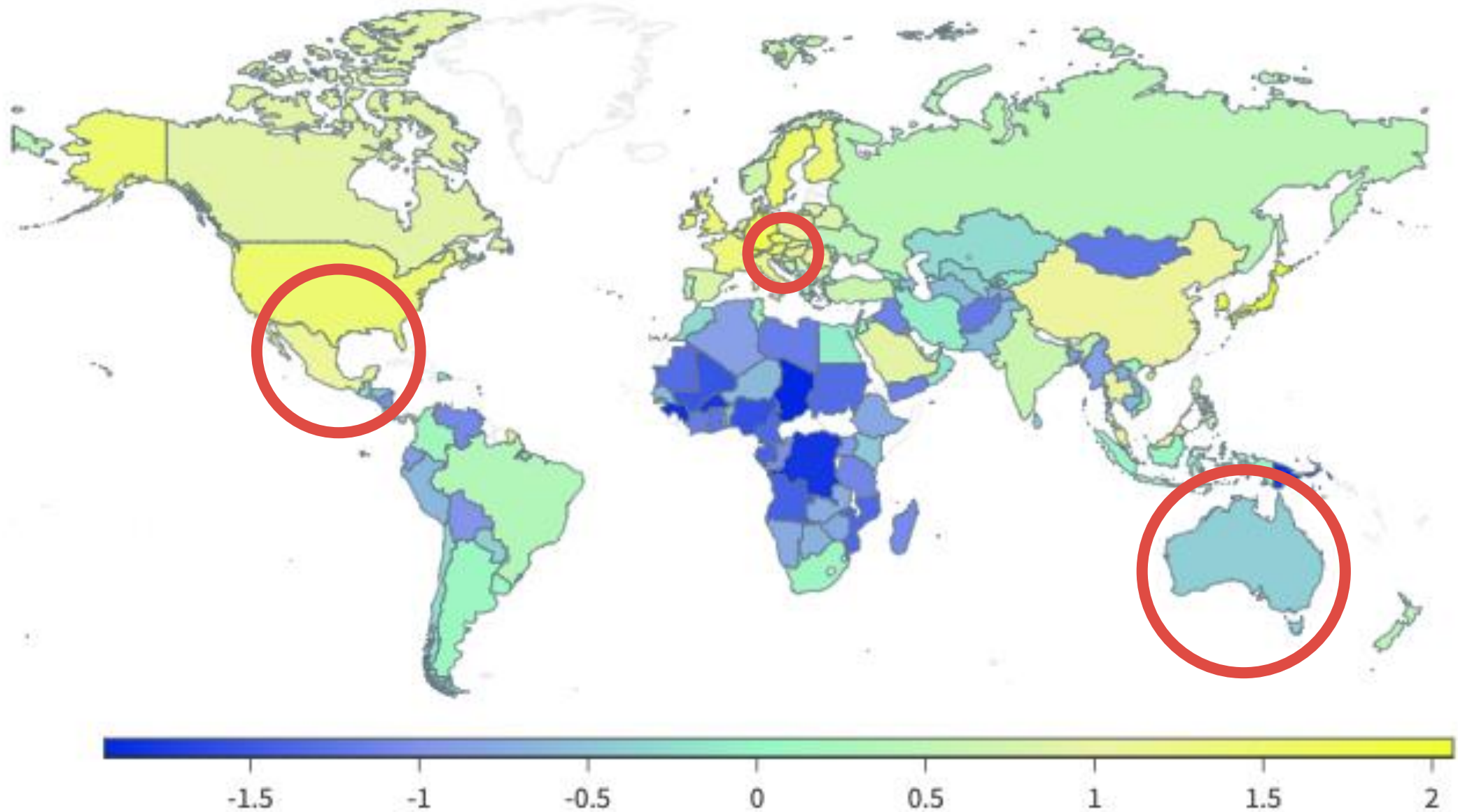
Basile, Roberto, and Gloria Cicerone. "Economic complexity and productivity polarization: Evidence from Italian provinces." *German Economic Review* 23, no. 4 (2022): 567-594.

Variable	Model 1	Model 2	Model 3
Coefficients and heteroskedasticity consistent s.e. (in parenthesis)			
Intercept	0.037 (0.100)	0.007 (0.113)	0.076 (0.090)
$\ln y_0$	-1.755*** (0.531)	-2.120*** (0.653)	-1.866*** (0.537)
$\ln(eci)$	0.259** (0.123)	0.221* (0.127)	0.300*** (0.116)
$\ln(open)$	0.321** (0.153)	0.349** (0.157)	0.401** (0.165)
$\ln(rv)$	0.525* (0.309)	0.467 (0.288)	
$\ln s_k$		-0.114 (0.425)	
$\ln(n)$		0.084 (0.120)	
$\ln s_h$		0.021 (0.028)	
$\ln s_A$		0.106 (0.121)	
Diagnostics			
Weak instr.- $\ln(eci)$	166.377*** (0.000)	128.993*** (0.000)	253.300*** (0.000)
Weak instr.- $\ln(open)$	85.596*** (0.000)	76.720*** (0.000)	130.441*** (0.000)
Weak instr.- $\ln(rv)$	52.356*** (0.000)	43.173*** (0.000)	
Wu-Hausman	6.684*** (0.000)	6.329*** (0.000)	6.795*** (0.001)
Sargan	0.927 (0.818)	0.753 (0.860)	0.104 (0.949)
Multiple R-Squared	0.523	0.520	0.538
Moran I	-0.132 (0.552)	-0.025 (0.510)	0.459 (0.323)

Our elaboration on ISTAT data.

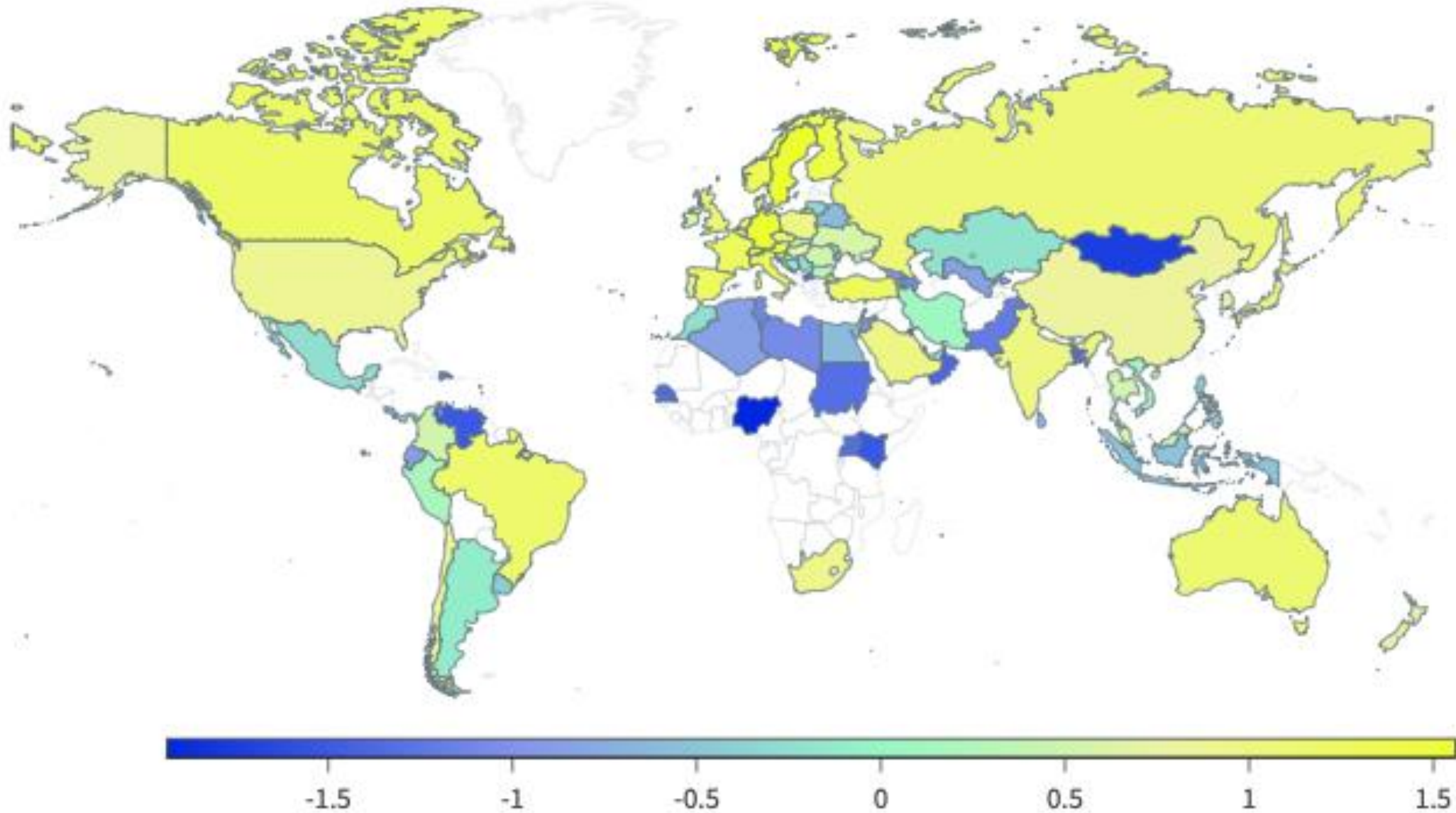
$$y = \beta_0 \mathbf{1} + \beta_1 \ln y_0 + \beta_2 \ln eci_0 + \beta_3 \ln open_0 + \varepsilon$$

Limitations of ECI trade

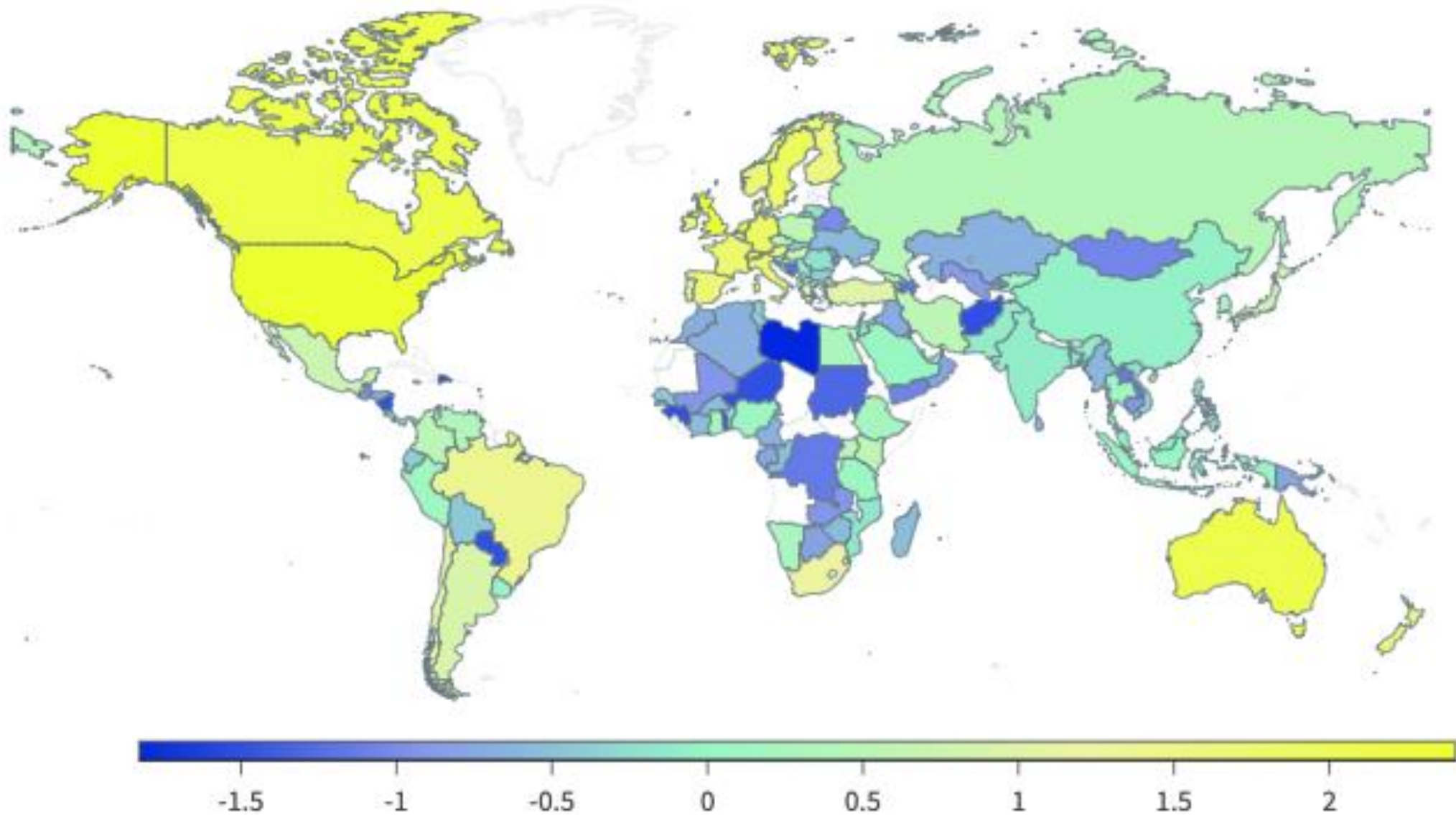


Stojkoski, Viktor, Philipp Koch, and César A. Hidalgo. "Multidimensional economic complexity and inclusive green growth." *Communications Earth & Environment* 4.1 (2023): 130.

ECI Tech (Patents)

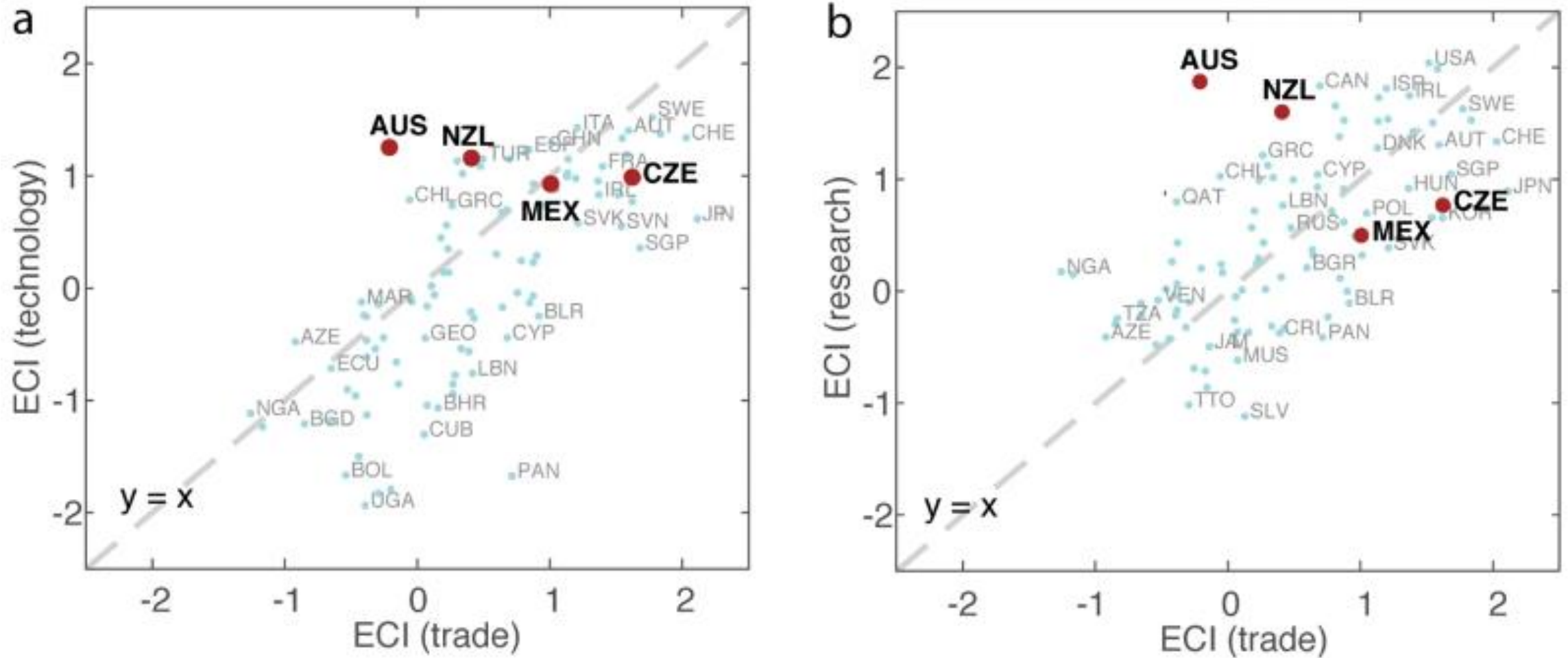


ECI Research



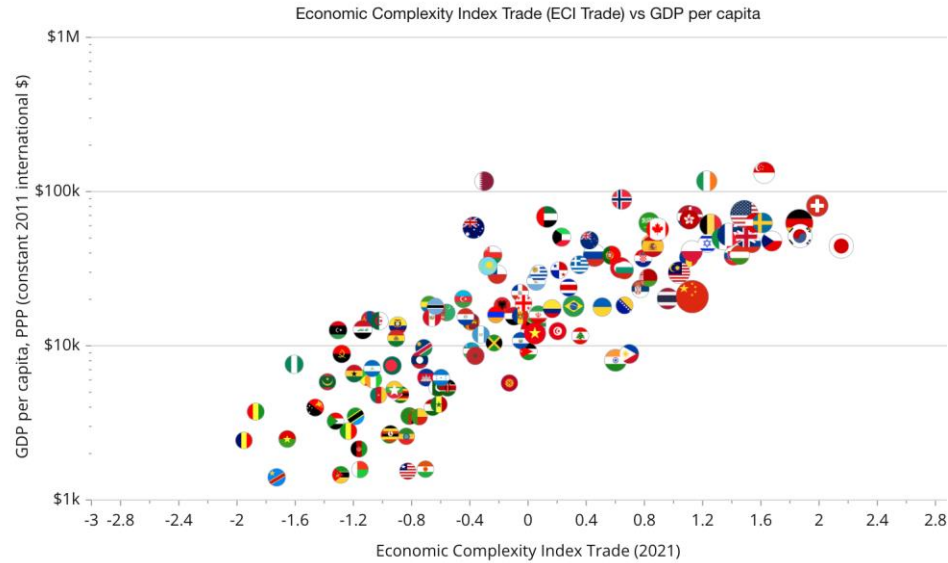
Stojkoski, Viktor, Philipp Koch, and César A. Hidalgo. "Multidimensional economic complexity and inclusive green growth." *Communications Earth & Environment* 4.1 (2023): 130.

ECI Trade, Tech, & Research are complements

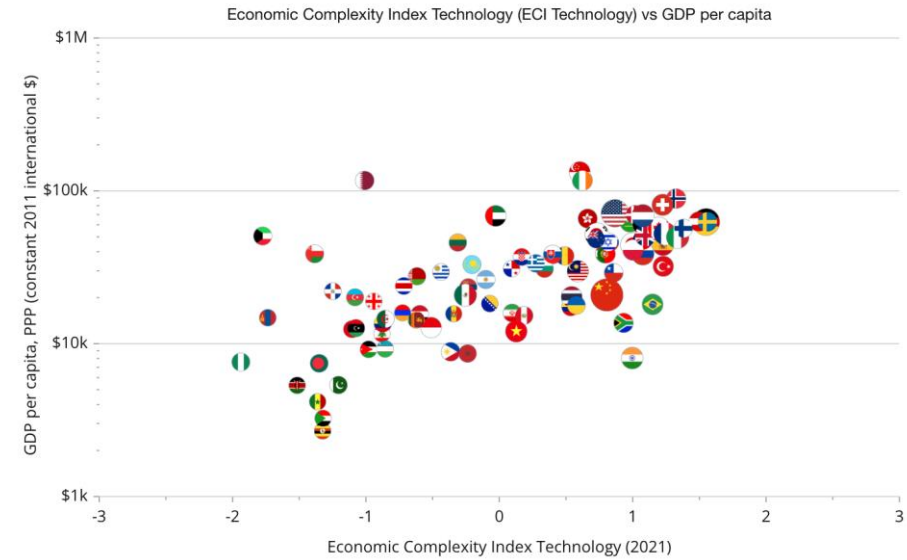


Stojkoski, Viktor, Philipp Koch, and César A. Hidalgo. "Multidimensional economic complexity and inclusive green growth." *Communications Earth & Environment* 4.1 (2023): 130.

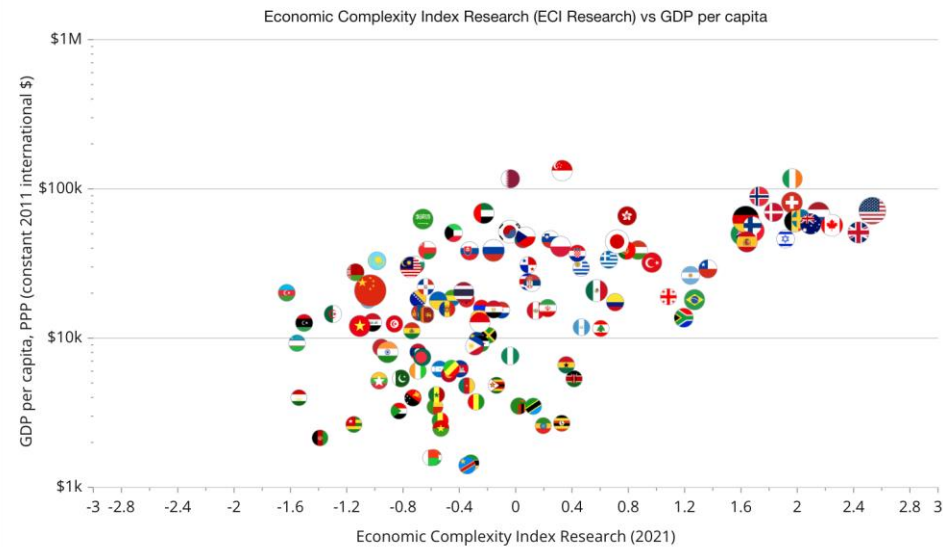
Trade (Product Exports)



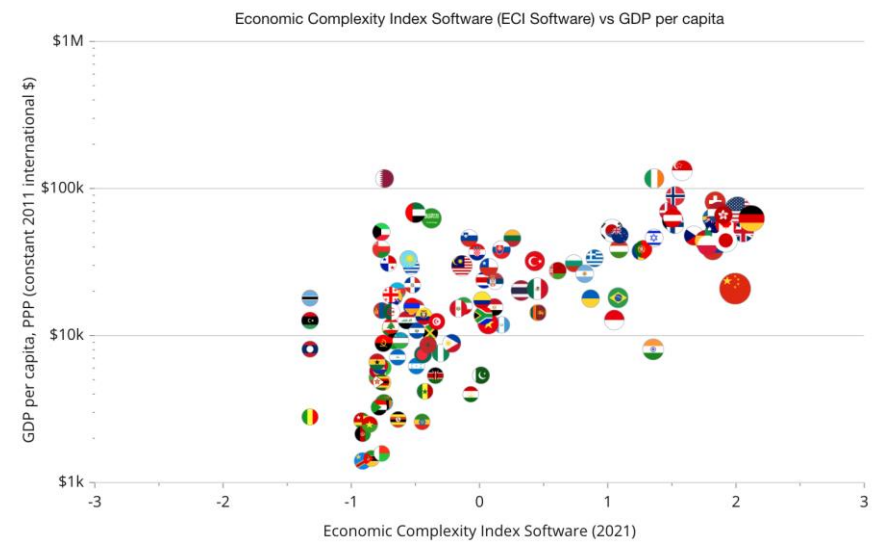
Technology (Patents)



Research (Papers)



Software (OSS Commits)



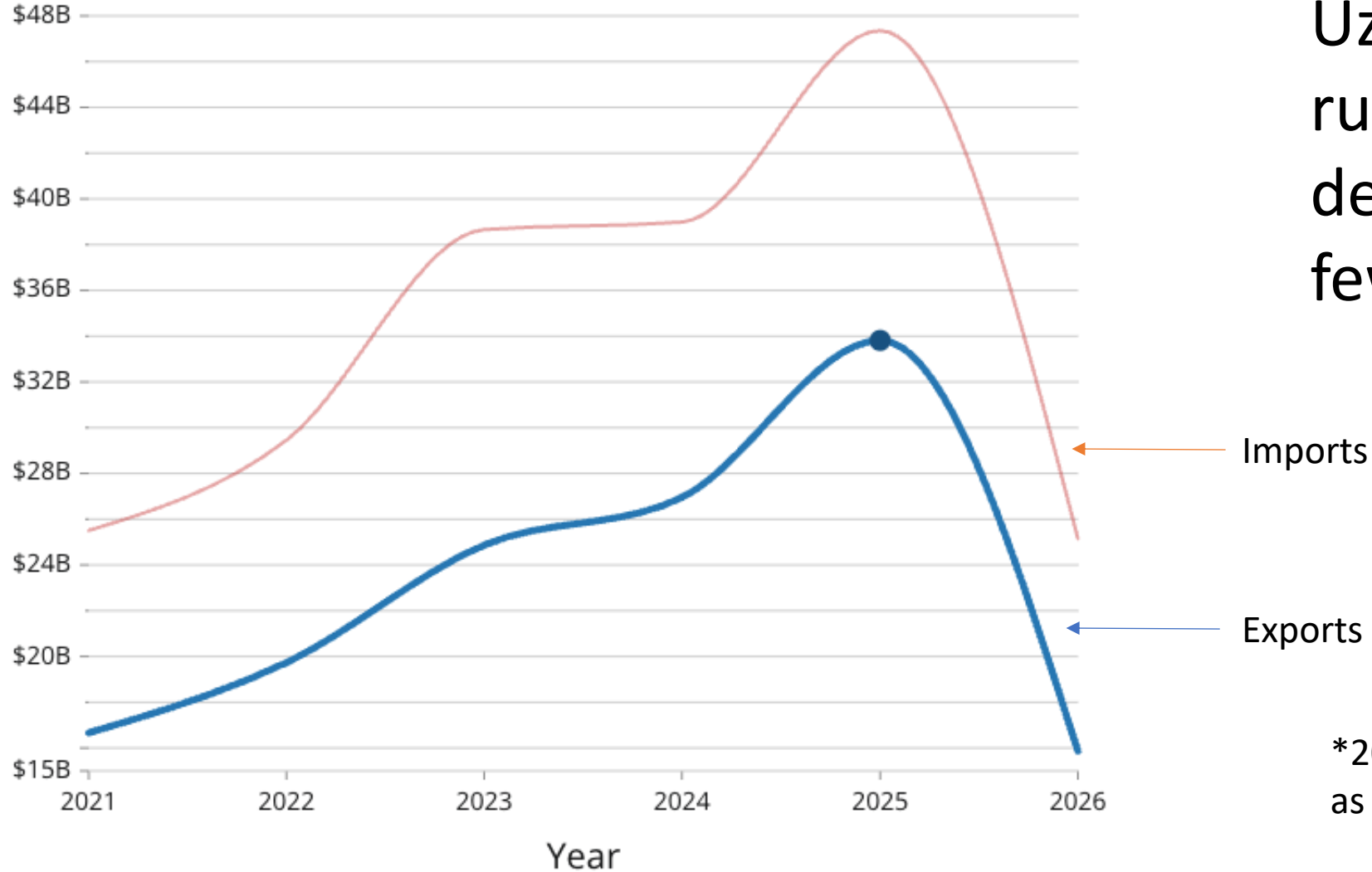


Let's take a quick look at Uzbekistan!



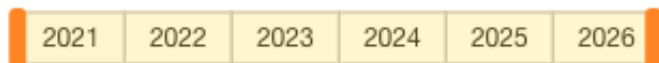
Uzbekistan Exports Over Time

[Click to select a Date]



Uzbekistan has been running a trade deficit in goods for a few years.

*2026 data is year to date as on June 2026



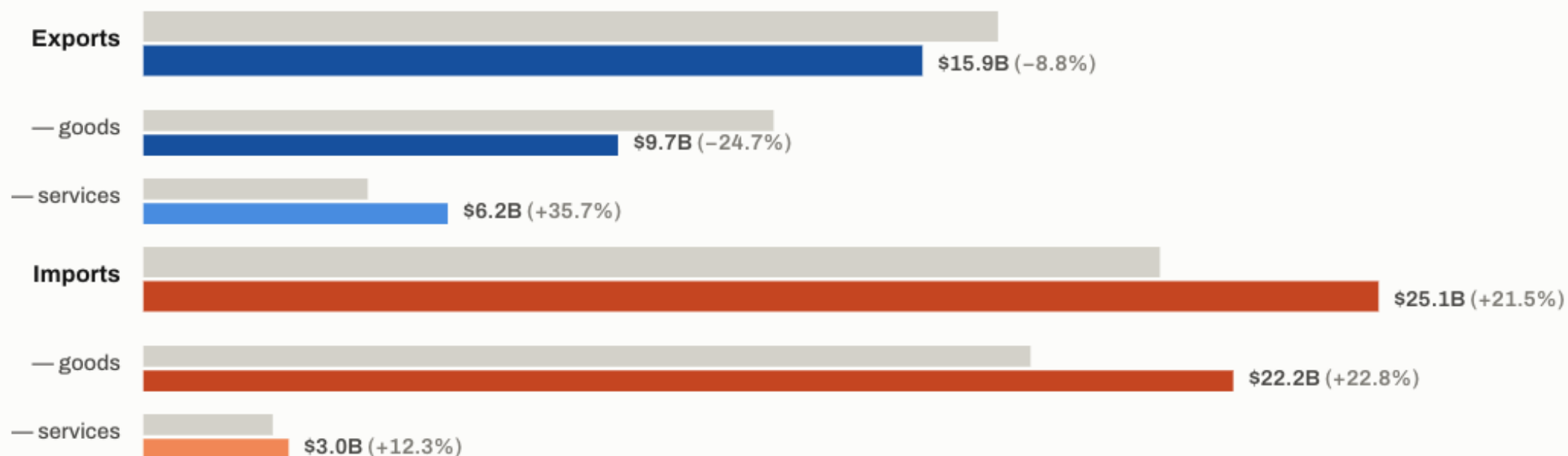
oec.world <https://stat.uz/>

And the imbalance continues to grow...

2026 so far: January–June vs a year earlier

Cumulative first-half figures, goods + services. Imports kept climbing; exports fell because gold-dominated “other goods” collapsed from \$6.6B to \$1.5B. Everything else — food, chemicals, industrial goods, and above all services — grew strongly.

■ H1 2025 ■ H1 2026 · exports ■ H1 2026 · imports



Own viz made from data at: <https://stat.uz/>

Total trade, 2025

\$82.8B

+23.1% vs 2024

Exports, 2025

\$34.2B

+25.6% vs 2024

Imports, 2025

\$48.5B

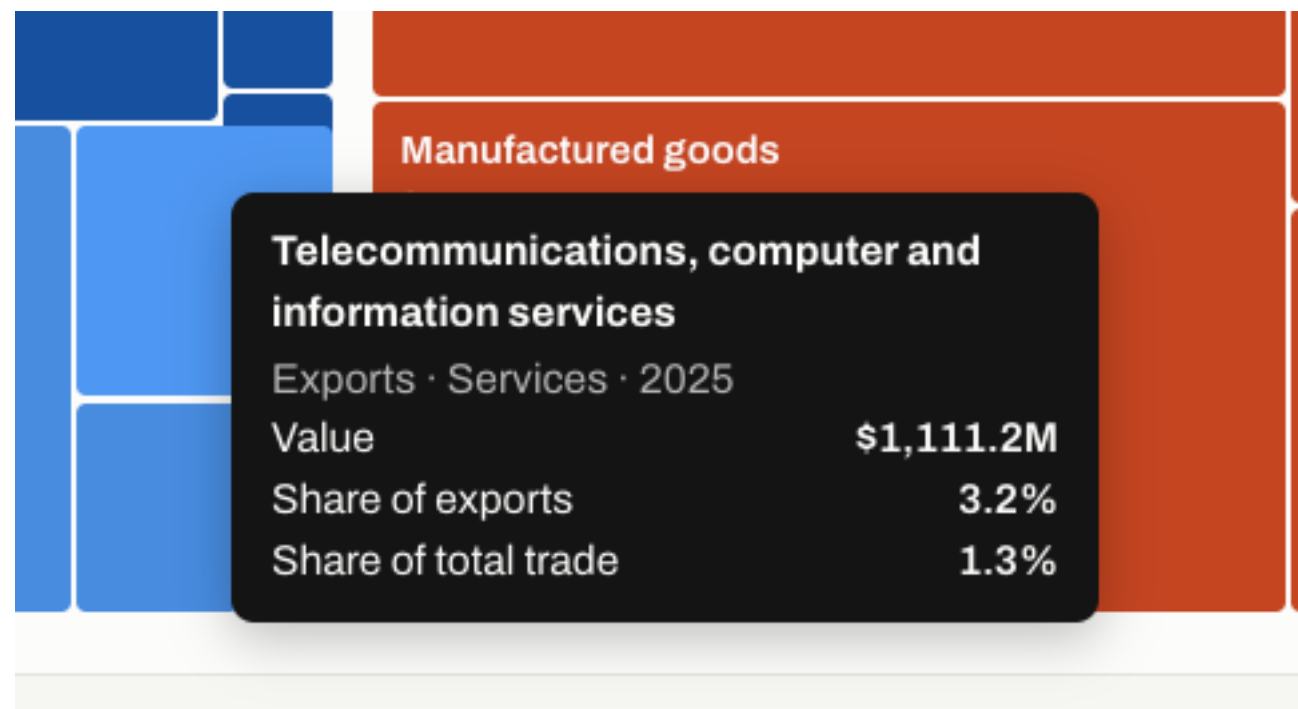
+21.4% vs 2024

ICT services exports, 2025

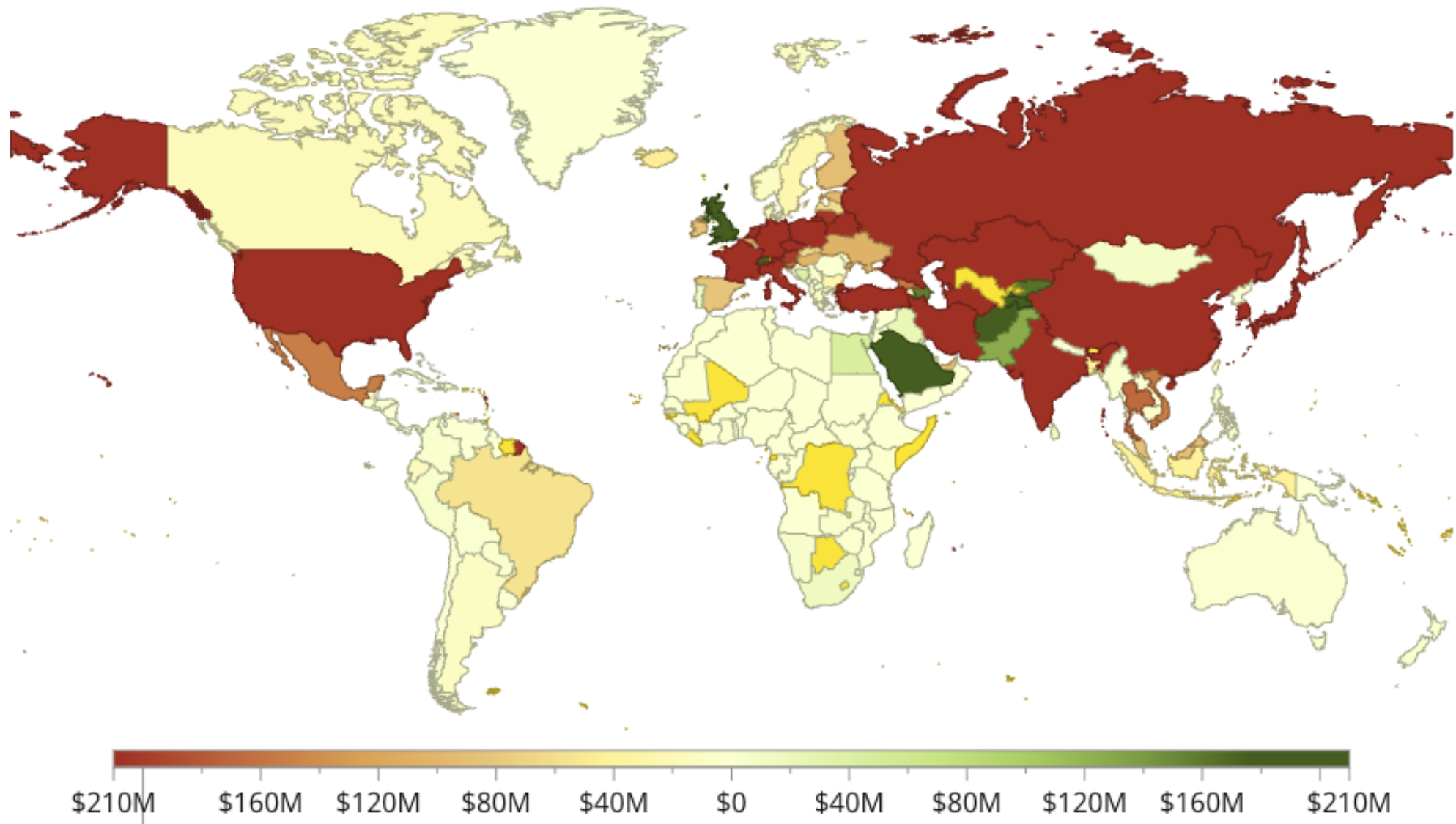
\$1.11B

+142% vs 2024

**ICT exports of
Uzbekistan are still
relatively small, but
are growing rapidly
(~140%).**



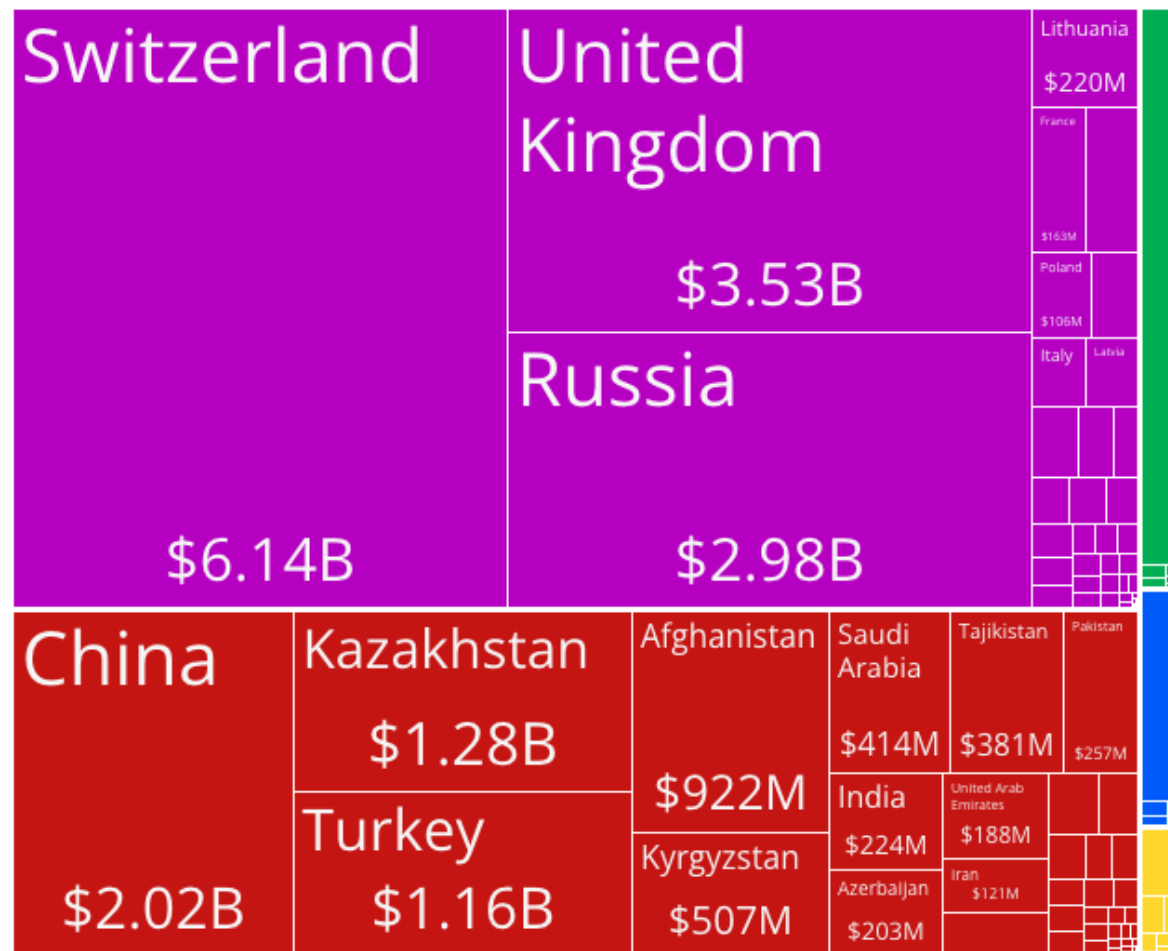
Trade Balance



Uzbekistan's Export Destinations

[Click to select a Country]

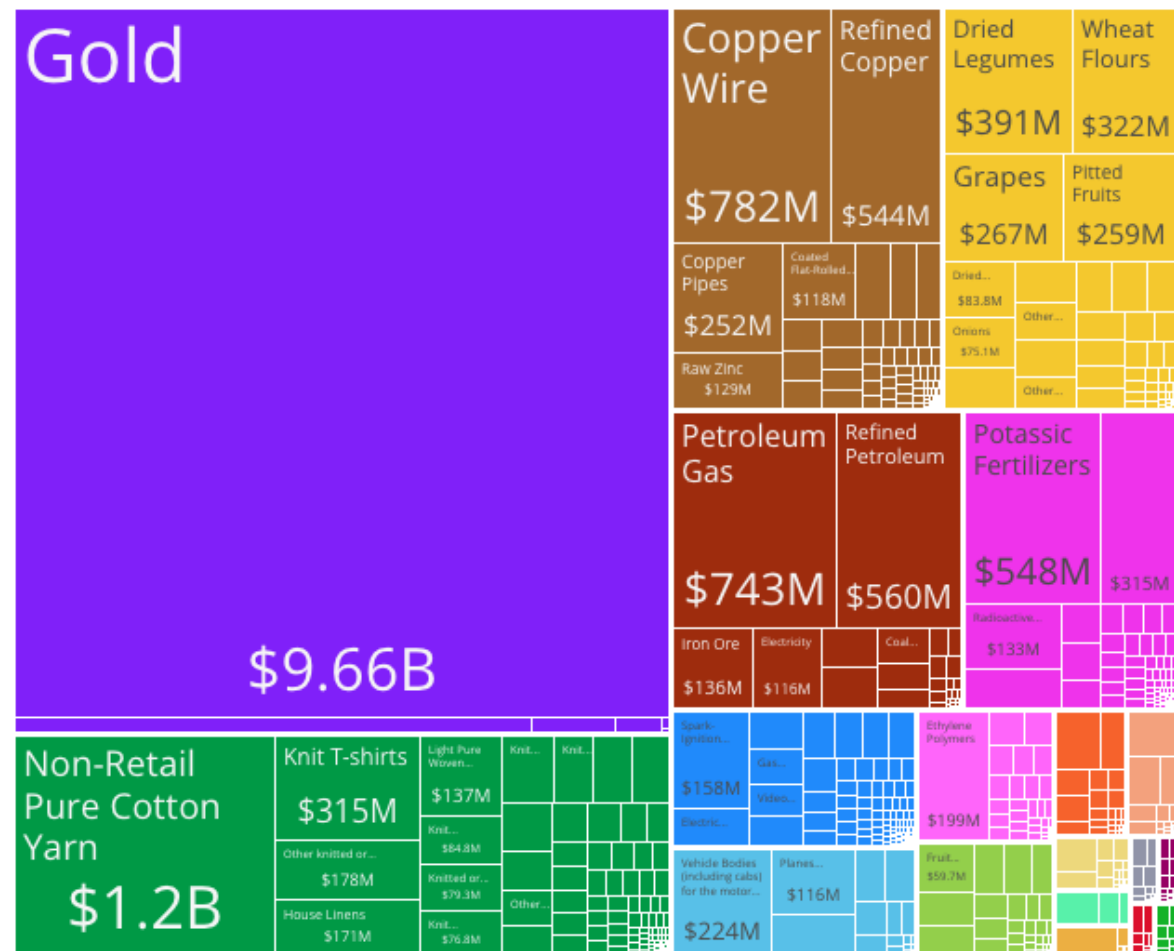
Total : \$22.8B



Uzbekistan's Exported Products

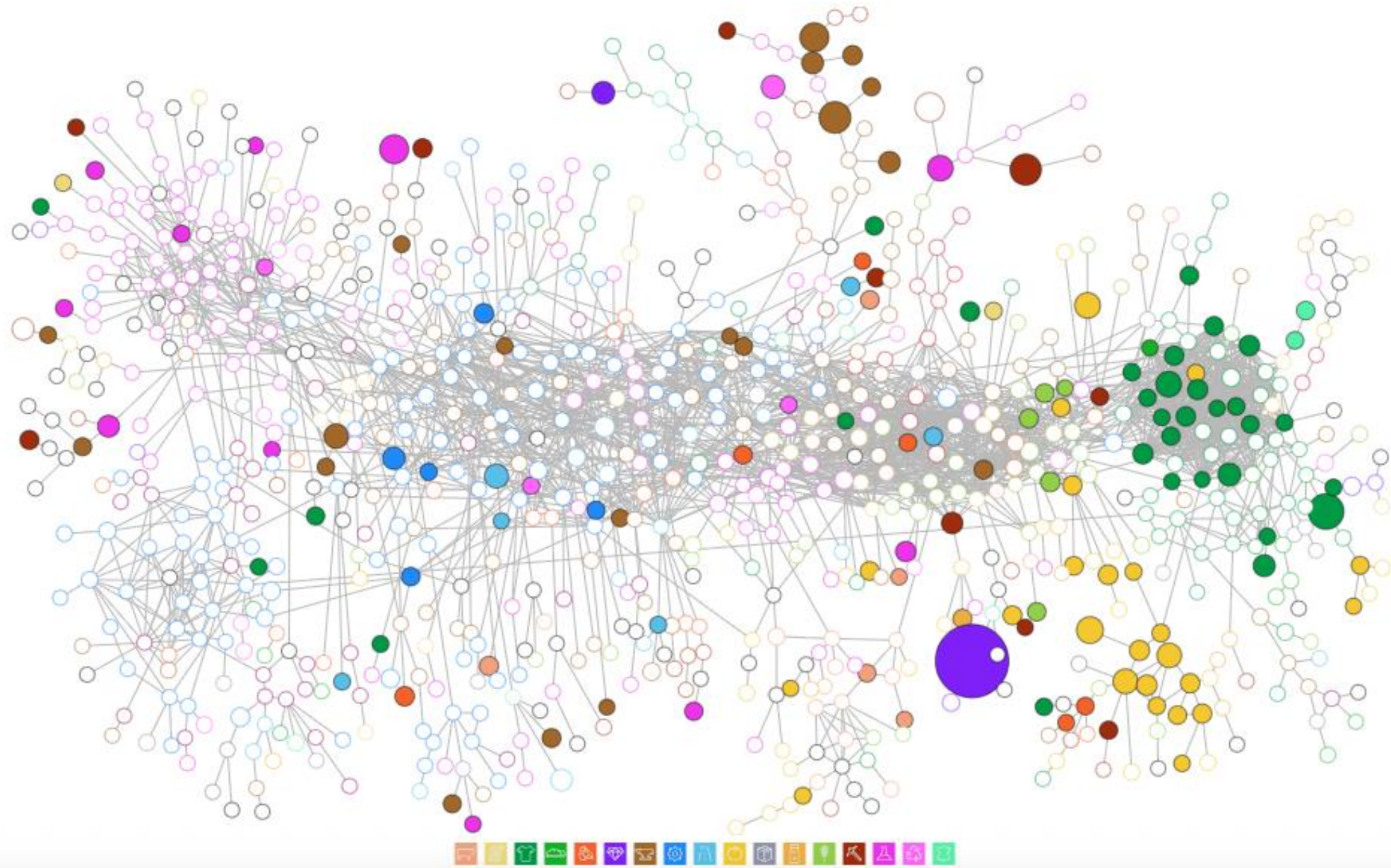
[Click to select a Product]

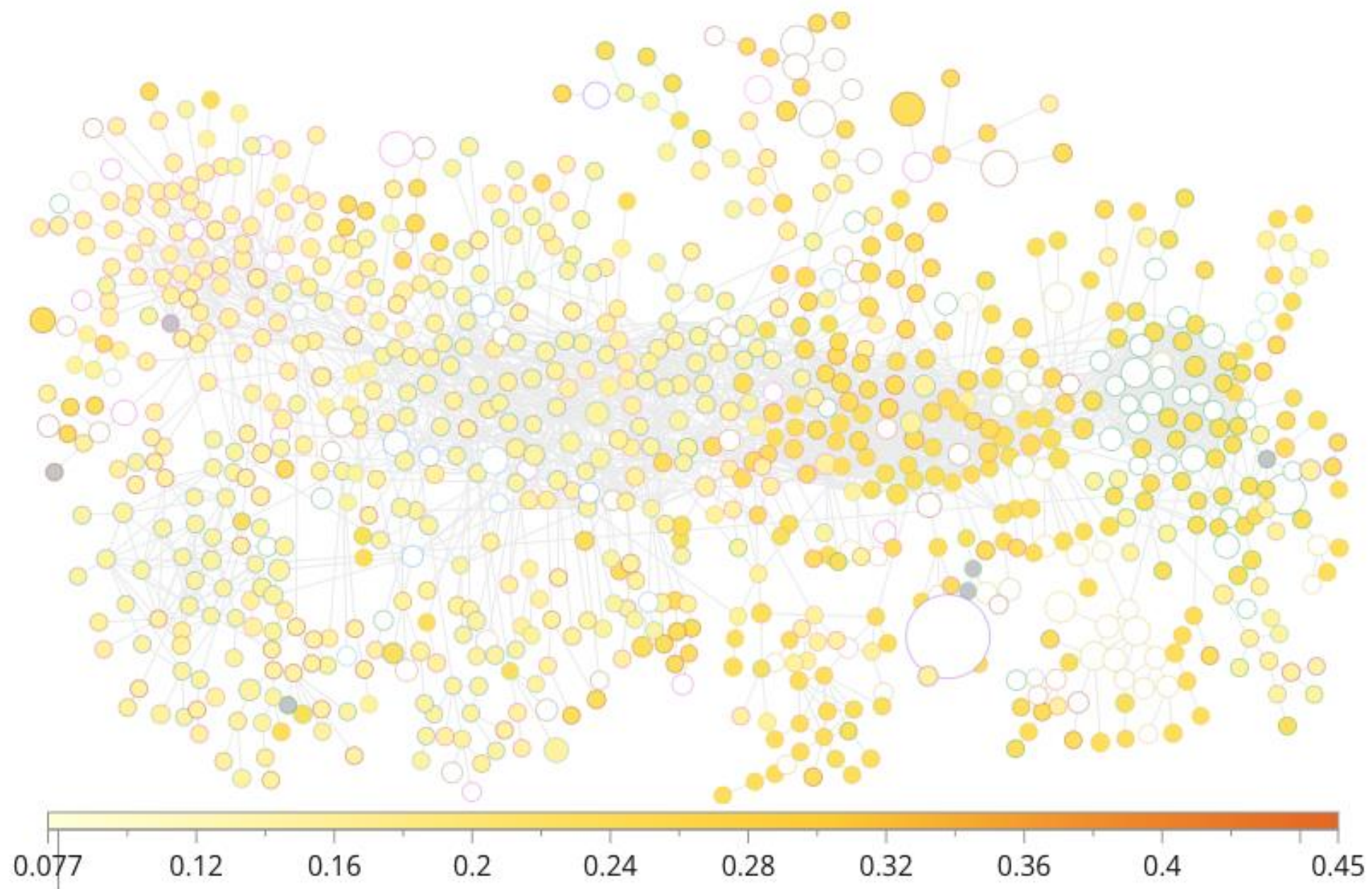
Total: \$22.8B

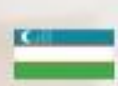


Source oec.world (large Gold exports usually represented unclassified transactions).

Product Space of Uzbekistan (2024)







Uzbekistan



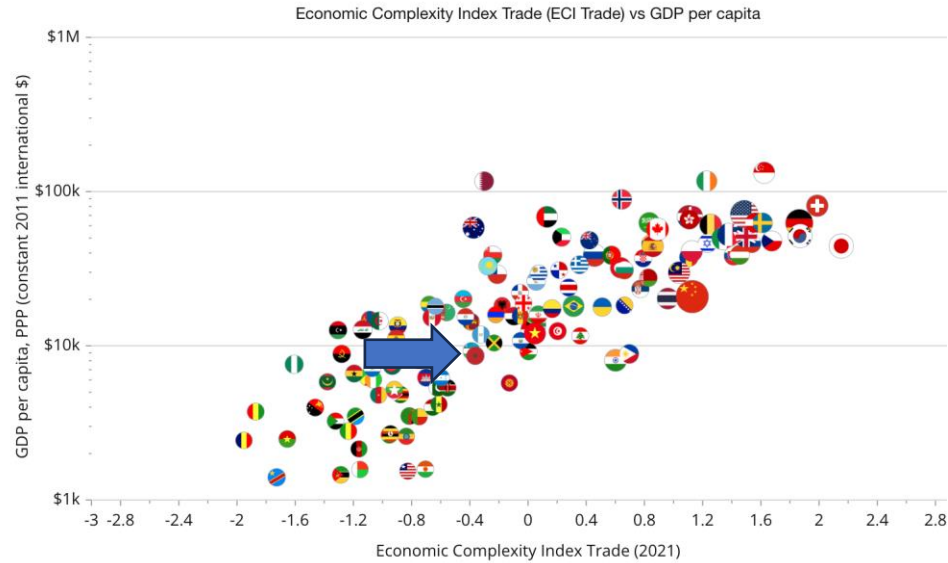
ECI
(TRADE)
77th
Out of 130
(ECI: -0.33, 2024)

ECI
(TECHNOLOGY)
71st
Out of 97
(ECI: -0.86, 2021)

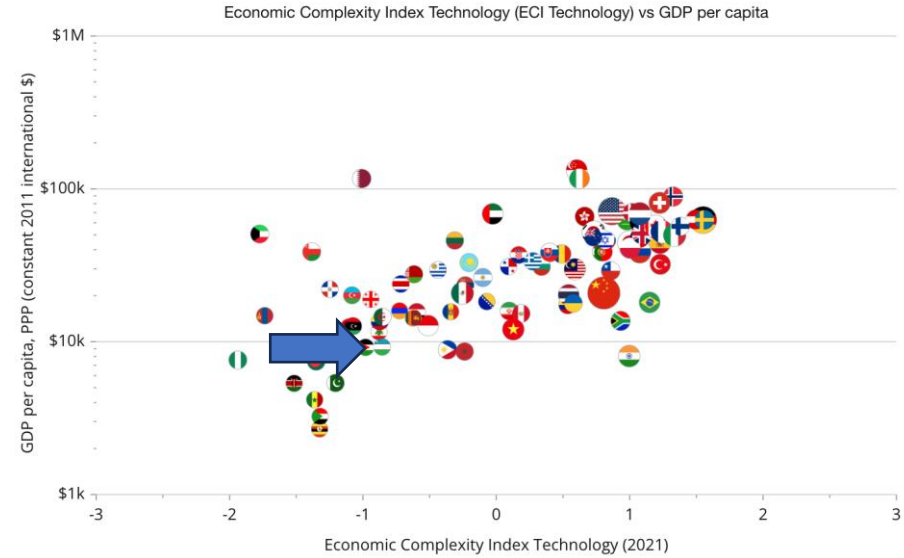
ECI
(RESEARCH)
138th
Out of 138
(ECI: -1.44, 2024)

ECI
(SOFTWARE)
83rd
Out of 172
(ECI: -0.33, 2025)

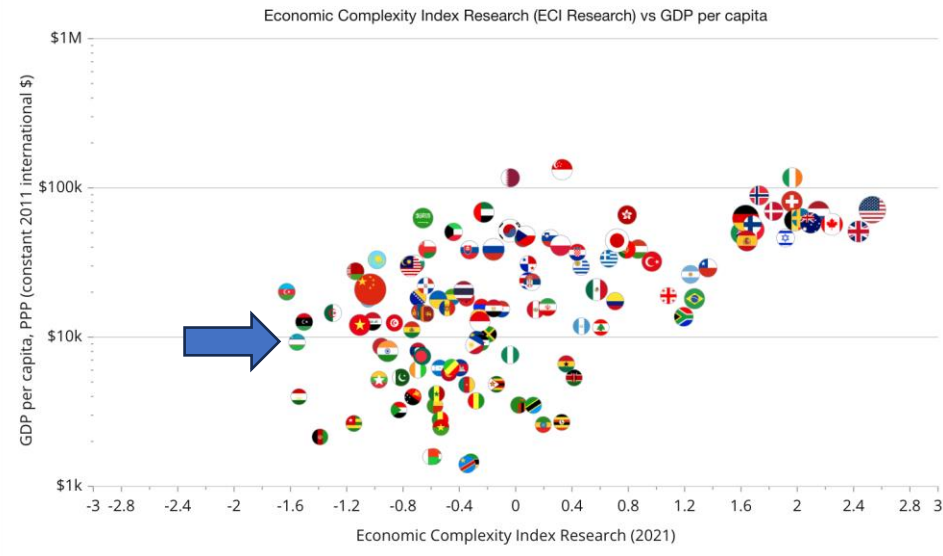
Trade (Product Exports)



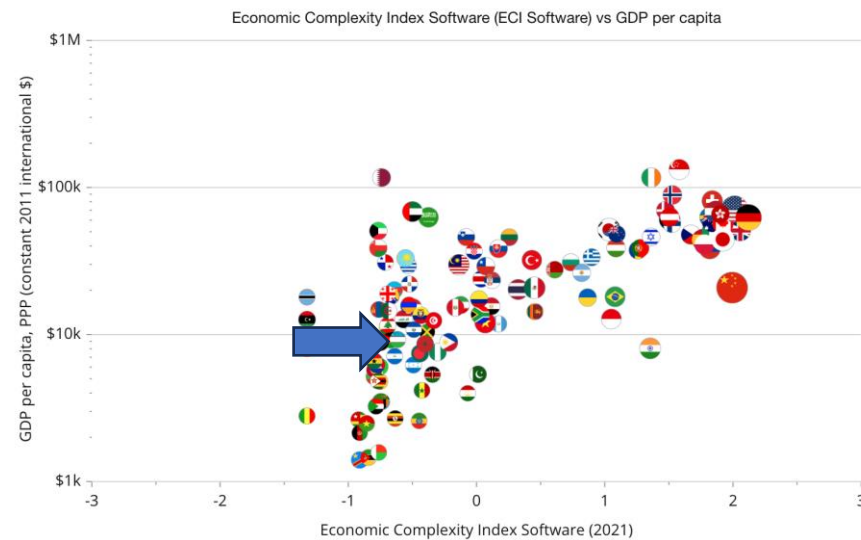
Technology (Patents)



Research (Papers)



Software (OSS Commits)





Putting these ideas into practice

DATA
SAUDI
بيانات السعودية



Kingdom of Saudi Arabia

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Explore our reports



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Ej. Ciudad de México, Monterrey

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PAÍSES ¡NUEVO!

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México



Minería



Estados Unidos



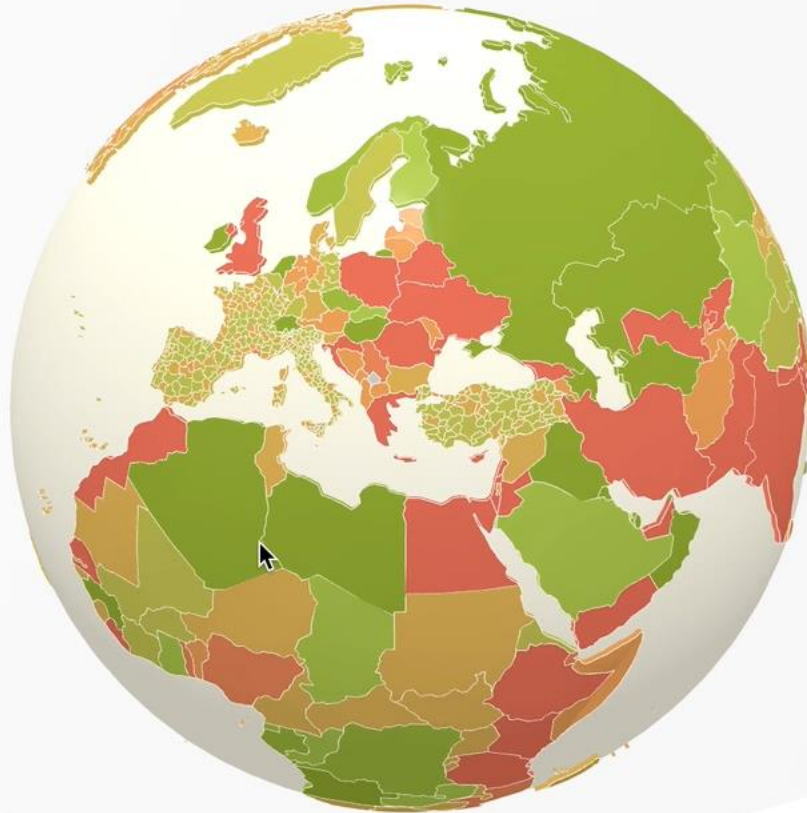
Médicos, Enfermeras y
otros Especialistas en
Salud



Combustibles Minerales,
Aceites, Etc



Universidad Nacional
Autónoma De México



The Trade Data You Need, When You Need It

🔍 Search a Country, Product, Region, ...

Chat ✨

Costco Wholesale

China

Amazon

Cement

2032 Growth Predictions

NEW



Chat OEC

NEW

What countries are the top importers of coffee?

The top importers of coffee are the United States, Germany, Italy, Belgium, and France.

Country	Trade Value	Quantity
United States	888793832	1577923
Germany	506402476	1205457
Italy	235945455	691120
Belgium	2382047044	506685
France	2178089617	297031

🔄

who was the top importer in 2022?



Create an Optimized Export Scenario

Target

Include Gold ☒ Include Oil ☒ Include Digital Products ☒

Growth and Complexity



📊 **Medium** (Growth: 1.166% - ECI: 1.100)



Simple input ☐

Create Portfolio



Home



AI Report



Summary



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Analytics



Markets



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journal homepage: www.elsevier.com/locate/respol

Optimizing economic complexity

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JEL classification:

O11

O25

O61

Keywords:

Economic complexity

Economic development

Policy

ABSTRACT

Efforts to apply economic complexity to identify diversification opportunities often rely on diagrams comparing the relatedness and complexity of products, technologies, or industries. Yet, the use of these diagrams, is not based on empirical or theoretical evidence supporting some notion of optimality. Here, we introduce an optimization-based framework that identifies diversification opportunities by minimizing a cost function capturing the constraints imposed by an economy's pattern of specialization. We show that the resulting portfolios often differ from those implied by relatedness–complexity diagrams, providing a target-oriented optimization layer to the economic complexity toolkit.

1. Introduction

To achieve sustainable economic growth economies must adapt to changes in markets and technologies (Soubotina, 2004; Stiglitz, 2002). Changes in economic structure, however, involve strategic considerations such as identifying diversification opportunities while facing the constraints imposed by an economy's existing productive structure (Chang and Andreoni, 2020; Juhász and Lane, 2024; Juhász et al., 2024a).

During the last couple of decades, economic complexity has become a common method to explore questions of strategic diversification

Yet, the practical implementation of these ideas often relies on the use of *relatedness-complexity diagrams*: a graphical method introduced more than a decade ago (Hausmann et al., 2014) to visually inspect an economy's diversification opportunities (Hausmann et al., 2014; Hidalgo, 2023; Balland et al., 2019; Hausmann and Chauvin, 2015; Hausmann et al., 2017).

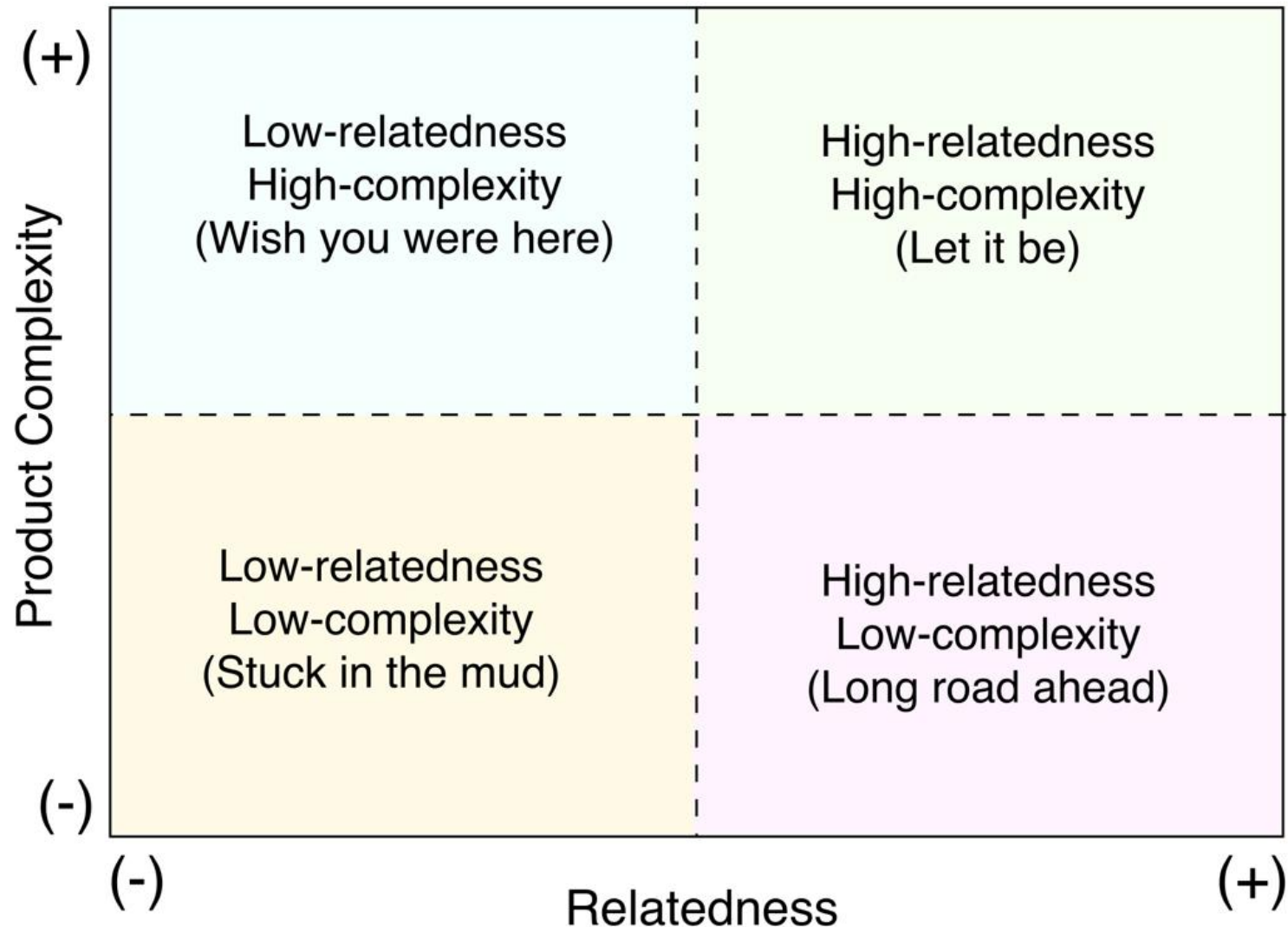
In a relatedness-complexity diagram, economic activities, such as product exports, industries, or technologies, are presented in a two-way scatter plot (Fig. 1) with relatedness in the x-axis and complexity on the y-axis. These diagrams are usually divided into quadrants to help facilitate their interpretation.

The Knee-Jerk reaction to Relatedness and Economic Complexity

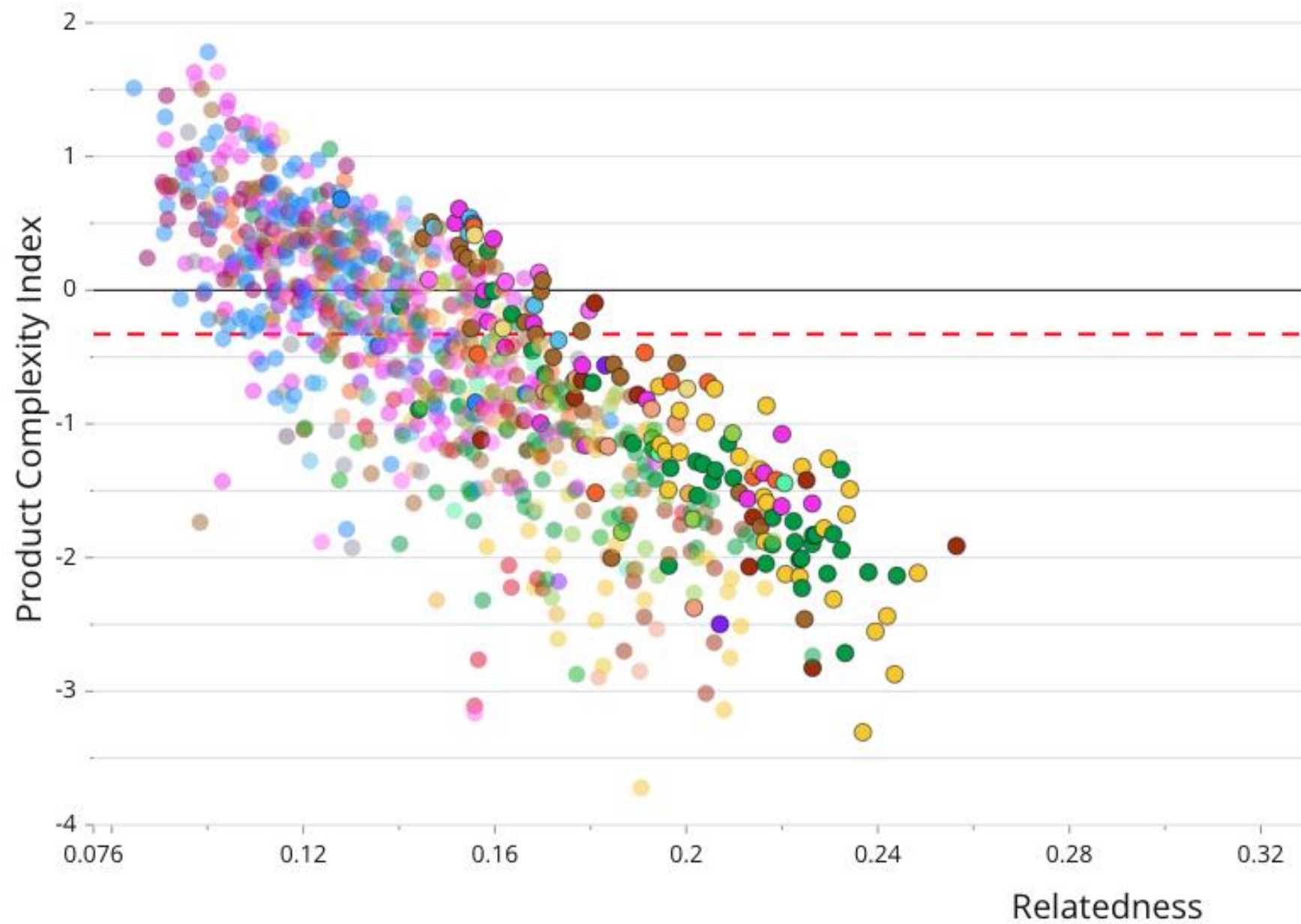
Economies should enter related & complex activities

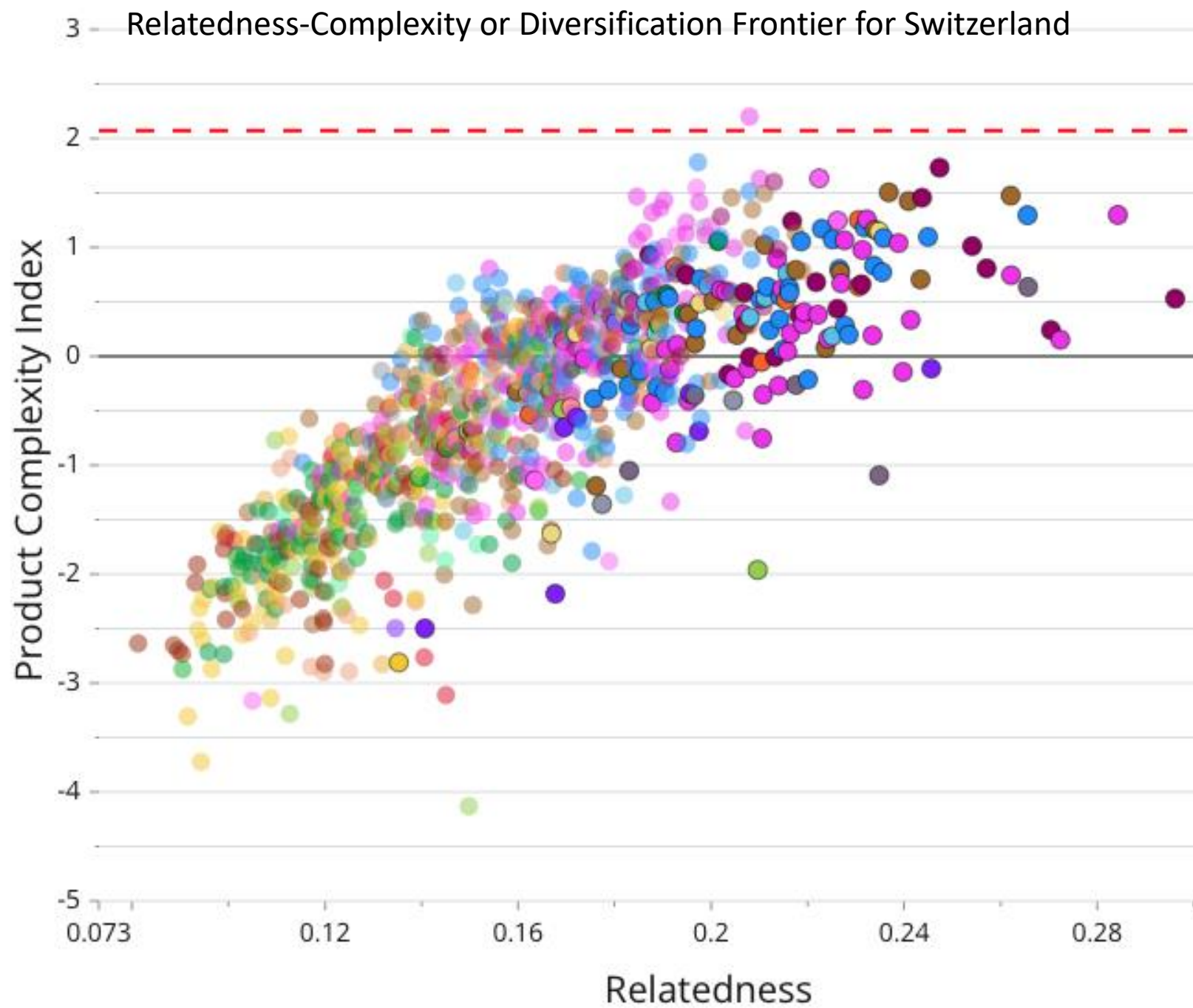


The Relatedness-Complexity (or Diversification Frontier) Diagram

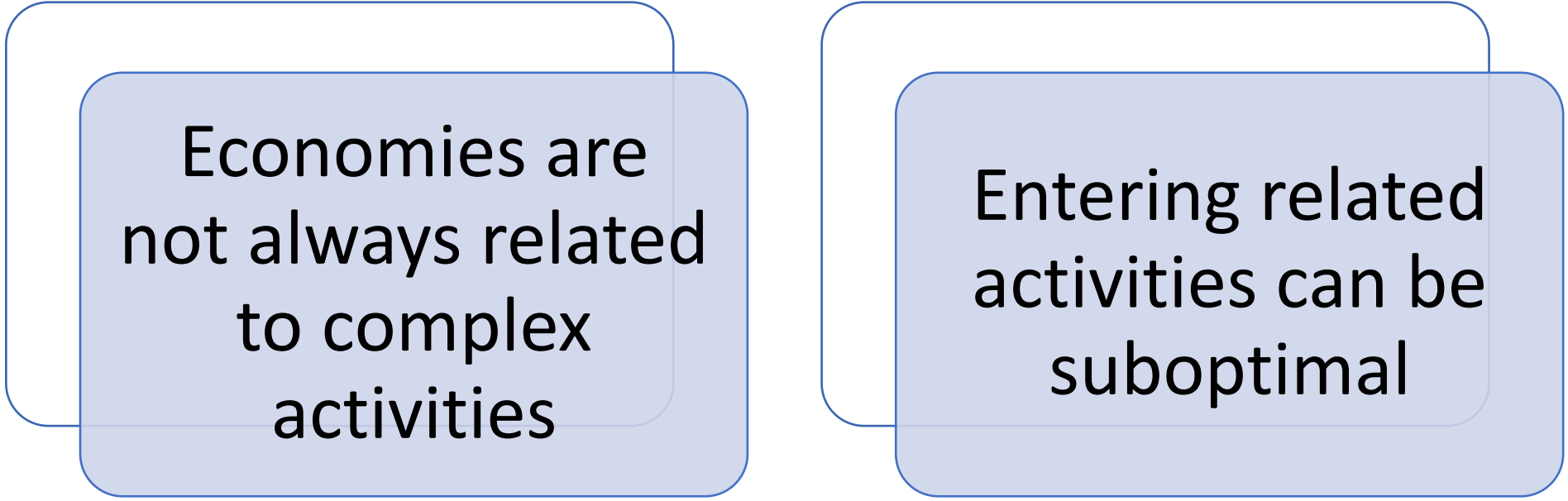


Relatedness-Complexity or Diversification Frontier for Uzbekistan





Two Problems

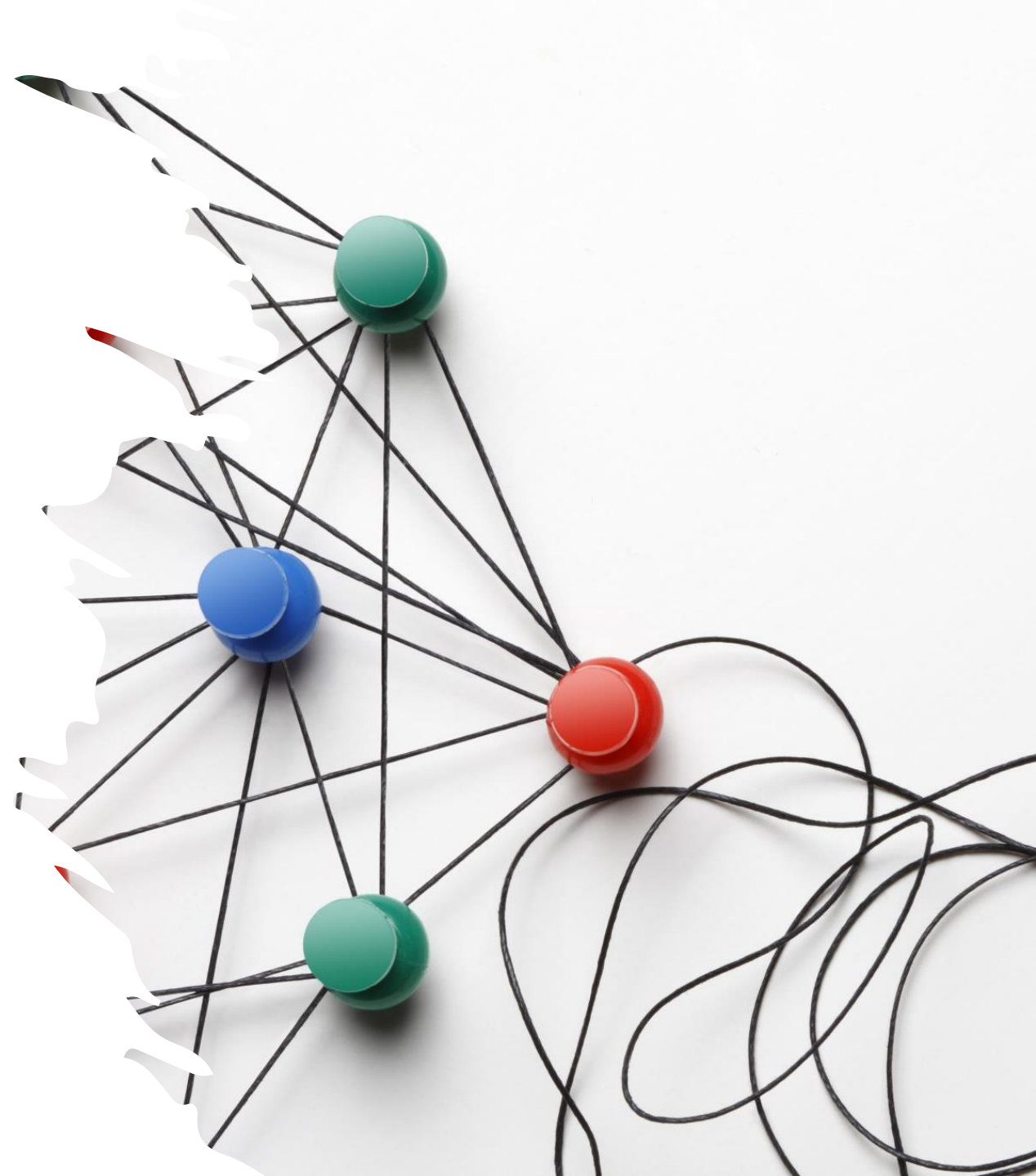
The diagram consists of two identical sets of overlapping rounded rectangles, one on the left and one on the right. Each set has a light blue outer rectangle and a slightly darker blue inner rectangle. The inner rectangles are offset to the bottom-right relative to the outer ones, creating a layered effect. The text is centered within the inner rectangles.

Economies are
not always related
to complex
activities

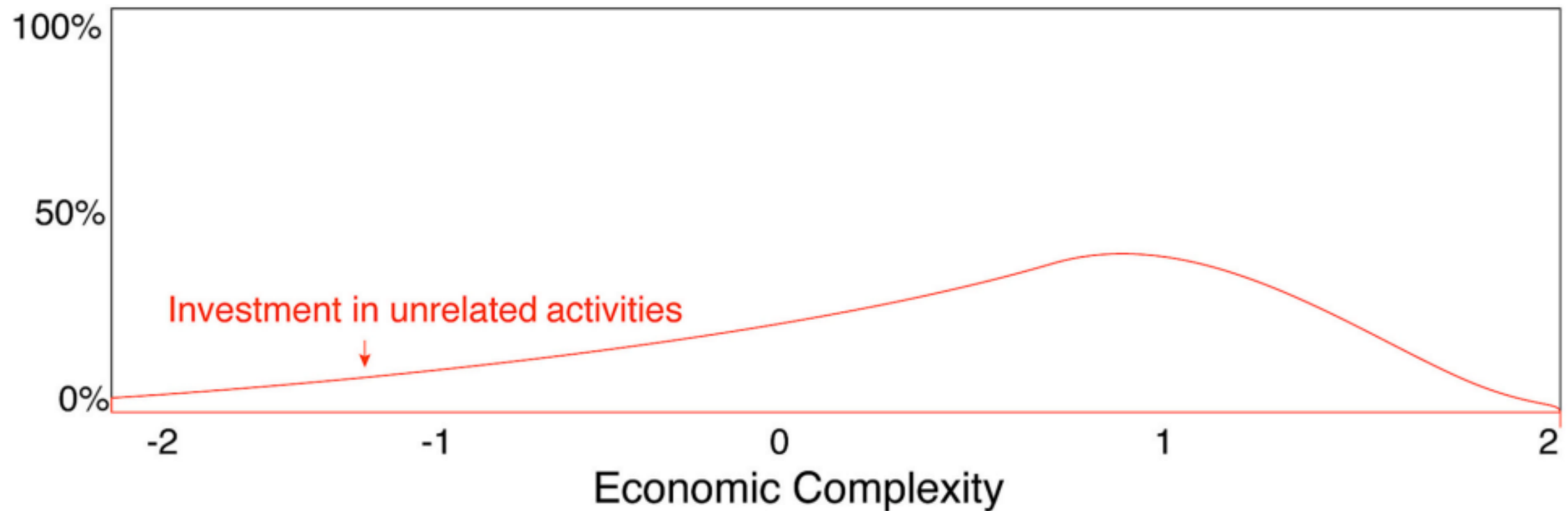
Entering related
activities can be
suboptimal

Problems with Relatedness-Complexity Diagrams

- First, relatedness is an incomplete measure of economic potential.
- Second, while complexity and relatedness change over time, it is common for applied work to only use current values.
- Third, in absence of complexity, relatedness is a suboptimal diversification strategy (Alshamsi et al. 2018)



These challenges point to a **portfolio strategy** that balances related and unrelated diversification attempts



The ECI Optimization Method

We are looking to optimize:

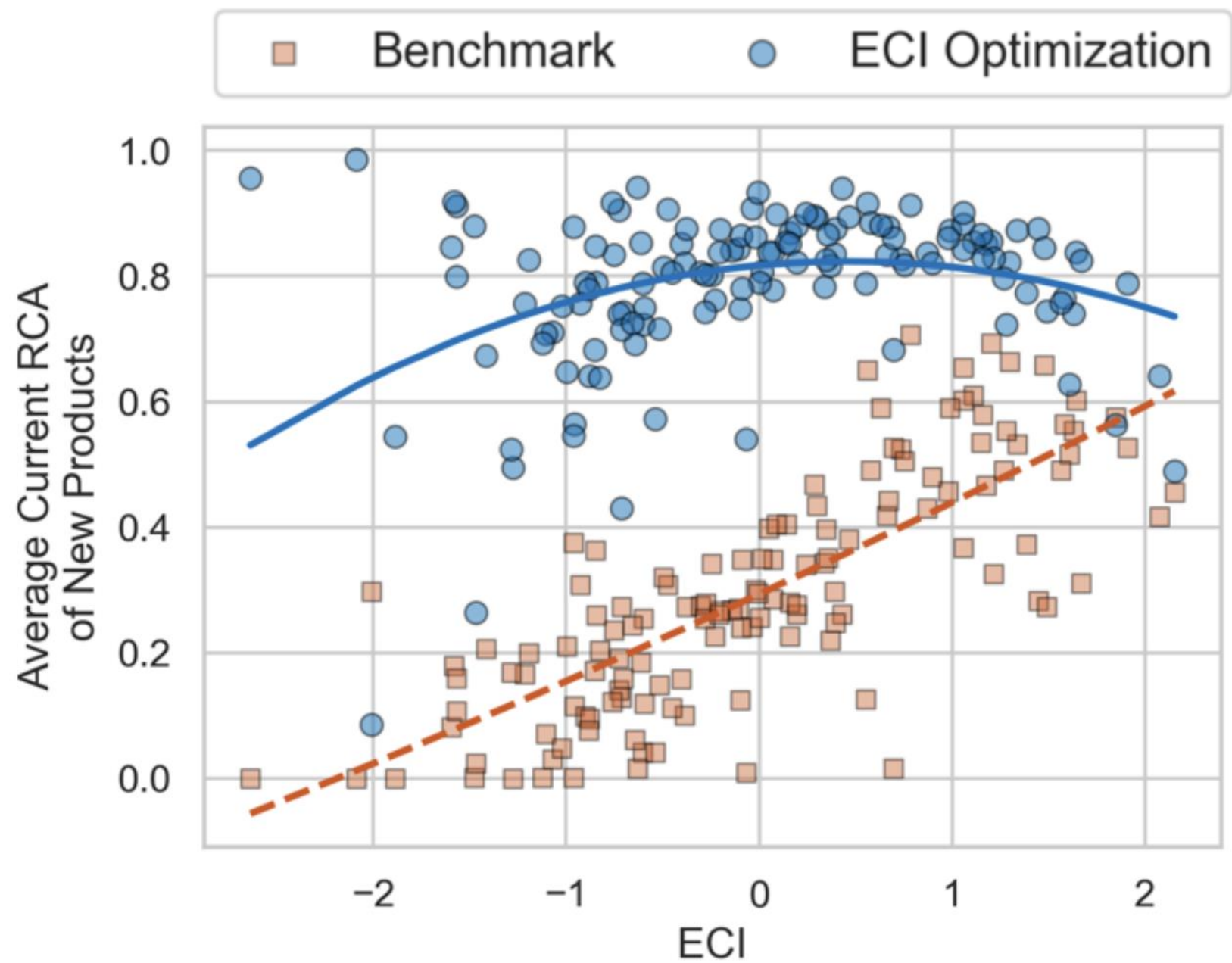
$$ECI_c^* = \frac{1}{M_c^*} \sum_p M_{cp}^* PCI_p$$

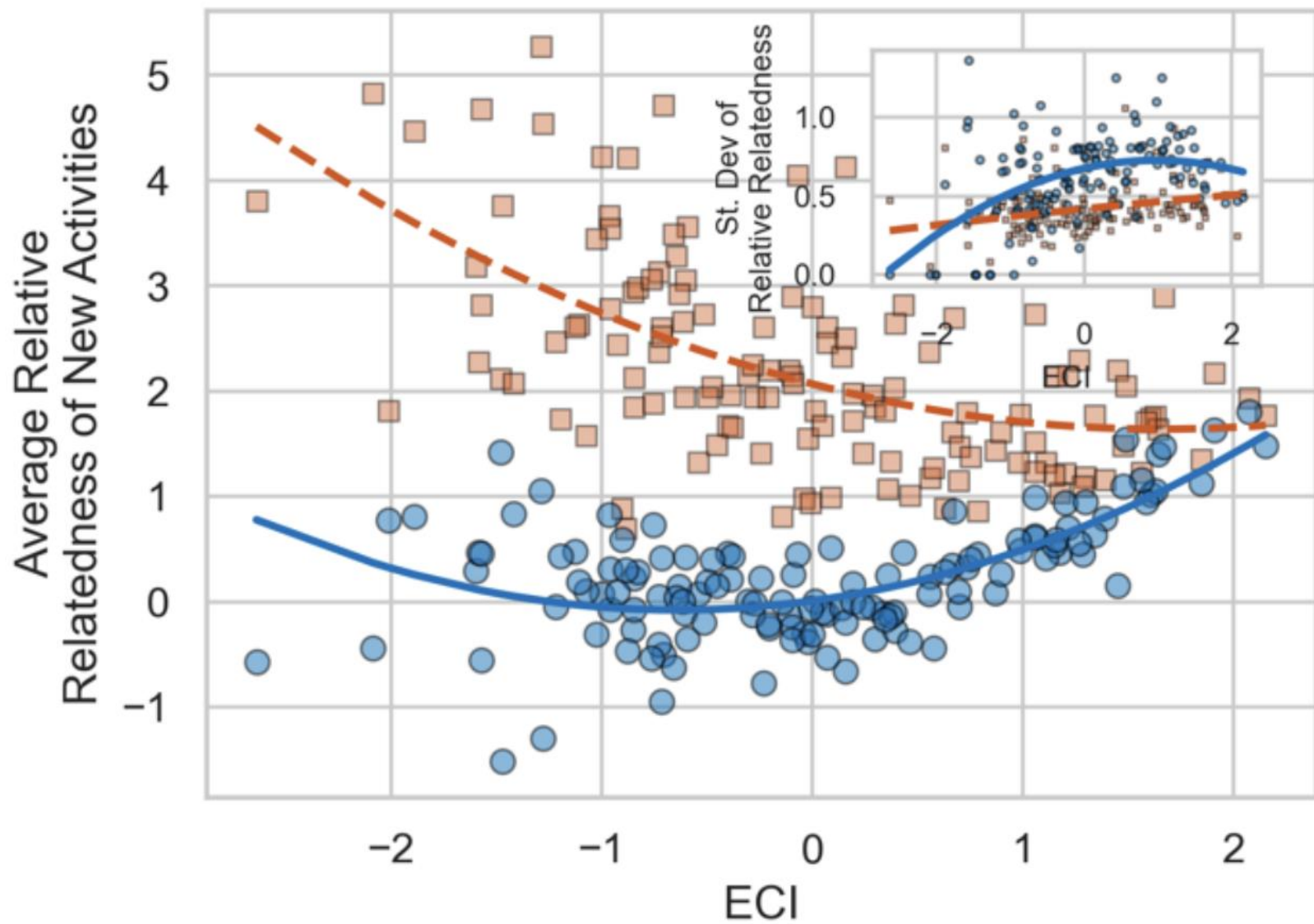
Economic complexity at time $T+\Delta T$
(TARGET)

Diversity at Time $T+\Delta T$

Specialization Matrix at Time $T+\Delta T$
(KEY VARIABLE)

Product Complexity at Time $T+\Delta T$







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Simple input ☒

Include Gold ☒ Include Oil ☒ Include Digital Products ☒

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📊 **Medium** (Growth: 1.166% - ECI: 1.100)



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To Conclude

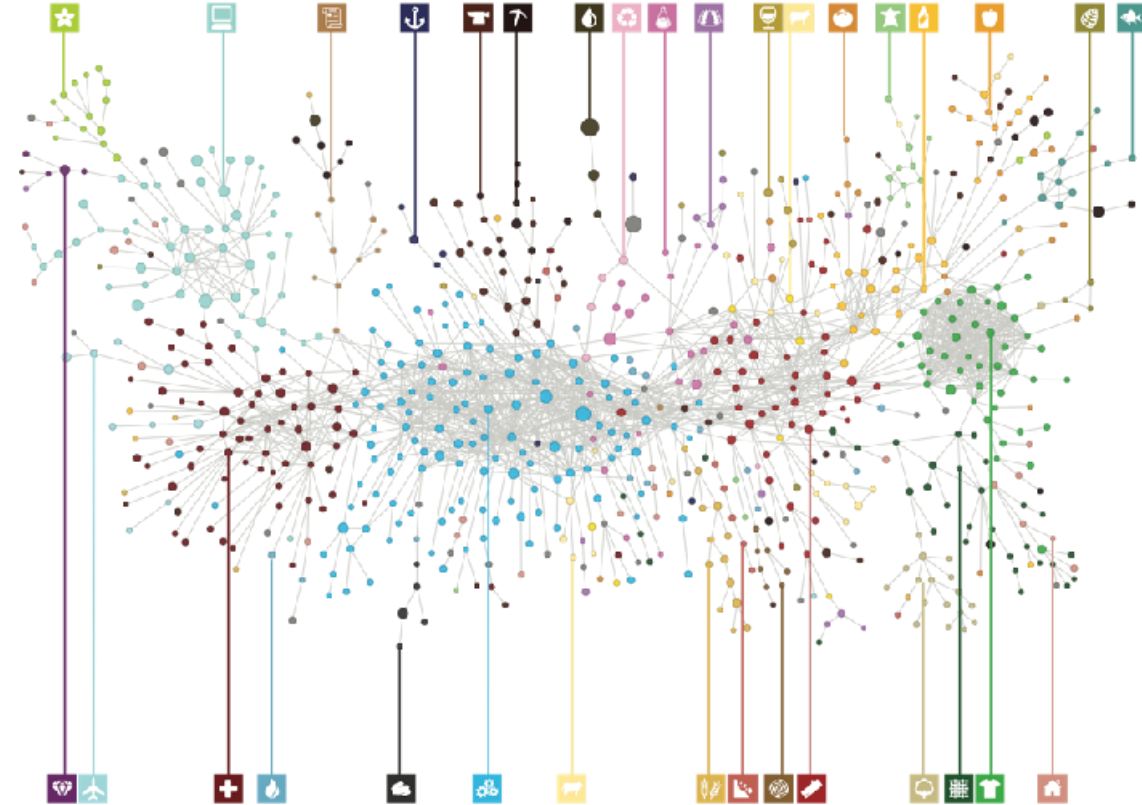
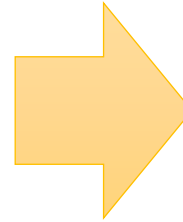
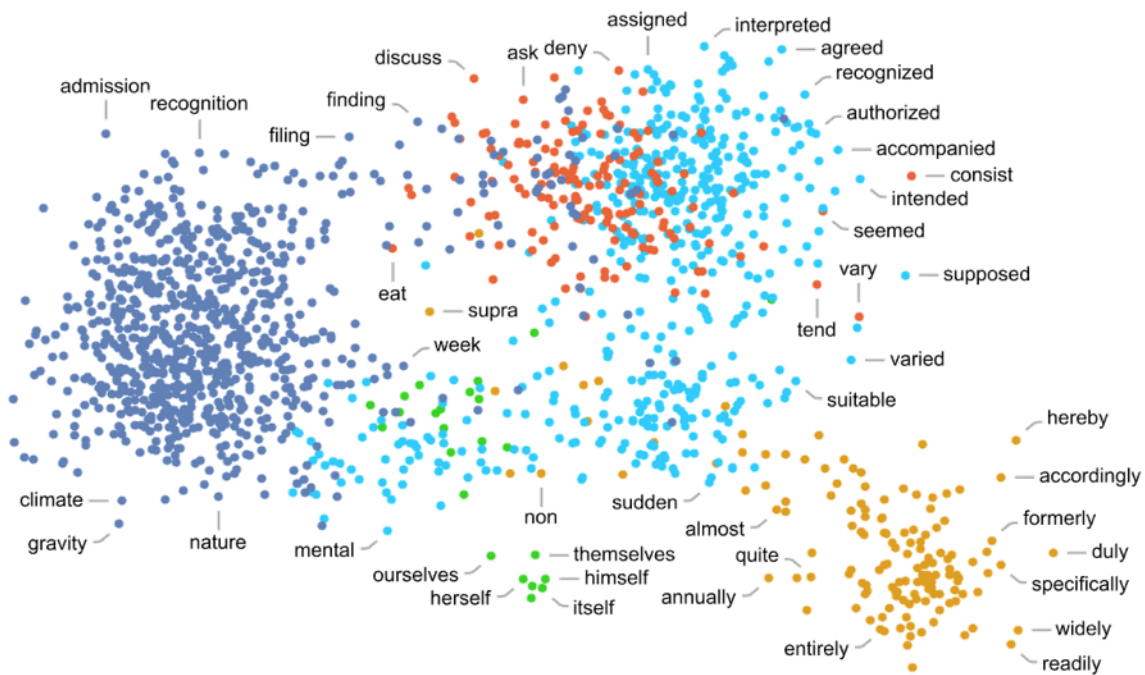


**Economies
are Complex**

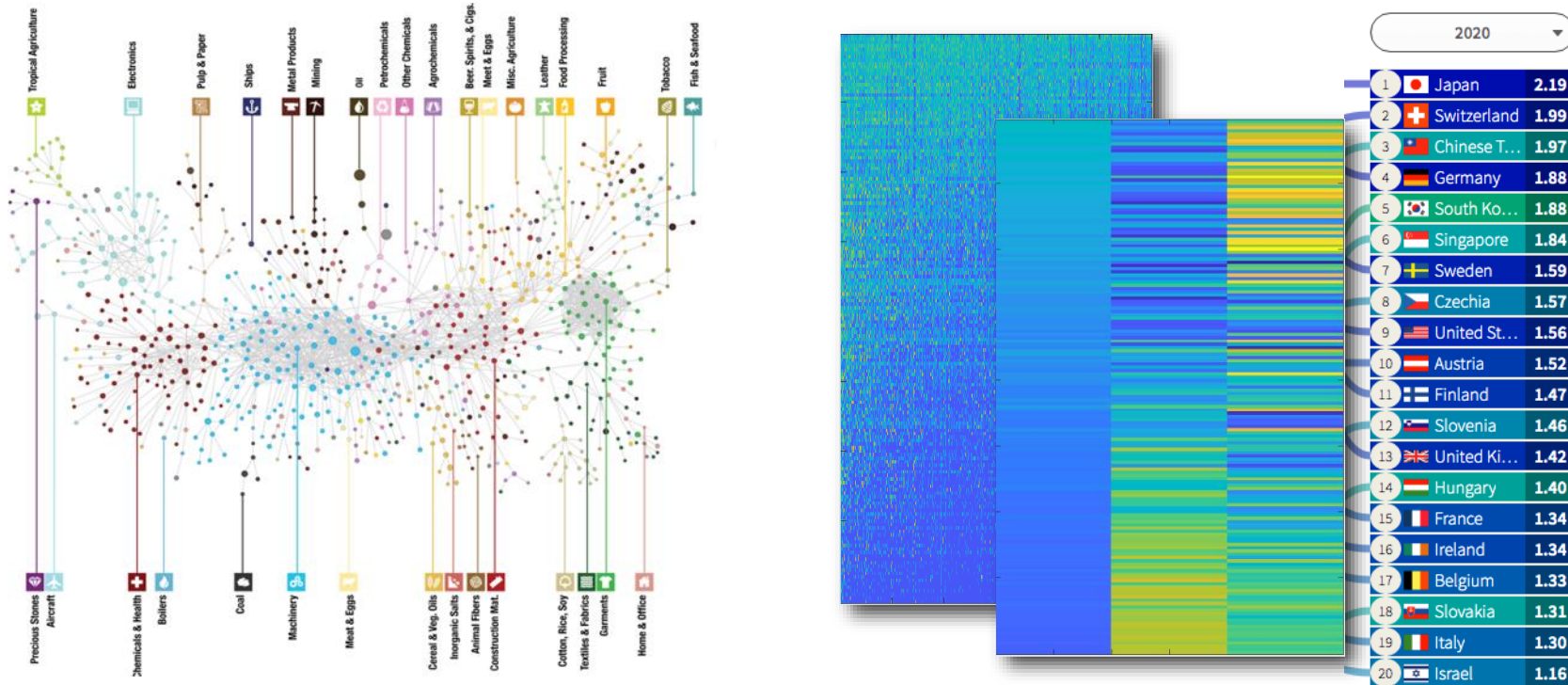
Because
Knowledge is
non-fungible !

Because
Knowledge is
non-fungible !

To understand the economy in detail we need granular representations



Economic complexity provides granular representations of economies that can help us understand where they are and where they are going.



Research: centerforcollectivelearning.org
Tech: datawheel.us